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ABSTRACT

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BSH23-PO02 | Novel HOXA3-HOXA9 fusion genes in acute myeloid leukaemia: The bride or the bridesmaid?

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The dysregulations of HOXA genes in acute myeloid leukaemia (AML) are evident in the immortalisation of leukaemic cells. Hitherto, little is known about fusion occurring among the HOXA genes and their clinical relevance in AML. In this study, the role of novel cryptic fusion genes HOXA3-HOXA9, which involves HOXA9 activation region detected using transcriptomic sequencing in the prognostication of AML-normal karyotype (AML-NK), was elucidated.
A total of 51 AML-NK subjects were subjected to high throughput deep transcriptomic sequencing using NovaSeq 6000 (200 M reads per sample, 150 bp paired-end). Customised transcriptomic pipelines were utilised to detect HOXA fusions and their expression profiles. HOXA3-HOXA9 fusions were correlated with their DEG profiles. In-silico validation and Sanger sequencing were performed on novel chimeric genes and DEGs. The prognostic significance of the HOXA3-HOXA9 fusion in quadrochotomised AML-NK patients based on NPM1 mutation (NPM1mut) and FLT3-internal tandem duplication (FLT3-ITD) or wildtype (FLT3wt) status was scrutinised. The Kaplan–Meier curve analysis and log-rank test were used in univariate analysis to assess potential prognostic factors in predicting patient survival. A prognostic scoring model was developed based on HOXA3-HOXA9/FLT3-ITD/NPM1 genotype, age groups below and above 60 years, and stem cell transplantation (SCT) using Cox regression multivariate analysis.
The novel HOXA3-HOXA9 fusion was detected in 16% (8/51) of the AML-NK patients, present with coexisting FLT3-ITD/NPM1mut and FLT3wt/NPM1mut in 50% of the cases, respectively. 50% of the AML-NK patients with HOXA3-HOXA9 fusion had poor overall survival (OS < 5 years). Patients with genotype negative for HOXA3-HOXA9/FLT3wt/ NPM1mut were assigned scores as prognostication predictors. Our prognostic scoring model designed for negative HOXA3-HOXA9/FLT3wtand NPM1mut showed that scores ≥2 with age < 60 as a confounding factor yielded to OS > 5 years. However, an in-depth analysis of the HOXA3-HOXA9/FLT3-ITD/NPM1 genotype and the DEG profiles did not show any statistical or clinical association.
The scoring model based on HOXA3-HOXA9/FLT3/NPM1 genotype developed in this study is applicable for AML-NK prognostication. We deduce that the novel HOXA3-HOXA9 fusion is a veiled bride on the aisle of AML-NK's dismal prognosis altar, necessitating the larger prospective cohort validation.

BSH23-PO03 | Demystifying acute myeloid leukaemia-normal karyotype by deep transcriptome sequencing through gene expression and survival analysis

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An insight into how signalling pathways govern cell fate is crucial in understanding the prognostication of acute myeloid leukaemia present with a normal karyotype (AML-NK), as aberrant cell signalling enhances the survival and proliferation of leukaemic cells. Hitherto, the transcriptomic landscape and gene expression profiles of the AML-NK patients that predisposes to their dismal outcome remain veiled. In this study, the differentially expressed genes (DEGs) of AML-NK patients with poor overall survival (OS) < 5 and good OS > 5 years were profiled and functionally enriched pathways were shed light on.
High throughput deep transcriptome sequencing was performed on 45 AML-NK patients using NovaSeq 6000 (200 M reads per sample, 150 bp paired-end). The AML-NK patients

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that were recruited consisted of 40% (18/45) having overall survival (OS) > 5 years and 60% (27/45) having OS < 5 years. Unsupervised clustering method was conducted on AML-NK patients with OS < 5 and OS > 5 years. A customised differentially expressed gene (DEG) pipeline was utilised for this study to compare the OS < 5 and OS > 5 years groups. DESeq2 was used to perform DEGs with the following filters: *p*-adjusted value (*p*adj) < 0.05, log fold change (lfc) >1 and <–1, and basemean >100. Functional enrichment analysis on DEGs using WebGestalt for gene set enrichment analysis (GSEA) for pathway enrichment by using the Kyoto Encyclopaedia of Genes and Genomes (KEGG) and Gene Ontology (GO) by filtering based on *p*-value < 0.05 and false discovery rate of ≤0.05. The Kaplan–Meier curve analysis and log-rank test were used to assess the OS < 5 and OS > 5 years groups, and prognosis-related genes were scrutinised by Cox regression univariate and multivariate analysis.

A total of 211 significant DEGs were discovered between the OS < 5 and OS > 5 years groups, consisting of 190 protein-coding genes, two pseudogenes, two processed transcripts, nine lncRNAs and two lincRNAs. Functional analysis revealed that the upregulated DEGs were significantly enriched in the JAK–STAT signalling pathway and cytokine-cytokine receptor interaction involving six upstream genes (INHBA, STAT4, IL15, IL2RA, IL12RB2, and TNFS8) that have been implicated with the poor OS using Cox multivariate regression analysis.

This study revealed that INHBA, STAT4, IL15, IL2RA, IL12RB2, and TNFS8 are prognosis-related genes in AML-NK with JAK–STAT signalling and cytokine-cytokine receptor interaction, precipitating poor OS and are potential biomarkers in AML-NK therapy targeting the immune pathways.

BSH23-PO04 | CD7 receptor expression as a therapeutic target in acute myeloid Leukaemia

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Introduction: The lymphoid receptor CD7 is aberrantly expressed in AML. Data from our group, and others, shows CD7 expression in up to a third of patients. It has been reported that CD7+ AML patients perform worse in key clinical outcomes including rates of relapse, remission, and relapse free, disease-free and overall survival. Therefore, CD7 expression represents a biologically and clinically relevant target in AML. Working with BiVictriX Therapeutics, a novel bispecific antibody-drug conjugate (ADC) against CD7 and CD33 is being developed for use in AML. Dual expression of CD7 and CD33 is unique to AML blasts; therefore, healthy cells are spared whilst malignant cells are eliminated. In this study, we aimed to elucidate the function of CD7 in AML to deduce how expression contributes to poorer outcomes. We also examined CD7-mediated internalisation of the bispecific-ADC by CD7/CD33 dual expressing AML cells, to gain better mechanistic insights into ADC processing by AML.

Methods: AML cell lines were stained with anti-CD7 fluorophore-conjugated antibody, washed and analysed by flow cytometry. CD7+ AML cell lines were stimulated with CD7 ligand SECTM-1 for 20 min, lysates were collected, the protein immunoblotted and probed for signalling proteins. Gemtuzumab and bispecific-ADC were conjugated to Alexa-fluor-647 for immunofluorescence. CD7+/CD33+ AML cells were incubated with indicated compound at 4°C, then for 1 h at 37°C. Cells were co-stained with LAMP-1-Alexa-fluor-488 and DAPI, and examined using the Zeiss Elyra 7 super-resolution microscope under the Structured Illumination Microscopy (SIM) setting at total 630× magnification.

Results: 4/7 AML cell lines tested showed expression of CD7. CD7+ cell lines showed increased phosphorylated ERK and AKT following incubation with SECTM-1. Co-culture of HS5 stromal cells, in which levels of SECTM-1 secretion is manipulated, with CD7+ AML cell lines has been used to examine paracrine signalling-mediated phenotypic changes of AML cells induced by CD7 activation or attenuation. Gemtuzumab, an ADC currently used in AML treatment, was included for comparison in endocytosis studies. Bispecific-ADC and Gemtuzumab were located at the surface at 0 min. Endocytosis of the bispecific-ADC and Gemtuzumab were similar, and co-localisation with LAMP-1 could be seen at 30 min.

Conclusions: CD7 is frequently observed in AML cell lines. Stimulation of CD7+ AML cells with SECTM-1 showed, for the first time, that CD7 is functional in AML, and that stimulation causes activation of downstream signalling. Rapid internalisation of bispecific-ADC was observed, validating the feasibility of targeting the combination of CD7 and CD33.

BSH23-PO05 | NRAS mutations contribute to drug resistance in acute myeloid Leukaemia

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Acute Myeloid Leukaemia (AML) is a highly heterogenous disease characterised by poor prognosis and high relapse incidence, with most diagnoses occurring in patients over 60 years old.[1] Patients are initially treated with induction chemotherapy, comprising cytarabine plus an anthracycline. Best outcomes are traditionally seen with allogeneic haematopoietic stem cell transplant; however in older patients, this is often unsuitable, partly due to comorbidities.

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Thus, targeted therapies are becoming increasingly trialled and utilised, including midostaurin, gilteritinib and quizartinib. These target FLT3, the most common leukaemogenic driver mutation in patients. However, whilst these are initially beneficial to FLT3-ITD mutation-positive patients, drug resistance remains an issue, which has been somewhat attributed to the emergence of NRAS mutations.[2] Here, we discuss our use of parental and NRAS-overexpressing AML cell line models to assess the extent to which emergence of NRAS mutants contribute to resistance phenotypes.

MOLM-13 (FLT3-ITD+/NRAS-WT) cells were made to overexpress NRAS wildtype, G12C, G12D or Q61K by lentiviral transduction. Successful overexpression was confirmed by Western blotting. Sensitivity to various AML-associated small molecule inhibitors was examined in these overexpressing cells, parental line and previously generated drug-resistant NRAS-mutant+ MOLM-13, using Annexin V/Propidium Iodide staining. Proliferation capacity was quantified through trypan blue counting and CFSE staining, over 96 h. Alterations in cycle status were measured by 7AAD staining. All fluorescent stains were quantified by flow cytometry. Colony forming capacity was measured after plating for 10 days in methylcellulose.

NRAS-G12C overexpression increased sensitivity to cytarabine, gilteritinib and quizartinib compared to parental cells, whereas NRAS-Q61K decreased sensitivity to cytarabine. Overexpression of the three NRAS mutants increased proliferative capacity, with colony forming capacity also significantly increased in MOLM-13-NRAS-G12C and MOLM-13-NRAS-G12D ($p < 0.05$), compared to parental, NRAS-WT overexpressing and drug-resistant controls. 7AAD staining also somewhat indicated a level of cell cycle arrest in G1 phase, compared to the NRAS wildtype overexpressing cells.

In summary, some NRAS mutations decrease sensitivity to AML therapies compared to a parental FLT3-ITD+ cell line. This resistance potential is greatest with NRAS Q61K, compared to G12 mutants, though the G12 mutants are perhaps more leukaemogenic. Overall, this suggests potential for treatment stratification based on a patient's NRAS mutational signature.

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BSH23-PO07 | Outcomes in B-cell precursor acute lymphoblastic leukaemia patients receiving inotuzumab ozogamicin prehaematopoietic stem cell transplantation

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Background: Inotuzumab ozogamicin (InO) is a CD22-directed antibody–drug conjugate for treatment of relapsed/refractory (R/R) B-cell precursor acute lymphoblastic leukaemia (ALL). This study examined posthaematopoietic stem cell transplantation (HSCT) outcomes in patients receiving InO prior to HSCT.

Methods: This US observational, postauthorization safety study reports outcomes in adults with ALL and R/R ALL who received InO and proceeded to first allogeneic HSCT (4-year interim data). Outcomes included disease status, overall survival, non-relapse (transplant-related) mortality (NRM), non-transplant-related mortality, and adverse events (Aes) of interest, including veno-occlusive disease/sinusoidal obstruction syndrome (VOD/SOS).

Results: 196 adults (median age 39 years, 54% male) received InO preallogeneic HSCT: 32% in first complete remission (CR1), 46% in CR2, 12% in CR ≥ 3, 5% in first relapse, 2% in third or subsequent relapse, and 3% with primary induction failure. Pre-HSCT, 32%, 45%, and 17% received 1, 2, and ≥3 cycles of InO, respectively.

Post-HSCT outcomes were analysed in 177 adults with ALL and 120 with R/R ALL. In ALL and R/R ALL patients, respectively, 12-month overall survival was 58% (95% CI 50–65) and 52% (42–61); 12-month NRM was 21% (15–28) and 24% (16–32) and non-transplant-related mortality was 22% (16–29) and 25% (17–33). Relapse rate (12-months) was 34% (27–42) and 36% (27–45), respectively. The most common causes of NRM were VOD/SOS (29%) and graft-versus-host disease (GVHD; 23%). Aes occurring in ≥30% patients ≤100 days post-HSCT were bacterial infection (46%), viral infection (41%), and acute GVHD (38%). VOD/SOS incidence ≤100 days post-HSCT was 15% and 19% in ALL and R/R ALL patients; corresponding mortality rates were 38%

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and 36% in patients who developed VOD/SOS ≤100 days (6% and 7% as a percentage of all patients). Median (range) time from HSCT to VOD/SOS was 0.4 (0.2–2.6) months in both groups. Of 26 adults with ALL who developed VOD/SOS, 10 were mild, 16 severe; 5/26 patients received prophylactic defibrotide; 12/26 received defibrotide alone or in combination. In those who died, median time to death was 4.5 (0.4–39.2) months.

Conclusions: In this real-world population of ALL patients who received pre-HSCT InO, including heavily pretreated patients, the incidence of VOD/SOS post-HSCT was similar to that observed in INO-VATE (phase 3 study) and a pooled analysis of 2 clinical trials of InO-treated patients with R/R ALL. Although VOD/SOS mortality was higher (36% vs. 26%), 12-month NRM was lower (24% vs. 38%) vs. the pooled clinical trial population.

BSH23-PO08 | Genes involved in Wnt and hippo pathways predict survival in adult B-acute lymphoblastic leukaemia

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Introduction: Adult B-cell acute lymphocytic leukaemia (B-ALL) is a rare cancer, and accounts for approximately 16% of all acute leukaemias in adults. Patients have a below-average survival at 10 years (currently around 45%) compared to all people diagnosed with cancer in England and Wales in 2010–11. Although most patients achieve first remission, aided by the recent introduction of tyrosine kinase inhibitors (TKI) for Philadelphia and Philadelphia-like subgroups that constitute 50% of B-ALL in adults, the development of TKI resistance suggests a need to find additional targets for treatment.

Methods: We identified B-ALL antigens through different techniques were examined to determine whether their above median expression was associated with patient survival as follows. Five methods including CTA database along with 180 genes/antigens identified in the cancer-testis antigen database,[1] SEREX using cleared sera from 5 B-ALL patients against testes cDNA library, DEG found using microarray (GSE38403) used gene expression data from 215 B-ALL patients prior to treatment in comparison to 12 healthy donor controls[2] to identify differentially expressed genes (DEGs; >1.5 fold and $p < 0.05$) and GSE13204 (MILE) containing 205 B-ALL patients compared 74 healthy control, 19 DEG were highly identified using nine patients pretreatment compared to eight healthy donor age and sex matched of 9000 protoarray,[3] have been used to identify target for B-ALL. Then the BloodSpot dataset was used to identify genes associated with the survival of patients based on their subtypes of B-ALL.[4] Those genes associated with survival of $p < 0.05$ and highly expressed in specific B-ALL subtypes compared with healthy bone marrow were then analysed using the gene enrichment program called DAVID.

Results/Discussion: We found the Wnt, and hippo pathways to be frequently subverted in B-ALL. We found 4 genes are highly expressed in B-ALL and associated with the median survival rate. T-cell factor (TCF), which regulates the transcription of Wnt-regulated genes and Dishevelled (DVL3) a downstream effector of the Wnt pathway were both associated with poor survival (p-value 0.002 and 0.02, respectively). TEAD4 is a transcriptional factor that acts downstream of the hippo pathway whilst TAOK1 is a coactivator factor that involves hippo, both are associated with poor survival $p = 0.02$ and 0.03, respectively.

Future Directions: We will examine the expression of these genes in B-ALL patient samples and determine their role in B-ALL initiation and progression using cell line models.

BSH23-PO09 | NUP98 rearrangements in AML: A regional review of patient outcomes to evaluate current testing strategies

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Background: Over 30 nucleoporin-98 (NUP98) pathogenic rearrangements have been identified in haematological malignancies, preventing differentiation and promoting self-renewal. Although most frequently associated with high-risk paediatric acute myeloid leukaemia, cases have been reported across a spectrum of both myeloid and lymphoid malignancies spanning all age groups. Despite small numbers within the literature NUP98 rearrangements are consistently associated with poor clinical outcome.

The Haematopathology and Oncology Diagnostic Service at Cambridge University Hospitals (CUH), which provides Specialist Integrated Haematological Malignancy Diagnostic Services for East Anglia, changed its NUP98 rearrangement testing strategy in 2018. NUP98 FISH was requested prospectively in patients up to 75 years. with new or relapsed acute myeloid leukaemia, no other suitable MRD marker and negative AML FISH panel.

Methods: Patients were identified from the CUH cytogenetics database. Between 2018 and 2022 43 patients underwent

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NUP98 FISH. Testing for this cohort was performed by the cytogenetics service at Nottingham University Hospitals, although this is now performed in-house at CUH. 88% were tested at diagnosis and 12% at first relapse.

Results: Six positive cases were identified. Two were identified through the paediatric MyeChild trial with local confirmatory testing, and one patient aged 89 was tested to confirm a suggestive abnormality on SNP array, rather than through the new testing algorithm. Median age at presentation was 55 years (range: 01–89). Five had acute myeloid leukaemia and one acute leukaemia of ambiguous lineage. Additional pathogenic mutations included FLT3-ITD in one case, FLT3-TKD in one case, WT1 and CEBPA in two cases, TP53 in one and TET2 in another. Clinical outcome was poor with 67% (4/6) not surviving beyond 2 years from diagnosis despite three of the four receiving high intensity treatment including allogenic transplant. The other two cases have not yet reached 2 years from diagnosis but are well following allogeneic transplant.

Discussion: Findings were consistent with historical literature that NUP98 rearrangements are associated with poor clinical outcome, such that management as high-risk disease is appropriate. Although total numbers were small, the new inclusive testing strategy at diagnosis allows identification of these high-risk individuals early, escalation of therapy and identification of molecular markers of measurable residual disease. Given the age range of positive cases, our data support NUP98 rearrangement testing for all AML patients of transplantable age and continuation of the new testing strategy.

BSH23-PO10 | Utilisation of wearable technology in an elderly population to improve health outcomes in MDS

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Background: There is much hope for the utilisation of data collected by wearable devices to improve health outcomes in a multitude of conditions. This has been used to good effect in many areas of healthcare. It is often overlooked in the elderly, perhaps due to perceived lack of engagement with technology. This unconscious discrimination may affect health outcomes. Myelodysplastic syndrome (MDS) is a condition, which predominantly affects the elderly population and hence potentially left behind with technological advances.

Methods: We conducted an experience study to assess the feasibility, comfort and usability of wearable accelerometer devices in a cohort of MDS patients. Participants wore an accelerometer for a period of 8 weeks. We assessed the utility of the data to understand changes related to transfusion and patients' experiences with the device and ability to engage appropriately with the technology. They undertook semi-qualitative surveys at week 4 and 8 as well and an in-depth qualitative interview at the conclusion of the study.

Results: Sample size was limited due to device availability. Three participants completed the study. All rated the device as 'very comfortable' or 'comfortable' and only removed them for cleaning and charging with 1 episode of removal across the total 24 weeks due to discomfort when sleeping. Participants stated they forgot the device was on and all would wear it again if it helped to determine transfusion need.

The data collected was complete, usable and repeatable, and therefore, likely to uncover a relationship between anaemia and changes in activity parameters if our hypothesis is correct.

Discussion: The use of technology to aid data capture and explore treatment efficacy is not novel or innovative. Yet it is often done poorly, if at all, in analysing existing treatments—more so in the elderly population. This small study shows that patients can engage effectively with technology and quality data can be obtained.

In MDS, optimisation of available therapies and prioritisation of patient-related outcome measures (PROMs) is of paramount importance. This will help move away from a numbers-led approach. With improving technology and access to large arrays of data biochemical, molecular, genomic and activity, it is important we focus on PROMs to guide research outcomes.

As highlighted by Patient Blood Management guidelines, there is a lack data to guide optimal transfusion practices in MDS. To better design future trials, we need to firstly identify a reliable and objective measure of the benefits treatment.

BSH23-PO100 | Experiences of flying in patients with sickle cell disease (SCD): A service evaluation project

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Introduction: Flight-related hypoxia, dehydration, and cold temperatures have potential to provoke sickle cell crises. We examined the experiences of people with sickle cell disease (SCD) around flights and advice given by clinicians to prevent flight-related crises.

Methods: Questionnaires were sent via email to consultants and nurse specialists working with SCD patients in a Haemoglobinopathy Coordinating Centre network on the advice they give SCD patients due to fly.

Telephone interviews were undertaken with a cohort of adult SCD patients attending one large tertiary centre within the

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network. Questions covered preflight communication with their clinicians, in-flight behaviours, access to oxygen and complications experienced during or after flights.

Results: 15 of 31 clinicians responded. 47% manage children and 53% adults, with over 70% managing >200 patients. 53% routinely enquired about flying, 93.3% knew of patients with complications after flights. All clinicians had experience of recommending in-flight oxygen. 6.7% advised all patients, 53.3% if history of chest crises or flights >6 h, and 80% if SpO2 < 94% on air. Additional free-text responses included: those on overnight oxygen, low baseline haemoglobin, prior flight complications, and on respiratory physician's advice. 66.6% of clinicians completed Medical Information Forms for patients needing oxygen. 40% of clinicians had access and were able to request hypoxic challenge tests—none were aware of any patient developing a crisis afterwards.

36 patients were interviewed. Median age was 40.5 years and 66.7% had genotype HbSS. 97.2% had flown before with 22.9% over 20 times. 45.7% of flyers had a crisis whilst flying or within two days, 56.3% of these required hospital review and subsequently 88.9% were admitted as inpatients. 54.3% inform their SCD team before flying, 44.4% knew oxygen was available, and only 8.3% had ever used in-flight oxygen—with two-thirds of those being unplanned. Flight behaviours were variable, with only 30.6% always carrying a clinic letter, 61.1% keeping warm, 66.7% maintaining hydration and 80.6% keeping analgesia in their hand luggage.

Discussion: Despite 93.3% of clinicians knowing patients with crises after flights, and 45.7% of patients experiencing this, only 53% of clinicians and 54.3% of patients communicated about flying. There was variability in patients recommended for in-flight oxygen as well as strategies to mitigate risks in-flight. Few patients are aware oxygen is available, and fewer have received it. Travel screening should be integrated into annual reviews to improve education. We highlight here the need for clinical guidelines.

BSH23-PO101 | Uptake of walk-in ophthalmology screening in sickle cell disease and iron chelation

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Background: The most common ophthalmic complication of sickle cell disease (SCD) is sickle retinopathy. This can result in visual loss although is often asymptomatic in the early stages, so identification of retinopathy most often occurs during specialist eye examination. The incidence and severity of proliferative sickle retinopathy is greater in sickle cell/haemoglobin C disease with peak prevalence around 20–35 years. Retinopathy laser photocoagulation therapy may prevent visual loss and vitreous haemorrhage. National guidelines call for all patients to have baseline retinopathy screening, with ongoing regular review for those with evidence of retinopathy. Treatment with iron chelation (deferasirox, desferrioxamine) is also associated with ocular disturbances and annual ophthalmology screening is recommended.

Methods: Ophthalmology screening for red cell patients was historically arranged following direct written referral to the ophthalmology department by haematology, with an ophthalmology appointment offered to the patient as per all other hospital areas. Patients rarely attended the offered appointment and new referrals and appointments were required on multiple occasions.

A switch to walk-in ophthalmology screening was implemented in November 2021. Generic invite letters asking red cell patients to attend the ophthalmology clinic weekdays 10–3.30 for screening was agreed between haematology and ophthalmology. The letter included instructions for the Ophthalmology Technician regarding required tests. Results were reviewed later by a dedicated Ophthalmologist and letter dictated, with follow-up arranged as necessary.

This retrospective audit and service evaluation reports on the impact of this change on screening uptake and use of NHS resources.

Results: Compliance with baseline retinopathy screening in SCD improved from 10% (2/21) to 55% (12/22) post introduction of walk-in screening appointments. Generic screening invite letters were given to patients at clinic, and resent at 4–6 months where the patient had not attended. This resulted in minimal administrative burden for haematology.

No change was seen in overall attendance for patients on iron chelation, which remained at 56% (5/9 although different patients). However, the average number of ophthalmology appointments booked per actual attendance reduced from 4 to 1 in this cohort following implementation. This change resulted in a significant reduction in NHS resources (appointment rescheduling, dictation of rereferral letters following non-attendance, sending appointment letters).

Only 1 patient had to reattend the department as the equipment was out of service the day he attended.

Conclusion: Walk-in ophthalmology screening for sickle retinopathy and complications of iron chelation improves both compliance with national standards and overall cost effectiveness for the NHS.

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BSH23-PO102 | Sickle cell health awareness, perspectives and experiences (SHAPE) survey: UK Physicians' perspectives on SCD

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Introduction: Sickle cell disease (SCD) adversely affects patients' physical and mental well-being; however, there is limited research on the impact of SCD on quality of life. In 2016, approximately 14 000 people in the UK (~1 in 4600 people) were living with SCD. The SHAPE survey is a multinational study examining the global burden of SCD.

Objective: This report analyses the experiences of physicians in the UK within the context of insights obtained internationally.

Methods: Physicians in the UK, Germany, France, the US, Brazil, Canada, Saudi Arabia, and the UAE answered closed-ended questions on their experiences treating patients with SCD. Physicians practicing for 3 to 35 years with ≥10 patients with SCD under their care were eligible.

Results: Of 219 global physicians surveyed ('all physicians'), 30 were from the UK. Physicians in the UK had an average of 65 patients currently under their care and an average of 12 years of experience treating patients with SCD. UK physicians identified bone aches as the symptom most commonly reported by patients aged ≤11 years (65%) and ≥18 years (69%) and fatigue/tiredness as the symptom most commonly reported by patients aged 12 to 17 years (59%), whereas all physicians surveyed identified vaso-occlusive crisis pain as the symptom most commonly reported by patients aged ≤11 years (63%) and fatigue/tiredness as the symptom most commonly reported by patients aged 12–17 years (68%) and ≥18 years (76%). Most UK physicians (90%) agreed that their patients often stated that 'fatigue or tiredness affect[ed] them a lot' and believed that early treatment of SCD can help manage or prevent severe symptoms (90%). Compared with all physicians, a greater percentage of UK physicians were concerned about their patients' non-adherence to treatment (67% vs. 46% total). A total of 87% of UK physicians wanted more support in educating their patients about the long-term effects of SCD. Most UK physicians (87%) believed that treating haemolytic anaemia can improve most aspects of SCD, including reducing end-organ damage. Although 80% of UK physicians agreed that their patients face financial difficulties due to SCD, only 43% of them agreed that the cost–benefit ratio factors into their decision to recommend a treatment.

Conclusions: The survey findings provide insight into areas of patient care for which UK physicians need more support and highlight the need for effective treatments that target underlying disease pathology.

BSH23-PO103 | Long-term complement inhibition and survival outcomes in paroxysmal nocturnal hemoglobinuria: Ravulizumab interim analysis

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Paroxysmal nocturnal haemoglobinuria (PNH) is a rare, chronic, hematologic disorder characterised by uncontrolled terminal complement activation, intravascular haemolysis, thrombotic events and significant morbidity and mortality. Ravulizumab is a treatment for patients with PNH, which offers complete complement C5 inhibition throughout the 8-week (q8w) dosing interval.

This analysis utilised pooled long-term survival data across the ravulizumab phase 1b, 2 and 3 trials (103 [NCT02598583], 201 [NCT02605993], 301 [NCT02946463] and 302 [NCT03056040]) with up to 4 years of open-label extension (OLE). Complement C5 inhibitor (eculizumab/ravulizumab) and dose received varied during the initial treatment period. During OLE, patients received weight-based dosing of intravenous ravulizumab q8w. Deaths reported, patient survival over time and adverse events resulting in patient death were analysed. Instances of patients that died owing to infection or sepsis were specifically analysed to provide insight into the events leading to death.

A total of 475 patients with PNH treated with ravulizumab over 1479.0 patient years of follow-up were included. Of these, 12 patients (2.5%) died during the initial treatment period and OLE (overall incidence of 0.8 per 100 patient years). Nine of these deaths occurred in Years 1–3, two in Year 4 and one in Year 5. Of the 12 deaths, six were attributed to infection or sepsis. These patients were predominantly male (83.3%),

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and white/Asian (50.0%, respectively), and median (range) age was 61 (43–75) years. In addition, of these six patients, two had aplastic anaemia and one patient had pancytopenia. Most (83.3%) of these patients were hospitalised owing to onset of infection; however, one patient (white male; 43 years old; ravulizumab to ravulizumab) was hospitalised owing to worsening of aplastic anaemia. During hospitalisation, this patient was diagnosed with acute respiratory infection and died owing to sepsis. One patient (Asian male; 62 years old; ravulizumab to ravulizumab) died owing to meningococcal sepsis (strain unknown). This patient was vaccinated against meningococcal groups A, C, Y and W-135, but had not received prophylactic antibiotics. Other causes of death included cardiac disorders (n = 1), neoplasms (n = 4; acute myeloid leukaemia, metastatic urothelial carcinoma of the kidney, metastatic malignant neoplasm of the lung and non-small cell lung carcinoma) and respiratory disorders (n = 1). This analysis reports the longest period of follow-up in 475 patients with PNH treated with ravulizumab. Long-term ravulizumab treatment was associated with a low incidence of death and few patients died owing to infection, supporting its long-term use for PNH.

BSH23-PO104 | Vemircopan (ALXN2050) monotherapy in paroxysmal nocturnal haemoglobinuria: Interim data from phase 2 Open-Label Proof-of-Concept study

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Paroxysmal nocturnal haemoglobinuria (PNH) is a rare haematopoietic stem cell disorder associated with intravascular haemolysis (IVH) and risk of thrombosis. Complement C5 inhibitor treatment has greatly improved clinical outcomes and survival in PNH; however, some patients taking C5 inhibitors develop clinically significant extravascular haemolysis (EVH) that may result in symptomatic anaemia and require blood transfusions. Inhibition of factor D, rate-limiting enzyme of the complement alternative pathway (AP), may control IVH whilst preventing EVH. The oral factor D inhibitor ALXN2050 (vemircopan) provides immediate, complete, and sustained AP inhibition with twice daily administration (BID).

An ongoing, phase 2, open-label, proof-of-concept study (NCT04170023) is assessing the efficacy, safety, and pharmacokinetics/pharmacodynamics of vemircopan monotherapy in patients with PNH. This interim analysis reports efficacy and safety data from a subpopulation of treatment-naïve patients. Patients receive vemircopan 120 mg BID, which can be escalated to 180 mg BID per investigator discretion based on guidelines in the protocol. Primary endpoint is change from baseline to week 12 in Hgb. Key secondary endpoints include change from baseline to week 12 in lactate dehydrogenase (LDH) and Functional Assessment of Chronic Illness Therapy—Fatigue (FACIT-Fatigue) score. Safety is reported as treatment-emergent adverse events (TEAEs).

As of data cut-off date of 30 April 2022, 11 treatment-naïve patients were enrolled. Nine patients completed 12 weeks and were included in this analysis. From baseline to week 12, mean (SD) Hgb increased by 3.9 (1.11) g/dL from 7.9 (1.29) g/dL to 11.8 (1.25) g/dL, LDH rapidly decreased by 81% from 7 × upper limit of normal (ULN) to 1.4 × ULN, absolute reticulocyte count decreased from 212.4 (86.47) × 10^3/μL to 120.0 (51.44) × 10^3/μL, and FACIT-Fatigue score improved by 13 points. One patient needed transfusion (on day 2, owing to low Hgb [5.1 g/dL]). As of data cut-off, 31 TEAEs were reported (n = 9 patients), most (25/31; 80.6%) considered unrelated to study drug. No serious TEAEs, grade ≥ 3 TEAEs, discontinuations, or deaths were reported. The most common TEAE was headache (n = 4; 36.4%). There were no thrombotic events, seizures, or meningococcal infections. Seven of 11 patients received dose escalation from 120 to 180 mg BID, which was associated with increased Hgb levels. This interim analysis of treatment-naïve patients with PNH suggests that oral vemircopan monotherapy controlled IVH and prevented EVH. No new safety signals were identified during the 12-week period. This analysis provides proof of concept of vemircopan in PNH and suggests that phase 3 trials are warranted.

BSH23-PO105 | Determinants of splenic preservation among patients with sickle cell disease in North-Eastern Nigeria

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Background: The spleen is one of the organs affected by sickle cell disease (SCD) early in life; repeated episodes of vaso-occlusion and infarction result in a reduction of both its structure and function. The transitions in the anatomic size of the spleen in SCD patients from normal to

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autosplenectomy and factors associated with this change have not been well-studied in areas where malaria is endemic. This knowledge may be useful in detecting factors inhibiting vaso-occlusion and may be of value for assessing determinants of splenic preservation among our SCD patients. The aim of this study was to explore clinical and laboratory variables associated with the preservation of the spleen determined by ultrasound among SCD patients.

Methods: Two hundred and four SCD patients (103 female; age 1–45 years) underwent abdominal ultrasonography at the University of Maiduguri Teaching Hospital, in NE Nigeria between October 2020 to November 2021. Imaging was performed by a single radiologist who assessed splenic visualisation and echotexture. Steady-state clinical parameters and blood samples for complete blood count, serum chemistry and high-performance liquid chromatography were obtained from all the patients.

Results: The spleen was visualised in 107 SCD patients (52.4%; 95% CI, 46%–59%) on ultrasound. There was a progressive decline in the visualisation of the spleen with increasing age; whilst the spleen was present in all the children less than 5 years of age, it was present in less than 24% of those aged >15 years. Visualisation of the spleen was significantly associated with low mean corpuscular haemoglobin concentration (MCHC) in all ages and high HbF in those less than 10 years. The odds of visualisation of the spleen on ultrasound scan increased by a factor of 1.17% for every 1% increase in HbF level. Only 32 (15%) patients were on regular hydroxyurea therapy; the HbF level was significantly higher among these patients on hydroxyurea (median 12.7 vs. 7.4; $p = 0.0001$).

Conclusion: In patients in Nigeria with SCD, autosplenectomy was not found in children less than 5 years old. Patients with visualised spleens had a higher level of HbF compared to those with absent spleens. HbF was significantly associated with visualisation of the spleen in patients less than 10 years of age. Early administration of hydroxyurea may be helpful in increasing HbF and consequently preservation of the spleen.

BSH23-PO106 | Prevalence, pattern, and predictors of sickle cell disease in children: Results of newborn screening

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Background: Nigeria continues to have the highest sickle cell disease (SCD) prevalence and mortality rates worldwide, despite increased advocacy and community education as well as improved diagnostic and therapeutic capacity. This is partly a result of the high prevalence of carriers of haemoglobin (Hb) S mutation in the population. Despite this, there is paucity of data from the country on newborn screening for SCD. This study documents the prevalence, pattern, and predictors of SCD in newborns at a secondary-level healthcare facility in Lagos.

Methods: This cross-sectional study enrolled mothers in the immediate postpartum period and their newborns. Sociodemographic and clinical data were retrieved from the mothers, following which a blood sample for Hb phenotype was drawn from their newborns via heel-prick. Blood samples were analysed using the HemoTypeSC point of care test, whilst statistical analysis was done using the statistical package for the social sciences (SPSS) version 28.0.

Results: A total of 417 mothers and 422 newborns were enrolled. The mean age of the mothers was 30.6 ± 5.2 (range: 18–43) years. There were slightly more newborn males than females (50.2% vs. 49.8%). The median (interquartile range) age of the newborns at the time of testing was 48.0 (24.0–72.0) h. The age or gender of the newborn was not statistically associated with a diagnosis of SCD.

Of the newborns, 75.1%, 21.8%, 1.4% and 1.7% had the haemoglobin AA, AS, AC, and SS phenotypes, respectively.

There was no statistically significant association between mothers' socio-demographic or obstetric characteristics and the prevalence of SCD in newborns. Lack of prepregnancy knowledge of Hb phenotype in the mothers [odds ratio (OR) (95% confidence interval (CI))] [0.044 (0.004–0.483), $p < 0.001$] or fathers [0.115 (0.025–0.526); $p < 0.001$] was significantly associated with SCD in the newborn.

Conclusion: The prevalence of SCD among newborns in this study was relatively low. The only form of SCD seen in newborns was the homozygous SCD form. Given that prepregnancy knowledge of Hb phenotype status was the only factor significantly predictive of SCD, there is a need for continued education to ensure prepregnancy and premarital Hb phenotype screening to reduce the likelihood of SCD in newborns in Nigeria.

BSH23-PO107 | Vaccination status of the adult sickle cell anaemia population: The South Wales experience

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The Sickle Cell Society have issued standards for additional immunisations that adults with sickle cell disease (SCD) require. These include annual influenza, 5-yearly pneumococcal conjugate vaccine (PPV23) and Hepatitis B vaccination. Patients who have not received their primary vaccination as part of the national schedule in the UK should also receive further additional vaccines. We reviewed whether adults with SCD in South Wales currently receive these.

49 adult patients were identified as having SCD under the care of the Hereditary Anaemia Service based in the University Hospital of Wales, Cardiff. GP records were not

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available for 5 patients leaving a final cohort of 44 patients to analyse. Average age was 33 years (range 17–67). Median age was 27 with the cohort predominantly lying in the 17–29 year category (52%).

Results showed good compliance with the annual influenza vaccine in those over 40 (>80%). However, compliance for the 17–29 category and 30–39 categories were 37.5% and 42.8%, respectively. The improved compliance in those >40 was not seen with the 5-yearly pneumococcal vaccine. Compliance was worse in all age groups compared to the annual flu vaccine with only 23% compliance overall. However, when looking at those who had received a single dose of PPV23, the numbers improved to nearly 60%.

Compliance with the SARS-CoV2 vaccination was highest at 61.3%. However, rates were lower in the 17–29 and 30–39 age groups in keeping with previous trends. Only 34.1% of patients had full hepatitis B cover. Again, trends in compliance mirrored previous with poorer rates in those under 40.

Assessing compliance for the remainder of the standards was more challenging given that we could not confirm retrospectively how many of our cohort had received their primary vaccinations in other parts of the UK, thought to be around half. However, most of the cohort had not received any additional vaccines suggesting high non-compliance regardless.

This review looked at data from 2020 and likely reflects the impact of the SARS-CoV2, whether positive or negative. The reduced compliance in 5-yearly pneumococcal compared to flu suggest better health-professional education is needed; if patients are attending for their annual flu vaccine, there is ample opportunity to administer other vaccines. The vaccination rate for our patient group is comparable to national rates by ethnicity although lower than the national average for age.

Vaccination rates for the SCD population of South Wales are not adequate. Better education and engagement is needed.

BSH23-PO108 | Incidence rates for acute complications in HbSS East London newborn sickle cell cohort (ELNSCS)

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Aim: To analyse the rates of acute complications in sickle cell disease (SCD) treated in the NHS over the past 30 years, compared with historical data.

Background: Management of SCD has evolved slowly over the past 30 years, with improvements in NHS care pathways and introduction of hydroxyurea (HU) and chronic transfusion therapy, simple (ST) or exchange (ET). The CSSD study analysed rates of complications in USA patients during the period 1979–1986, before these treatments were in widespread use.

Methods: The ongoing ELNSCS includes patients born in the London Boroughs of Hackney and Tower Hamlets, diagnosed by universal newborn screening programmes from 1983, and treated in our centre since diagnosis. All clinical data has been entered in a bespoke database and validated. Follow-up was from birth until the end of 2018. Subjects were censored on moving to another clinic, BMT, or at death. We calculated incidence rates (IR) per 100 patient year follow-up (+/−95% Confidence interval [CI]) and compared these with published rates from the CSSD study.

Results: Of 404 cohort subjects, 267 (66.1%) were HbSS with 3733 patient years of follow-up. Median age was 14.0 (range 3 months to 35) years. 21.3% were treated with HU, 21.5% with ST, 5.4% with manual ET and 9.8% with automated ET. In the paediatric age range (0–19 years) we confirmed reduced rates of mortality, acute ischaemic stroke and pneumococcal sepsis compared to the CSSD study. Some complications (dactylitis, aplastic crisis, splenic sequestration, hepatic sequestration, girdle syndrome) were only seen in the paediatric population. Rates of acute pain, priapism, acute ischaemic stroke and mortality progressively increased in the adult population, and pain events were more frequent in adults compared to the CSSD study.

Conclusion: Improvements in outcomes are seen during in childhood but not sustained in the adult population. Although uptake of HU has been suboptimal during follow-up, and increased uptake may have long-term benefits, there remains a large unmet need justifying development of effective new therapies to prevent acute complications of SCD.

BSH23-PO109 | Using wearables and web-based self-reporting to explore deep sleep and patient-reported-outcome correlations in sickle cell disease

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Background and Aim: Patients with Sickle Cell Disease (SCD) and their clinicians flag sleep quality as an important factor in daily well-being and the potential onset of a pain crisis. Previous relationships between sleep quality and both acute and chronic pain, as well as quality of life (QoL), overall physical health, functional disability, and clinical depression, have been identified. Nonetheless, there is limited data on the detailed impacts of specific sleep metrics in SCD. This research investigated associations between real-time sleep metrics and patient-reported outcomes, to measure their impact upon QoL.

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Methods: A CE-marked wearable monitoring device was provided to 85 participants with SCD and worn daily, automatically recording biometrics including sleep quality and heart rate. Participants manually self-reported EQ-5D-5L scores via a digital patient-reported outcomes (PRO) portal. Data for a final cohort of 54 patients (64%) who had both recorded sleep and EQ-5D-5L scores was analysed, and stratified into low, average, or high categories based on their sleep metric means compared to cohort terciles. Mean EQ-5D-5L scores were compared across these categories, and direct correlation analysis was performed to identify any links between sleep metrics and QoL.

Results: The mean hours of sleep per night was 7h10m (SD 56 m; taken over mean 109 days, SD 92 days). The mean proportion of deep sleep was 49% (SD 12%).

Of the wearable-captured sleep metrics analysed, statistically significant differences in EQ-5D-5L were seen only at different 'deep sleep' levels. Analysis of the associations between patients' mean percentage of deep sleep (as a proportion of total sleep duration) and their mean EQ-5D-5L scores found that patients within the lower range (0%–39%) had statistically significantly higher mean EQ-5D-5L scores than those in the high ranges (50%–83%) of deep sleep ($p = 0.030$). Moreover, direct correlation analysis identified a statistically significant association between increasing deep sleep levels and both decreasing haemoglobin levels ($p = 0.019$) and increasing pain scores ($p = 0.016$).

Conclusions: Our data identified an association between the amount of deep sleep patients with SCD have and both their EQ-5D-5L and pain scores, suggesting longer periods of deep sleep may be linked to lower patient-perceived well-being.

Further investigation is required to explore potential causal or confounding relationships, such as medication, in order to establish whether poorer QoL may see patients enter longer periods of deep sleep, or whether this widespread excess in deep sleep itself exerts an impact upon patient QoL in SCD.

BSH23-PO11 | Scoping review: The impact of ethnicity on acute myeloid Leukaemia (AML) outcomes and reporting consistency

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Black, Asian and other minority ethnic (BAME) patients experience significantly worse AML survival outcomes compared to Caucasian counterparts globally. The National Institute for Health and Care Research (NIHR) states that within haematology research there is a 'neglect impacting Black African, Asian and Caribbean patients'. This lack of research affects all aspects of AML-related healthcare, from guidelines influencing referral, investigative procedures (scans and biopsies), management decisions including end-of-life care and pain management.

A literature search was undertaken via Medline, the search terms included were; 'AML, Male, Demographic data, population, ethnicity' with additional limits being a 'phase 3 trial, English language, in the last 5 years'.

A total of 207 titles were found via Medline literature search, after duplicates were extracted 84 abstracts were reviewed and a total of 62 full texts were included in this rapid review.

Of the 62 papers: (Trends in mortality (23), research focus (2), racism (9), importance of demographic data in research (3), socio-economic factors (9), genetic differences (4), personalised care (10), and genetic mutation profiles (2)).

It should be noted that the proportion of UK AML research papers including racial demographic data was (5/176).

Investigation of the causes of this disparity in survival outcomes shows that the main barriers in BAME groups include; language and cultural barriers to research participation, job sector employment of ethnic minorities, and lack of trust in research institutions due to historical mistreatment. This causes a wider cycle of ethnic exclusion causing systemic racism within the healthcare system perpetuating further lack of inclusion and mistrust.

Other outcomes specific to healthcare and differential clinical treatment of BAME groups include; later presentation of symptoms in BAME patients, a higher rate of pre-existing comorbidities complicating diagnosis and treatment and significant barriers black males experience in receiving a tertiary referral. A shorter transformation time from MDS-SLD (myelodysplastic syndrome with single lineage dysplasia) to sAML (secondary acute myeloid leukaemia) is seen in black patients; a very aggressive form of AML with poor prognosis. Black patients also have a higher prevalence of the CBF-AML (core-binding factor) karyotype, associated with poorer survival. Lastly the lack of ethnic bone marrow donors, and the higher rate of graft rejection in BAME groups (16% higher) significantly affects survival outcome.

In light of the disparities identified in the literature review ethnicity needs to be consistently reported in UK trials and further discussion needs to be had surrounding ethnic survival outcomes in haematology.

BSH23-PO110 | Mortality and clinical complications among patients with sickle cell disease with recurrent VOCs in England

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Background: Sickle cell disease (SCD) is characterised by vaso-occlusive crises (VOCs) and a progressive clinical course leading to end-organ damage and early mortality. SCD complications are multifactorial and driven by vaso-occlusion, haemolysis, and vasculopathy associated with the disease.

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Objective: To describe the mortality and clinical complications among patients with SCD with recurrent VOCs in England.
Methods: This longitudinal, retrospective cohort study used the Clinical Practice Research Datalink linked with secondary care data (Hospital Episode Statistics) in England to identify patients with a diagnosis of SCD between 1 July 2008 and 30 June 2018. Eligible patients with SCD were required to have ≥2 VOCs per year in any 2 consecutive years. A VOC was defined as SCD with crisis, priapism, or acute chest syndrome. Patients were required to have ≥1 year of follow-up data after their index date (second VOC in the second year of 2 consecutive years). Those with hereditary persistence of foetal haemoglobin, HbSC genotype, or evidence of haematopoietic stem cell transplantation were excluded. Patients were matched to up to 5 controls without SCD by age, sex, region, and ethnicity. The pseudo-index date for controls was the same as the index date for patients. Patients were followed from index until a censoring event (e.g. death or study period end [30 June 2019]). Demographics were assessed at index. Mortality (proportion of total population and rate [deaths per 100 person-years; overall rate and rate by index date]) and clinical complications (proportion of total population) were summarised descriptively during the study period. A Z-test was used for significance testing ($p < 0.05$) for mortality proportion.
Results: Overall, 1117 patients with SCD met inclusion/exclusion criteria and were matched with 5585 controls. The mean age of patients at index was 24.96 years, 51.39% were female, and 91.58% were Black. Controls had similar demographics. Mortality proportion (SCD: 41/1117 [3.67%] vs. controls: 38/5585 [0.68%]; $p < 0.001$) and rate per 100 person-years (0.78 vs. 0.16, respectively) were higher for patients with SCD than for controls. Mortality rates were similar for patients with a more recent or older index date (2008–2013: 0.77 vs. 2014–2018: 0.81). The most prevalent chronic complications in patients were cardiopulmonary complications (30%), bone/joint problems (26%), retinopathy (19%), mental health complications (16%), and chronic pain (15%).
Conclusion: Patients with SCD with recurrent VOCs had significantly higher mortality than controls, with no notable improvements over time. Despite the best available care, patients experience significant SCD-related clinical complications.

BSH23-PO111 | Mode of Nipocalimab action and pharmacology in cells and preclinical models

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Background: The neonatal Fc receptor (FcRn) is an endosomal receptor responsible for maintaining the long half-life of immunoglobulin G (IgG), including IgG autoantibodies. Warm autoimmune haemolytic anaemia (wAIHA) is a rare, life-threatening autoimmune disorder caused by haemolytic anti-erythrocyte IgG autoantibodies. Nipocalimab is a monoclonal antibody that selectively binds and blocks FcRn to decrease IgG autoantibodies. This study demonstrates the in vitro and in vivo pharmacology of nipocalimab confirming concentration-dependent FcRn engagement, inhibition of IgG recycling, and increased clearance of IgG consistent with the pharmacodynamics (PD) of FcRn engagement and IgG clearance observed in Phase 1/2 studies.
Methods: Primary human endothelial and villous trophoblast cells incubated in the presence of fluorescently tagged IgG were treated with nipocalimab to assess its impact on IgG recycling. After in vitro or in vivo nipocalimab administration, FcRn receptor occupancy (RO) was detected after cell fixation-permeabilization with labelled nipocalimab. In vivo PD effects were assessed in human transgenic FcRn mice or cynomolgus monkeys. RO was measured over time in circulating monocytes in both species.
Results: Nipocalimab treatment of primary human target cells increased cellular IgG content by increasing lysosomal IgG consistent with inhibition of IgG recycling. Nipocalimab also demonstrated similar concentration-dependent inhibition of IgG recycling (IC50, 0.03 μg/mL) and FcRn saturation (EC50, 0.03 μg/mL) demonstrating a high degree of correlation between target engagement and PD effect. Saturation of FcRn binding by nipocalimab occurred within 30′ to 2 h at concentrations consistent with pharmacokinetic (PK) at full RO and the period of IgG lowering observed in clinical studies. In vivo, full RO also occurred rapidly within 2 h of single intravenous (IV) infusions (dose ≥2 mg/kg). PK and PD evaluation of single IV doses of nipocalimab in cynomolgus monkeys and human FcRn transgenic mice established consistent dose, exposure-, and time-dependent relationships between PK, the PD effect on FcRn RO and the lowering of serum IgG concentrations. Duration of RO in peripheral blood monocytes in both preclinical species was dose-dependent, as was the duration of IgG clearance, suggesting a relationship between RO and maintenance of an increased IgG clearance rate.

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Conclusion: These in vitro and in vivo preclinical studies demonstrate the in vitro and in vivo mechanism of action for nipocalimab. The close mechanistic relationship between nipocalimab concentration, FcRn occupancy and inhibition of IgG recycling resulting in rapid, deep and sustained IgG lowering in vivo is consistent between preclinical and clinical studies.

BSH23-PO112 | Designing ENERGY: A phase 2/3 study of Nipocalimab in warm autoimmune haemolytic Anaemia

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Background: Warm autoimmune haemolytic anaemia (wAIHA) is a rare condition characterised by the premature destruction of red blood cells (RBCs) mainly in the presence of pathogenic immunoglobulin G (IgG) autoantibodies that preferentially bind to RBCs at 37°C, resulting in extravascular haemolysis of these RBCs in the spleen (or liver). Patients with wAIHA can experience acute or chronic worsening of anaemia, increased risk of thromboembolic events, and premature death. Innovative targeted therapies that address the underlying pathogenesis of wAIHA are needed, as there is currently no approved treatment for this disease. Nipocalimab is a high affinity, fully human, aglycosylated, effectorless IgG1 monoclonal antibody that targets the neonatal Fc receptor (FcRn) to lower circulating IgG levels, including pathogenic autoantibodies. Nipocalimab is being evaluated across multiple IgG-mediated diseases. A completed phase 2 study in patients with generalised myasthenia gravis demonstrated rapid, durable total serum IgG and pathogenic IgG autoantibody reductions and meaningful clinical response.
Aim: Describe the rationale and study design of ENERGY, an ongoing, adaptive, phase 2/3 multicentre, randomised, double-blind, placebo-controlled study that will evaluate the efficacy, safety, tolerability, pharmacokinetics, and pharmacodynamics of nipocalimab compared with placebo in patients with wAIHA (NCT04119050).
Methods: Subjects ≥18 years of age who have been diagnosed with primary or secondary wAIHA and are currently receiving treatment for wAIHA or have previously received treatment for wAIHA will be included in the study. Subjects with cold antibody AIHA, cold agglutinin syndrome, mixed type (i.e. warm and cold) AIHA, or paroxysmal cold haemoglobinuria will be excluded. Stable doses of corticosteroids or immunosuppressants will be allowed. Approximately 111 patients will be randomised 1:1:1 to receive nipocalimab at two different dose schedules or placebo. Following completion of 24 weeks of double-blind treatment, patients may enter an open-label extension period to receive nipocalimab for 144 weeks with a follow-up period of 6 weeks after last assessment.
Results: ENERGY will include over 170 sites across nearly 20 countries. The primary endpoint is percentage of participants achieving durable response of improvement in haemoglobin (Hgb). The secondary endpoints include change from baseline in the total score from the Functional Assessment of Chronic Illness Therapy-Fatigue Scale, corticosteroid dose reduction from baseline, and normalisation of haemolytic markers.
Conclusion/Summary: The results of ENERGY have the potential to identify a novel treatment option to address the significant unmet needs of patients with wAIHA. Enrolment is ongoing in this clinical trial.

BSH23-PO113 | ActIVe study: Impact of Voxelotor on sleep quality, physical activity, and patient-reported outcomes

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Introduction: Voxelotor is a first-in-class sickle haemoglobin polymerisation inhibitor approved in Great Britain for the treatment of sickle cell disease (SCD) in patients aged ≥12 years. The safety and efficacy of voxelotor in increasing haemoglobin and decreasing haemolytic markers have been demonstrated in clinical trials.
Objective: To examine patient-reported quality of life (QOL) and safety data from the phase 4, multicentre, open-label ActIVe study (NCT04400487).
Methods: Patients aged 12–55 years with SCD received voxelotor 1500 mg/day for up to 24 weeks across 10 US sites. Outcomes were evaluated using the Patient Global Impression of Change (PGI-C), the National Institutes of Health Patient-Reported Outcome Measurement Information System-43 (PROMIS-43) for adults (aged >17 years), PROMIS-37 for adolescents (aged ≤17 years), a 10-point Numeric Rating Scale (NRS) for pain intensity, and the Clinical Global Impression of Change (CGI-C).

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Results: Twenty-five patients were enrolled: median (range) age was 20.0 (12–51) years, 64.0% were female, and 92.0% were Black or African American. At week 24, haemoglobin increased in 78% of patients (14/18); the mean (SD) increase was 0.53 (0.839) g/dL.

Clinicians reported improvement in health status among 84.6% (11/13) of patients at week 24: 7.7% (1/13) were 'very much improved', 46.2% (6/13) were 'much improved' and 30.8% (4/13) were minimally improved; 2 patients had no change. At week 24, 92.9% (13/14) of patients self-reported improved health status: 21.4% (3/14) were 'very much improved', 42.9% (6/14) were 'much improved', and 28.6% (4/14) were minimally improved; 1 patient reported minimally worse health status.

In adults, a meaningful change from baseline (defined as a ≥5-point group-level change) for fatigue [mean (SD) of −7.9 (5.12)] and pain interference [mean (SD) of −6.5 (5.29)] was found at week 24, with near-meaningful improvements for anxiety, sleep disturbance, and ability to participate in daily activities. Whilst some PROMIS domains improved among adolescents, none reached a meaningful level.

A total of 18 patients (72.0%) had ≥1 non-SCD-related treatment-emergent adverse event (TEAE); the most common (>10%) were diarrhoea (32.0%), headache (20.0%), nausea (20.0%), and vomiting (12.0%). Three patients (12.0%) had ≥1 grade 3 non-SCD-related TEAE.

Conclusions: At week 24, most voxelotor-treated patients experienced improvements in haemoglobin and overall health status, with meaningful changes in fatigue and pain interference found among adults. The safety profile was consistent with that of previous studies. Overall, results suggest that voxelotor-treated patients may experience improvements in QOL, including in physical and emotional well-being.

BSH23-PO115 | Clinical features and management of sickle cell intrahepatic cholestasis: Retrospective analysis of 36 cases

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Sickle cell intrahepatic cholestasis (SCIC) is a rare but severe variant of sickle cell hepatic crisis with traditionally poor outcomes, especially in adults. SCIC involves sickling in the hepatic sinusoids leading to vascular stasis. Such a process results in hypoxia, which causes the hepatocytes to swell up causing intracanalicular cholestasis. Symptoms include right upper quadrant pain, nausea and vomiting, fever, and jaundice. This picture could be complicated by renal impairment and encephalopathy. Herein, we present 36 paediatric cases (<18 years old) of SCIC from the published literature spanning 1965 to November 2022, including a successfully treated case at our hospital. Due to overlap in syndromes related to sickle cell hepatopathy and lack of an agreed definition for SCIC, studies which used the term 'intrahepatic cholestasis' were primarily included. Furthermore, other terms such as 'acute cholestasis' and 'diffuse intrahepatic sickling' were permitted as long as other diagnoses such as viral hepatitis and choledocholithiasis were excluded. From the available data, median age was 8 (IQR 6.75–12.25) with a male preponderance (61%, 17/28). The commonest clinical feature was hepatomegaly, identified in 85% (28/33) of patients. Other features included clinically observed jaundice (82%, 27/33), abdominal pain (61%, 20/33), nausea and vomiting (21%, 7/33), and fever (18%, 6/33). The primary diagnosis included 'sickle cell disease' (17), HbSS (15), HbS/β+ (2), HbSC (1) and HbSD (1). The median total bilirubin was 512 (IQR 347.5–911.9) μmol/L with coagulopathy (prolonged prothrombin time) observed in 64% (14/22) of patients. Forty-two percent (15/36) of patients received supportive care, which included intravenous fluids, analgesia, coagulopathy correction and/or electrolyte monitoring. In total, 14% (5/36) received blood transfusion as treatment, whereas 42% (15/36) received red cell exchange transfusion (RCET), of which 93% (14/15) survived after RCET. Furthermore, including our case, plasma exchange (PEX) was used effectively in three cases (two patients had prior RCET). In total, 17% (6/36) deaths were observed, despite 89% (32/36) of patients recovering after initial treatment. From the 6 patients who died, available data on 4 patients showed the presence of coagulopathy. Furthermore, all 6 patients who died received either supportive care and/or blood transfusion, except for one patient who received RCET and died as a consequence of sepsis after receiving his third liver transplant. In conclusion, SCIC in the paediatric population can be effectively treated with RCET in the majority of patients with the potential of using PEX in more difficult-to-treat cases that do not respond to RCET.

BSH23-PO116 | The effects of nigella sativa supplements on Sudanese children with sickle cell disease

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Introduction: Sickle cell disease (SCD) is the most common monogenic disorder worldwide and is associated with high morbidity and mortality. It is characterised by persistent pro-inflammatory state, high level of oxidative stress, intravascular haemolysis and lifelong haemolytic anaemia. Nigella Sativa (NS) is known for its therapeutic properties, including anti-inflammatory, immunomodulatory, and antioxidant effects.

Objectives: The aim of this study was to investigate the effects of NS supplements on patients' clinical and laboratory parameters.

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Methodology: 33 SCD patients aged between 6 and 18 years were given NS capsules at a dose of 2 g per day for 12 weeks. The patients were assessed clinically, and by laboratory investigations at baseline and following 12 weeks of supplementation.

Results: NS supplementation was well-tolerated with no reported side effects. There was a reduction in the number of hospital admissions ($p = 0.038$), vaso-occlusive episodes ($p = 0.008$) and clinically detected pallor ($p = 0.01$) using Wilcoxon rank test. Comparing pre- and postintervention assessments, the following were found: the Peak Expiratory Flow Rate (PEFR) increased by 15.6 L/min ($p = 0.026$), Forced Vital Capacity (FVC) increased by 0.106 L/min ($p = 0.017$), Forced Expiratory Volume 1 (FEV1) increased by 0.077 L/min ($p = 0.026$), Platelet-Large Cell ratio (P-LCR) increased by 3.9% ($p < 0.001$), Mean Platelet Volume (MPV) increased by 0.6 fL ($p < 0.001$), reticulocytes percentage decreased by 3.35% ($p = 0.037$), Lactate Dehydrogenase (LDH) level decreased by 160 U/L ($p = 0.004$), creatinine level decreased by 0.04 ($p = 0.005$), urea level decreased by 6.5 mg/dL ($p < 0.001$), Iron level decreased by 0.24 mg/dL ($p < 0.001$), Copper level increased by 0.17 ($p < 0.001$), Magnesium level decreased by 9.7 mg/dL ($p < 0.001$) and Zinc level increased by 0.25 mg/dL ($p < 0.001$) using paired sample t-test.

Discussion and Conclusion: Statistically significant reduction in the number of hospital admissions, vaso-occlusive episodes and clinically detected pallor were demonstrated following 12 weeks of NS supplementation. PFT showed statistically significant improvement in PEFR, FVC and FEV1. Blood tests showed statistically significant reduction in reticulocytes percentage, LDH, creatinine, urea, iron and magnesium and significant increase in MPV, platelets LCR, copper and zinc. NS supplementation resulted in clinical and PFT improvement, reduction of haemolysis markers and improvement in the redox state as demonstrated by the changes in trace elements levels. These promising findings need to be replicated in randomised clinical trials.

BSH23-PO117 | Partnership working with citizens advice Leicestershire: Setting up a bespoke welfare Service for Haemoglobinopathy patients

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The East Midlands Sickle Cell and Thalassaemia Network (EMSTN) hosted by University Hospitals of Leicester (UHL) acts as the Haemoglobinopathy Coordinating Centre for the East Midlands. Following the introduction of the service specification in 2019, it became apparent that the complex welfare needs of the patients living in the East Midlands were a significant burden to them; as a result the Clinical Nurse Specialists reported an increased need to offer support relating to welfare issues detracting from clinical work. We looked at differing models to address this recognising the specialist nature of this service and approached Citizens Advice Leicestershire to propose a partnership model. Following discussions, a decision to proceed with a pilot of a UHL funded post working in Citizens Advice to offer a bespoke service to our patient population was agreed. This service went live in May 2022. Initially referrals were received from Clinical Nurse Specialists via an online portal. Self-referrals were introduced November 2022.

Data to date for Q1 and Q2 has indicated that 36 referrals were received, 29 of which were in Q2 and included 11 self-referrals. In Q1, there was a total of 26 issues raised equating to 3.7 issues per client. In Q2 there were 82 issues, equating to 2.8 issues per client. This compares to an average of 1.5 issues for persons accessing help via the Leicestershire generalist service, indicating the higher burden and interconnecting social welfare issues affecting our patient cohort. The three most commonly raised issues were benefits, housing & debt.

Modelling indicates that in Q1 there was an estimated financial benefit for clients accessing help of £4626.00 (mean of £660 per person); for Q2 this was £22 675 (mean of £758 per person).

Feedback from the clinical team is that the burden of this workload has been taken off them. They feel able to concentrate on their clinical duties, and we aimed to monitor the impact on outcomes and experience. Patient feedback collection is ongoing, but initial reports are positive.

Conclusion: The Q1 and Q2 reports indicate a rise in the number of patients accessing the service and, therefore, benefitting from the expert advice they receive. We can already demonstrate the positive impact the service has had on both the patients and clinical staff. Early indication is that the Q3 numbers are significantly higher as service promotion allows us to advertise the service more widely.

BSH23-PO118 | A comprehensive haemoglobinopathy and rare inherited anaemia education package for children, adolescents and their carers

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A major element of Haemoglobinopathy and Rare Inherited Anaemia's (RIA) care is education around the condition. Education has an important long-term role in improving quality of life of individuals with chronic conditions. National guidelines recommend that every outpatient contact should be an opportunity for education, which will be provided in a systematic way, varying with age. With this in mind we looked at the education that we provided for patients and their carers within our centre.

We have been running a successful education package for adolescents for many years. A self-assessment of their

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knowledge and well-being is undertaken at 3 stages of adolescence, this builds the basis for their education plan. Education is delivered through a number of different formats—education days, condition-based workbooks and teaching in clinic. This structured programme of education has been well received so has been expanded to other age groups and parents.

We found that education delivered in the clinic setting was often limited and unstructured. The self-assessment questionnaires that are used for adolescents offer the opportunity for structure and a more meaningful encounter. Using these as a starting point they were adapted for carers of children aged 0–5 years and 7–12 years and for children aged 7–12 years. These annual assessments will build a basis on which to direct education and are the start of an educational toolkit, which has been designed to give carers as much information as possible about their child's condition. This toolkit includes a mixture of locally developed and national leaflets to be given as each topic is discussed. An educational record is kept for when each leaflet is given and each time a topic is discussed. We are looking at expanding carer education further by holding monthly virtual talks.

For children with Sickle Cell a number of resources have been developed based around a character called Little George. Cartoon based leaflets about their condition and treatment have been developed. Workbooks and educational sessions are in development. We are currently leading on the development of a computer game, which provides a more interactive approach to education. Siblings are also encouraged to participate in the education programme. We have started to develop the children's education programme for Thalassaemia and RIA utilising the same framework.

With a more structured approach to education we are hoping to empower patients and their carers to be more confident in managing their condition.

BSH23-PO119 | Management of VTE and adherence to the NICE VTE Guideline (NG158) 2020

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In March 2020, NICE updated its VTE Clinical Guideline (NG158) 'Venous thromboembolic disease: diagnosis, management and thrombophilia testing', recommending that people with a confirmed proximal DVT or PE are offered anticoagulation with either apixaban or rivaroxaban. If neither apixaban or rivaroxaban are suitable, physicians are guided to offer LMWH for at least 5 days followed by dabigatran, edoxaban or LMWH concurrently with a VKA.

We performed an observational, retrospective, descriptive study that evaluated acute VTE treatment patterns in England between 2013 and 2020 using the Clinical Practice Research Datalink (CPRD), which collects anonymised patient data from a network of GP practices across the UK. The study was approved by the CPRD's Independent Scientific Advisory Committee. We included de novo patients aged 18+ with a first non-cancer VTE diagnosis on/after 1 Jan 2013–31 Dec 2020 and treated with apixaban, dabigatran, edoxaban, rivaroxaban, or warfarin within 30 days of VTE diagnosis.

A total of 13 653 patients were identified [warfarin: 5487 (40.2%); apixaban: 2605 (19.1%); dabigatran: 58 (0.4%); edoxaban: 121 (0.9%); and rivaroxaban: 5382 pts (39.4%)]. Over 40% of patients presented with an incident PE (5907 pts (43.3%)); over half with an incident DVT (8019 pts (58.7%)); and just 2% with both PE and DVT (273 pts (2.0%)).

The percentage of incident patients prescribed warfarin decreased over time, from 87.7% in 2013 to 7.4% in 2020, dropping below 50% in 2016. However, apixaban use increased from 0% in 2013 to 49.5% in 2020; rivaroxaban use increased from 12.3% in 2013 to peak at 52.6% in 2016 then reduced to 40.4% in 2020. Dabigatran and edoxaban use did not exceed 2.3% for any year in the study period.

We observed that the NICE CG158 recommendation that proximal DVT/PE patients are first offered apixaban or rivaroxaban, mirrors current practice rather than influencing it, such that by the end of 2020, 90% of patients with incident VTE were prescribed one of these two DOACs. It is anticipated that the observed DOAC prescribing trends will continue beyond 2020 into the next decade. The convenience of prescribing a DOAC without the need for a parenteral anticoagulant, as is the case with apixaban and rivaroxaban, will also help facilitate ambulatory DVT and PE pathways and result in shorter hospital stays, which are key factors in managing patient flow. This is of particular importance with the additional pressures faced in the national health care system following the pandemic.

BSH23-PO12 | Assessing Bionano optical genome Mapping's ability to detect structural variation in AML and MDS

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The West Midlands Regional Genetics Laboratory currently investigates structural and copy number variation in acute myeloid leukaemia (AML) and myelodysplastic syndrome (MDS) using G-band karyotype, fluorescence in situ hybridisation (FISH) and single nucleotide polymorphism (SNP)-array analysis. Bionano Saphyr Optical Genome Mapping (OGM) technology has the potential to replace these techniques with a reduced workload, turn-around-time and cost whilst increasing resolution and diagnostic yield. OGM can detect structural variants over a wide dynamic range (500 bp

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to multiple megabases) and is more precise than G-band analysis with >3000× higher resolution. It can reveal complex rearrangements and, unlike SNP array, detect balanced translocations.

OGM technology relies on the isolation of ultra-high molecular weight DNA. These long strands of DNA are fluorescently labelled at specific six nucleotide motifs, which are spaced regularly throughout the genome. Labelled DNA is linearized in massively parallel nanochannels and imaged with a digital microscope in repeated cycles following electrophoresis. The assembled image reads are parsed with bioinformatic tools specific to structural orientation and interpreted with the aid of proprietary analysis software. Bionano software automatically identifies variants and produces copy number plots, structural breakpoint locations and confidence scores.

We performed OGM on five AML and two MDS cases and detected all structural and copy number abnormalities already identified by standard of care (SOC) testing. Notably, OGM detected additional abnormalities, some with potential clinical relevance. In a case of MDS transformed to AML, OGM refined an unidentified marker chromosome as a homogeneously staining region (hsr) with c-MYC gene amplified. For an MDS-EB1 case in which deletions of 2q and isochromosome 17 were established with SNP array, OGM detected additional deletions of 9q, 14q, 19q, and 22q. Furthermore, we were able to reclassify insertion events as translocations and clarify the origin of double minutes as amplification of KMT2A.

The aim of this study was to evaluate OGM as a potential replacement for aspects of current SOC testing in a diagnostic laboratory environment. OGM can detect and further resolve novel and rare structural aberrations of potential clinical relevance in AML and MDS. With automation and improvements to capacity on the Bionano Saphyr instrument, and assuming costs will reduce in line with Bionano's ambition, OGM has the potential to replace the majority of G-band, SNP array and FISH analysis. OGM could facilitate quicker results, novel variant discovery and improve patient care.

BSH23-PO120 | RAPIDO: Results from a UK-wide audit of reversal agents for direct oral anticoagulant associated bleeding

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Despite their relative safety compared to warfarin, direct oral anticoagulants (DOACs) are associated with a risk of major bleeding. There is much interest in establishing the optimal treatment of patients who present with major bleeding and in particular the use of haemostatic reversal agents. In the UK, there are two licensed reversal agents: andexanet alfa (apixaban & rivaroxaban) and idarucizumab (dabigatran), as well as prothrombin complex concentrates (PCC), Beriplex and Octaplex, which are widely used off-licence. Despite the DOACs having been in widespread use for over a decade, there is no randomised control trial evidence comparing a reversal agent to a control arm or standard of care that can guide clinicians and as such, practice is likely to be highly heterogeneous. RAPIDO is an ongoing, retrospective, UK-wide audit of anticoagulant reversal agents. The project is being delivered by HaemSTAR, a UK-wide network of trainee haematology doctors.

As of December 2022, over 200 HaemSTAR collaborators have submitted data from 38 hospitals, comprising 837 patients who received a reversal agent between 1 October 2020 and 31 January 2022. Median age was 81 years (IQR 73–87), 487 patients (58%) were male, 699 (84%) had atrial fibrillation and 155 (18%) had a history of venous thromboembolism. 778 (93%) fulfilled the study criteria for major haemorrhage. 431 (51%) had intracranial haemorrhage, 269 (32%) gastrointestinal bleeding, and 137 (16%) had bleeding at other sites. 448 (53%) patients were taking apixaban, 260 (31%) rivaroxaban, 77 (9%) edoxaban, and 52 (6%) dabigatran. 414 (49%) received Octaplex, 327 (39%) Beriplex, 49 (6%) andexanet alfa, and 47 (6%) idarucizumab. Median times to administration of a reversal agent were: from last dose of anticoagulation (where known) 15.8 h (IQR 9.8–23.3), from bleed onset 6.3 h (IQR 3.3–12.5), and from admission (in those who were outpatients at bleed onset) 4.3 h (IQR 2.1–7.9). At 90 days

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postadministration of any reversal agent, 350 (41%) patients had died and 33 (4%) had had a thrombotic event.
These results provide important information about the real-world use of reversal agents and also highlight delays in identification and treatment of bleeds.

BSH23-PO121 | Utility of falls risk assessment tool and clinical frailty score to risk-stratify patients requiring anticoagulation

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Introduction: Falls are the second leading cause of accidental injury resulting in death worldwide, with 37.3 million falls requiring hospital admission. With patients living longer, falls are more prevalent with fall-associated intracranial haemorrhage (ICH) in older adults increasing. Patients who are on anticoagulation are at greater risk of fall-associated ICH and are more likely to suffer more extensive haemorrhage. In practice, HAS-BLED and ORBIT scores (both predictors of bleeding risk in anticoagulated patients) have been used to aid clinicians in decisions to commence anticoagulation; however, their utility in practice is an area of contention. Identifying risk factors of fall-associated ICH in anticoagulated patients is essential to assist clinicians' management of anticoagulation and help reduce patient mortality.
Methods: 30 adults using anticoagulation who were admitted to hospital for fall with radiologically confirmed ICH between February 2019 to March 2021 were included in a retrospective audit. The mean age of patients was 78 (range 51–93) and patients were anticoagulated with either warfarin, direct oral anticoagulants, or low molecular weight heparin. ICH was confirmed within 30 days of the fall. Demographics, comorbidities, HAS-BLED score, ORBIT score, clinical frailty score, falls risk assessment tool (FRAT), blood tests and reversal agents used for each patient were reviewed. Survival curves were subsequently generated for each variable to determine predictors of mortality using GraphPad Prism.
Results: When establishing bleeding risk through HAS-BLED and ORBIT scores, neither were good predictors of mortality in this cohort. Patients with high-risk FRAT had significantly poorer survival outcomes 90 days post ICH compared to those with lower FRAT (medium risk vs. high risk: $p = 0.05$, low risk vs. high risk: $p = 0.029$). Interestingly, age did not correlate with FRAT in this cohort. FRAT was better at differentiating mortality compared to previous falls requiring admission. Patients with high-risk clinical frailty scores had poorer survival outcomes 90 days post ICH compared to those with lower scores; however, no significance was established. As expected, patients with liver failure ($n = 8$) had poorer survival outcomes than those that did not; however, no significance was established.
Conclusion: We have demonstrated that whilst HAS-BLED and ORBIT scores were poor predictors of survival in 30 anticoagulated patients who had fallen and sustained an ICH, FRAT and the clinical frailty score were good differentiators of mortality. This preliminary but exciting data demonstrates the potential utility of FRAT and the clinical frailty score when risk-stratifying patients who require anticoagulation.

BSH23-PO122 | Evaluating a weight-based unfractionated heparin infusion protocol in adult patients during the initial dosing period

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Background: Unfractionated heparin (UFH) has been used in clinical practice for many years and remains the agent of choice in critically unwell patients, those patients with severe renal dysfunction, and periprocedurally during cardiac procedures. At King's College Hospital, a new IV UFH adult guideline was introduced in Oct 2021, recommending a weight-based dosing schedule and a calibrated anti-Xa assay to monitor UFH, in place of Activated Partial Thromboplastin Time (APTT). This audit aimed to review how well the new dosing protocol performed in clinical practice during the initial dosing period.
Methods: A retrospective audit was conducted of all adult patients commencing UFH infusion from the launch of the guidelines—13 October 2021 to the 27 of January 2022. In addition to the demographic information on the patients commenced on UFH, the indication and setting of the infusion were recorded, along with the initial dosing received and whether the guidelines were followed. Descriptive statistics were used to analyse the data.
Result: Ninety-seven patients were commenced on a UFH infusion during the audit period with a mean age of 60.7 years [SD ±14.9] and a mean first anti-Xa result of 0.63 IU/mL [SD = 0.43]. Most patients were in the ICU [$n = 37$, 38%], renal [$n = 22$, 23%], and cardiac ward setting [$n = 14$, 15%] when the UFH infusion was initiated. Sixty-seven percent [$n = 65$] were male, with 40% prescribed the UFH infusion for venous thromboembolism. The results show that patients who had been prescribed the UFH bolus dose and followed the weight-based UFH guideline [$n = 16$, 41%] were above the target therapeutic range. However, patients treated

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according to the guideline but not administered a bolus dose [n = 38] were more likely to have the first anti-Xa level in the range [n = 27, 71%]. Female patients were more likely to be above the range [n = 11, 34%] compared to male patients [n = 12, 18%].

Conclusion: The new protocol using a weight-based dosing schedule and anti-Xa performed well in ensuring therapeutic levels of anticoagulation with UFH. The findings suggest that patients who were not administered the bolus and followed the UFH infusion guideline were more likely to achieve the anti-Xa therapeutic range. We have now amended our UFH guideline from mandating the bolus to considering the administration of a bolus if clinically indicated. We plan to reaudit, following this change.

BSH23-PO123 | How will UK hospitals use andexanet alfa? A review of local protocols

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Andexanet alfa is a reversal agent for patients who present with major haemorrhage related to anti-factor Xa direct oral anticoagulants (DOACs). In 2021, the National Institute of Health and Clinical Excellence (NICE) recommended andexanet alfa as an option for patients with uncontrolled or life-threatening gastrointestinal haemorrhage only. This recommendation was based on data from a single-arm trial, ANNEXA-4, and from indirect comparisons against historical controls. ANNEXA-4 found reported that 82% patients achieved excellent or good haemostatic efficacy at 12 h, but 14% had a thrombotic event within 90 days. However, without randomisation, whether there is clinical benefit over standard of care is uncertain. Moreover, andexanet alfa is a high-cost drug and the cost effectiveness is unproven. Given these uncertainties, risks, and high cost we set out to better understand how NHS haematology teams would implement this drug. We wanted to investigate how different hospitals would approach decision-making and gatekeeping within their local protocols, as these will be important influences on the use of this drug.

We contacted 120 acute care trusts in England, Wales, and Northern Ireland. Of 89 (74%) respondents, we found that 52 (58%) had a protocol in place at a median time of 8.1 months from NICE recommendation to local protocol approval. We acquired 49 protocols and found that only 13 (31%) included criteria to define major bleeding. 23 (47%) stated that the storage location for the drug was in an emergency drugs fridge. 24 (47%) required authorisation by a haematologist and 10 (20%) by a gastroenterologist. Protocols requiring consultation with a gastroenterologist were more likely to include a stipulation that endoscopy should be arranged (p = 0.0076). Additionally, only 44% of guidelines mentioned thrombosis as a significant risk—and 70% did not give include information on restarting anticoagulation following the use of andexanet alfa.

In summary, local protocols for andexanet alfa in the UK are heterogeneous. NICE recommendations should be enacted by hospitals in England within 3 months, which may compel hospital trusts to encourage the use of andexanet alfa through the development of local protocols. This data is important as andexanet alfa is a high-cost drug with limited evidence of clinical benefit and evidence of harm; widespread, unrestricted use could inflict significant costs on the NHS. Robust protocols with appropriate gatekeeping are important to ensure consistent use that is appropriately restricted whilst avoiding delay.

BSH23-PO124 | Can ultrasound shear wave Elastography reliably measure clot stiffness in patients with thrombosis?

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Introduction: Being able to distinguish between acute and chronic thrombus is essential in the optimal management of a patient with deep venous thrombosis (DVT). Ultrasound Doppler is currently used to determine the presence of thrombus; however, differentiating between an acute or chronic nature of the thrombus can be subjective using this technique. Research has shown that the pathological composition of thrombus changes over time (10) and the elastic properties decrease gradually causing the thrombus to become stiffer as it ages (11, 12). A novel ultrasound technique known as Shear Wave Elastography (SWE) can provide numerical measurements related to tissue stiffness. The stiffer the structure, the faster the shear wave will travel (12).

Study Aim: This study was developed to determine if SWE could be used to reliably measure clot stiffness in patients with a DVT.

Method: 19 patients who had been confirmed as having a DVT on Doppler ultrasound were recruited and consented to the study with a symptom duration ranging between one day and 15 months. Three patients were excluded due to an incomplete SWE data set. SWE measurements were performed three times by two observers at the proximal and distal ends of the thrombus using a high frequency linear probe.

Results: Interclass coefficient correlation (ICC) was used to measure intra and interobserver reliability of the measurements obtained from the 16 participants. Results showed good intra and inter-observer reliability (0.96–0.99).

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Correlation analysis between symptom duration and thrombus stiffness showed no significant relation between the two factors (0.43–0.96); this is likely due to the small sample size. **Conclusion:** The results showed that reproducible and reliable SWE readings can be obtained from a blood clot in the deep veins. This supports the proposal for a further pilot study to track the changes in the stiffness of thrombus over time, with a view to helping improve clinical practice for patients with a DVT. A limitation of the study is the small number of participants with prolonged symptom duration included, which, therefore, made correlation between symptom duration and thrombus stiffness unreliable.

BSH23-PO126 | UK NEQAS for blood coagulation evaluation of the point of care qlabs® FIB monitoring system

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UK National External Quality Assessment Schemes for Blood Coagulation (UK NEQAS BC) provide external quality assessment (EQA) for haemostasis tests across both laboratory and point of care (POC) setting. Currently we have over 1200 laboratory sites registered for a range of haemostasis tests. We also have over 4600 sites registered for POC tests, which include INR, ACT, D-dimer and viscoelasticity testing. When new tests are introduced into laboratories or POC we endeavour to provide EQA for these tests.

One such test recently introduced and distributed by Stago is measurement of Fibrinogen using the qLabs® FIB Monitoring system. The level of fibrinogen is critical in obstetrics whereby centres commonly use a 2 g/L threshold to trigger replacement product therapy. A rapid and reliable measure of fibrinogen using a POC device would enable faster provision of blood products to this group of patients.

Qlabs® FIB monitoring system measures functional fibrinogen giving a result within 10 min. UKNEQAS BC has evaluated this device to determine whether EQA could be provided. 6 samples that had previously been distributed to over 800 laboratory sites were tested on the qlabs® FIB monitoring system. For each sample, the results obtained by duplicate testing on the qlabs® FIB monitoring system were compared to the median Clauss fibrinogen result calculated when the EQA samples were circulated. All qlabs® FIB system results were within 0.1 g/L of this median value.

Two samples were chosen from the panel of 6 for further precision testing. 10 tests were completed by 3 operators and on different days. Coefficient of variance (CV) for these 2 samples were 4.7% and 3.1% showing good agreement.

Further testing was completed to compare the citrated whole blood samples tested on the qlabs® FIB monitoring system to citrated plasma tested in the local laboratory for Clauss fibrinogen using Siemens thrombin and the Sysmex CS5100. EQA is recommended for all POC testing where available. We can conclude that UK NEQAS BC could provide EQA for the qlabs® device for fibrinogen measurement. Good precision was seen on lyophilised samples with both samples recording a CV of <5%.

BSH23-PO127 | Clinical implications of cigarette smoking on platelet and inflammatory markers of newly diagnosed tuberculosis patients

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Tuberculosis (TB) infection and smoking habit are clinically and epidemiologically linked. Smoking is an important risk factor for tuberculosis infection and the World Health Organisation has called for integration of anti-smoking efforts as a key strategy towards TB elimination. There is also strong evidence that smoking and TB infection could affect systemic immunity and inflammation. Similarly, tuberculosis infection and smoking have been shown to affect platelet activity and both have also been linked to haemostatic and thrombotic abnormalities. Differences in markers of platelet activity and inflammatory cytokines between cigarette-smoking and non-smoking TB subjects and its clinical implications were assessed in this study. This cross-sectional study involved a convenience sample of 100 newly diagnosed, treatment-naïve, age-matched male TB subjects. These comprised 48 cigarette smokers who were infected with TB and 52 TB-infected subjects who had never been smokers. The platelet indices namely; platelet factor-4 (PF4), Platelet activating factor (PAF), thrombopoietin hormone (TPO), P-selectin, Platelet glycoprotein Iib/IIIa (GP Iib/IIIa) and cytokines namely; interleukin-6 (IL-6), interleukin-2 (IL-2), tumour necrosis factor alpha (TNFα), interleukin-10 (IL-10), transforming growth factor beta (TGF-β) were assayed by sandwich Enzyme-linked immunosorbent assay (ELISA) technique. Platelet count was carried out by the standard manual method. Mann–Whitney *U* statistics was used to compare values between groups. PAF (pg/ml), TPO (pg/ml), GP Iib/IIIa (ng/ml) and platelet-lymphocyte ratio were significantly lower in TB-smokers compared to TB-non-smokers ($p < 0.001$) but PF4 (ng/mL) was higher in TB-smokers compared to TB-non-smokers ($p < 0.001$). The cytokine values indicates that TGF-β (pg/mL) was significantly higher in TB-smokers compared to TB-non-smokers ($p = 0.001$) whilst IL-10 (pg/ml) was lower in TB-smokers

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compared to TB-non-smokers ($p < 0.001$). However, there was no change in platelet count, *p*-selectin, IL-6 (pg/mL), IL-2 (pg/mL) and TNFα (pg/mL) levels ($p > 0.05$). Clinically, the findings suggest that smoking results in a reduced GP Iib/IIIa and PAF action as well as thrombopoietin hormonal activity in TB subjects. These define a downregulation in platelet adhesion and aggregation tendencies and a negative effect on the hormonal regulation of platelets, which reflected in no significant change in platelet count. Conversely, PF4 level was potentiated in TB-smokers implying an enhanced platelet release reaction and a procoagulant inducement. The major pro-inflammatory markers (IL-6, IL-2 and TNFα) were not affected but the anti-inflammatory cytokine (IL-10) was downregulated, whilst TGF-β was upregulated in the smokers.

BSH23-PO128 | NEQAS BC and WBS use cold-stored platelets to launch EQA programme for light transmission aggregometry

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Light transmission aggregometry (LTA) for bleeding disorders is a valuable diagnostic clinical service. LTA is complicated by: short timeframes for testing; (<4 h after phlebotomy), difficulty providing abnormal controls and internal quality controls (IQC) material limited to normal donors. Currently external quality assessment (EQA) does not audit the analytical process. NEQAS BC in conjunction with the Welsh Blood Service (WBS), has provided cold-stored pooled platelets for a pilot EQA exercise for LTA.

Pooled platelets were prepared at WBS (day 0) with platelet counts between $833–1045 \times 10^9$/L. Aliquots (15 mL) were dispatched to 27 sites plus FFP in temperature regulated transport boxes, to arrive within 24 h. Sites were required to adjust the platelet count, by diluting with FFP. LTA was performed on days 1, 2 or 3 using platforms including: Helena Aggram; PAP-8, Sysmex CN-6500 and Chronolog 700. The platelet count at the 27 sites was adjusted to $299–408 \times 10^9$/L prior to aggregation with agonists including: ADP; arachidonic acid; collagen; epinephrine; ristocetin; TRAP and U46619. Sites were grouped based on if they could perform agonist concentrations aligned to BSH or ISTH recommendations.

1. 98% of centres performing LTA with ADP at concentrations of: 10; 5 or 2 μM interpreted responses as abnormal.
2. 87% of centres performing LTA with arachidonic acid at a concentration of 1 mM interpreted responses as normal.
3. 98% of centres performing LTA with collagen at concentrations of: 2.5; 2; 1.25 or 1 μg/mL interpreted responses as abnormal.
4. 100% of centres performing LTA with epinephrine at concentrations of: 10 or 5 μM interpreted responses as abnormal.
5. 93% of centres performing LTA with ristocetin at concentrations of: 1.25; 1.2 and 0.5 mg/mL interpreted responses as normal/expected.

Greater than 86% consensus interpretation for results with recommended BSH or ISTH agonist concentrations, suggests platelets, delivered at 4–8°C, are a viable source of EQA material for LTA. Fifteen sites performed the testing on day 1, 8 on day 2 and 1 on day 3. The degree of consensus with the interpretative data demonstrates that the testing time window of 1–3 days, when the platelets are stored at 4–8°C, does not affect agonist response on the aggregometer.

LTA testing at multicentre locations is logistically possible. Therefore, centres can audit the majority of the analytical process, ensuring testing and interpretation is standardised when employing recommended BSH or ISTH agonist concentrations. This may aid clinical laboratories to gain accreditation for LTA under ISO standard 15189.

BSH23-PO129 | DOAC EQA scheme assay results and interpretations from UKNEQAS BC DOAC surveys 21–24, in 2021–2022

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The DOAC EQA scheme requires participants to return results plus it encourages participants to provide interpretations. Participants are performance assessed on results returned against an overall median for each of the DOAC they are registered for. Data was collected as part of the DOAC exercise surveys 21–24 (2021–2022). Participants were asked to provide interpretations for results returned from the following options: below level of detection (BLD); anticoagulant present at sub-therapeutic level (ApsubTL); anticoagulant detected, unable to state if therapeutic (ADUST); anticoagulant present at therapeutic level (APTL) or anticoagulant present at supra-therapeutic level (ApsupraTL).

Exercise 21 comprised of normal plasma without DOAC interpretations are as follows: dabigatran 0–22 ng/mL 51 BLD and 16 ApsubTL; rivaroxaban 0–52.1 ng/mL 74 BLD, 29 ApsubTL, 1 ADUST, 1 APTL; apixaban 0–39.8 ng/mL 75 BLD, 27 ApsubTL, 1ADUST, 1 APTL and edoxaban 0–25.9 ng/mL 41 BLD, 17 ApsubTL, 1 APTL.

Exercise 22 comprised of plasma with DOAC interpretations are as follows: dabigatran 93.1–161 ng/mL 5 ApsubTL, 15 ADUST, 49 APTL, 3 ApsupraTL; rivaroxaban 195–509 ng/mL 1 BLD, 24 ADUST, 33 APTL, 40 ApsupraTL; apixaban 50.18–165 ng/

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mL, 21 ApsubTL, 43ADUST, 33 APTL and edoxaban 200.5–287 ng/mL 11 ADUST, 40 APTL, 7 ApsupraTL.

Exercise 23 comprised of plasma with DOAC interpretations are as follows: dabigatran <8–72.19 ng/mL 2 BLD and 38 ApsubTL, 9 ADUST, 12 APTL; rivaroxaban 62-233 ng/mL 9 ApsubTL, 28 ADUST, 61 APTL, 3ApsupraTL; apixaban 36–119.89 ng/mL 27 ApsubTL, 20 ADUST, 46 APTL, 3 ApsupraTL and edoxaban 8–96.5 ng/mL 26 ApsubTL, 14 ADUST, 20 APTL and 3ApsupraTL.

The interpretations for the DOAC assays for the four surveys 21–24 2021–2022 were provided by either: anticoagulant nurses <1%; biomedical scientists 53%; clinicians 32% or clinical scientists 8%. There are a large number of centres, that do not provide any interpretation 7%.

DOAC monitoring is not required for the vast majority of patients receiving these anticoagulants. However, measuring DOAC levels is indicated in some settings including but not exclusively: bleeding risk; thrombotic risk; compliance and or accurate interpretation of other laboratory assays. The data returned in these exercises indicate that there is substantial variability in how the same levels of DOACs are interpreted. The need for clear on therapy levels is required to assist interpretation of DOAC dose adequacy following laboratory measurement. Monitoring these drugs in selected cohorts may help with establishing therapeutic windows or on-therapy ranges.

BSH23-PO13 | Real-world transfusion independence among newly diagnosed acute myeloid leukaemia patients receiving venetoclax combinations versus other therapies

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Background: This study is part of the ongoing Acute myeloid leukaemia (AML) Real-world evidenCe (ARC) Initiative in the US and Israel, which aimed to provide insight into the use and relative efficacy of venetoclax (VEN) -based combinations among newly diagnosed ND patients (pts) with AML.

Aims: To describe transfusion independence (TI) among ND AML pts treated with VEN-based combinations vs. non-VEN regimens in clinical practice.

Methods: The ARC Initiative is a multicentre chart review study of adult pts with AML who received VEN ≥April 2016 (VEN cohort) or non-VEN regimens ≥May 2015 (control cohort). Pts in the VEN cohort were matched 1:1 to pts in the control cohort based on age category (<60; 60–74; ≥75) and European Leukaemia Net (ELN) risk classification. Outcomes, including TI, haematopoietic cell transplantation (HCT), and best response, were assessed during first-line therapy over a transfusion observation period. TI was defined as no red blood cell (RBC) or platelet (PLT) transfusions during any consecutive ≥56-day period during the transfusion observation period. Kaplan–Meier (KM) analyses were conducted to assess the TI rate at 3 months after treatment initiation.

Results: A total of 119 VEN and 119 matched control pts with ND AML were included in this analysis. During a median transfusion observation period of 4.0 months in the VEN cohort and 3.6 months in the control cohort, 87.4% of VEN and 93.3% of control pts received ≥1 RBC or PLT transfusion. Pts in the VEN cohort received a median of 2.0 RBC and/or PLT transfusions per month and 5.0% received HCT; pts in the control cohort received a median of 3.5 RBC and/or PLT transfusions per month and 13.4% received HCT (all $p < 0.05$). Among 103 pts in the VEN cohort and 99 pts in the control cohort, 61.2% of VEN pts and 50.5% of control pts achieved TI during first-line therapy ($p = 0.17$), and 60.6% of VEN pts and 55.8% of control pts with response data achieved response ($p = 0.59$). The KM rate of achieving TI at 3 months was 45.0% in the VEN cohort and 35.9% in the control cohort (log-rank $p = 0.12$). Pts in the VEN and control cohort achieved TI after a median of 2.3 and 2.8 months, respectively ($p = 0.08$).

Conclusion/Summary: Interim results show pts receiving VEN-based combinations receive statistically fewer transfusions and are trending towards achieving TI more and more quickly after initiation of treatment compared to matched control pts.

BSH23-PO130 | Systemic snakebite envenoming causes histone-induced thrombocytopaenia

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Introduction: There are 1.8 million cases of snakebite envenoming each year causing 138 000 deaths and 400 000–500 000

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cases of long-term morbidity. WHO classifies snakebites as a priority neglected tropical disease with urgent need for new therapies beyond the limitations of current anti-venom strategies.

Snake envenoming can directly lead to haemotoxicity, neurotoxicity, or cytotoxicity. Haemostatic abnormalities, such as thrombocytopaenia play important pathological roles following systemic envenoming but we still do not understand why it happens. In other conditions associated with extensive cell death, we now know that the released histones can induce thrombocytopaenia and contribute to mortality.

The aims of this study were to determine (1) whether extracellular histones are released following systemic envenoming and (2) if they then play an important role in thrombocytopaenia associated with cytotoxic venoms.

Methods: Four venoms from three cobra species (genus Naja) endemic to Sub-Saharan Africa were used in this study. Mice were injected intravenously with either N. nigricollis (east and west African forms) or N. pallida (cytotoxic venoms) or N. haje (neurotoxic venom) venom, without or with anti-histone antibody (ahscFv). Circulating histones were quantified by Western blot and platelet counts determined by blood film. Washed human platelets were also incubated in vitro for 20mins with either venoms (25 μg/mL) or histones (0–100 μg/mL). Platelet activation was determined using anti-CD62P antibody and monitored by flow cytometry.

Results: Systemic envenoming with cytotoxic venoms resulted in significant elevations of circulating histones within 90mins (*N. nigricollis*; 74.6 ± 10.0 μg/mL and 88.0 ± 25.5 μg/mL, *N. pallida*; 129.0 ± 25.0 μg/mL), which was not observed with systemic neurotoxic envenoming (*N. haje*; 0.0 ± 0.2 μg/mL). Thrombocytopaenia was associated with elevated circulating histones following envenoming with cytotoxic venoms, and platelet counts remained normal post neurotoxic envenoming. Thrombocytopaenia associated with cytotoxic envenoming with *N. nigricollis* and *N. pallida* was abrogated with anti-histone antibody infusion. Incubating pathological levels of either cytotoxic (*N. nigricollis*; 7.8 ± 4.6% and 9.6 ± 3.4%, *N. pallida* 7.5 ± 4.2%) or neurotoxic (*N. haje*; 8.5 ± 2.5%) venoms with washed platelets had no effect on platelet activation compared to untreated platelets (UT; 5.0 ± 0.5%). Pathological levels of extracellular histones (25 μg/mL; 37.4 ± 12.0%, 50 μg/mL; 50.2 ± 3.0%, 100 μg/mL; 77.2 ± 11.6%) directly caused significant platelet activation.

Discussion: These results suggest that the thrombocytopaenia of cobra envenoming is caused by circulating histones and not by the venoms directly. This provides a potential therapeutic target, which is additional to anti-venom strategies in reducing haemostatic abnormalities and improving outcomes.

BSH23-PO131 | Thrombocytosis and acquired bleeding disorders in patients with myeloproliferative neoplasms

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Introduction: Patients with myeloproliferative neoplasms (MPNs) are at substantial risk of thrombosis, but can also develop bleeding due to acquired von Willebrand syndrome (AVWS). It is not clear what tests can best assess the bleeding risk and there is considerable variation in practice.

Aims: To assess:

1. Frequency of AVWS in patients with MPN and extreme thrombocytosis.
2. How best to assess their bleeding risk.
3. The relationships between platelet count, laboratory measures of VWF and bleeding events.

Methods: This is a single-centre observational study in a tertiary hub for MPN in London. Twenty-one consecutive patients with platelet count ≥800 × 10^9/L that attended clinic from May to December 2022 were included. All patients had full blood count, coagulation screening, liver and renal profile at baseline. Laboratory assessment comprised of platelet function analyser (PFA-100), von Willebrand factor antigen (VWF:Ag), ristocetin cofactor assay (VWF:Rco), collagen binding assay (VWF:CB) and VWF multimer analysis.

Results: Median age was 39 (17–82) years and M:F ratio was 17:21.

Essential thrombocythaemia was the predominant diagnosis (16/21), with 2 chronic myeloid leukaemia and 3 polycythaemia rubra vera. Sixteen patients were given antiplatelet treatment. Two had temporary anticoagulation (during pregnancy and primary thromboprophylaxis, respectively); 13/21 (61.9%) required cytoreduction and 2/21 (9.5%) required venesections. Median platelet count was 1064 (range 800–2048) × 10^9/L. An elevated platelet count was not always associated with reduced VWF:Rco. However, the VWF:Rco/VWF:Ag ratio was <0.7 in 16/21 subjects and showed a clear inverse correlation with platelet counts (r = −0.5442, p = 0.01). VWF:CB was mostly within normal range, except for three patients with levels <0.45 IU/mL (0.45–1.5). VWF multimers were normal. Flow obstruction limited PFA results to only 6 cases, where it showed either prolongation or non-closure.

Eight patients experienced bleeding events, mainly mucocutaneous manifestations, except two with significant bleeding (quadriceps haematoma following exercise and severe menstrual bleeding requiring emergency admission). Those who had bleeding were mainly young and otherwise healthy females with no significant comorbidities (5/8). All patients were on aspirin. Two had a platelet count <1000 × 10^9/L; three

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had normal VWF:Rco and four had a VWF:Rco/VWF:Ag ratio <0.7. PFA-100 results were available only in 3 of these subjects and were all prolonged.

Conclusions:

1. In MPNs with extreme thrombocytosis, bleeding can occur below the canonical platelet count of 1000×10^9/L.
2. All bleeding patients were on aspirin at the time of the event.
3. VWF:Rco/VWF:Ag ratio showed an inverse correlation with platelet count.
4. VWF multimers were normal and did not detect reduced VWF:Rco/VWF:Ag ratio.

BSH23-PO132 | Interim results of the LEAP pilot: Low molecular weight hEparin versus aspirin postpartum

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Expert opinion from the American Society for Haematology (ASH), the RCOG, UK, as well as the 'Society of Obstetricians and Gynaecologists of Canada' guidelines recommend a minimum of 6 weeks of prophylactic LMWH for patients with a history of venous thromboembolism (VTE) or high-risk thrombophilia. There are no RCTs determining the optimal duration of anticoagulation postpartum. A 2014 Cochrane review concluded that there is insufficient evidence to base thromboprophylaxis in the postpartum setting.

The principal question to be addressed in a large definitive RCT is whether a combination of prophylactic LMWH for 3 weeks followed by low dose ASA for the following 3 weeks (Group A) is non-inferior to the standard of care of prophylactic LMWH for 6 weeks postpartum (Group B). This single-centre pilot RCT addresses enrolment, adherence and feasibility.

Inclusion criteria: (1) Personal history of VTE prior to pregnancy and not prescribed therapeutic anticoagulation (2) Family history of VTE and antithrombin deficiency, proteinC or proteinS deficiency (3) Combined thrombophilia/homozygous for factor V Leiden mutation/prothrombin gene mutation, regardless of family history. Sample size: 50 patients, 25 patients treated according to each strategy. Primary outcomes: (1) Enrolment (2) Adherence (3) Loss to follow-up. Secondary outcomes: (1) VTE rate (2) Bleeding rate three months postpartum (3) Quality of life assessment (QoL) (Duke Anticoagulation Satisfaction Scale).

The interim results of our single-centre pilot feasibility RCT (https://clinicaltrials.gov/ct2/show/NCT05058924) demonstrated an enrolment rate of 63.6%, with 21 of 33 patients agreeing to participate. The adherence to treatment at 3 weeks was 97.4% and 95.9% in Group A and B, respectively ($p = 0.7$). At 6 weeks PPM, there was no difference in compliance to treatment in Group A vs. Group B (97.4 vs. 97.3) ($p = 0.52$). With regards to secondary outcomes, there was no VTE documented in either treatment arm. One patient in each arm reported mild gingival bleeding with tooth brushing. The QoL assessment score in Group A improved by 51% following the switch from LMWH to ASA (32.6 to 16.1), whilst it remained stable in Group B (24.5 vs. 24.6). At 6 weeks postpartum, the QoL score was 35% improved in the ASA group ($p = 0.01$).

Conclusion: Our data shows promising enrolment, adherence, loss to follow-up and feasibility of our ongoing pilot trial. Demonstration of non-inferiority of ASA and LMWH to LMWH will lead to a shorter course of LMWH postpartum, potentially reducing bleeding risk, need for subcutaneous injections, as well as cost, whilst improving quality of life.

BSH23-PO133 | Fibrinogen protects against histone-induced Thrombocytopaenia

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Introduction: Thrombocytopaenia, frequently observed in critical illness and predisposes to poor outcomes, is induced by high cell-free histone (CFH) levels (>30 μg/mL). Fibrinogen (FBG), a coagulation factor, electrostatically binds and precipitates CFH. We investigated (1) if this interaction protects against CFH-induced platelet activation, and (2) does CFH:FBG ratio have predictive utility for the development of clinically significant thrombocytopaenia in critical illness.

Methods: Isolated washed platelets were incubated for 15 min with CFH (50 μg/mL), with or without FBG preincubation (4 g/L). Platelet activation was assessed by fold change in CD62P-PE mean fluorescent intensity by flow cytometry. Platelet aggregation was monitored turbidometrically (405 nM).

CFH:FBG predictive utility was assessed through retrospective study of 278 patients admitted to the intensive care unit (ICU) at the Royal Liverpool University Hospital. Serial daily measurements across study duration (Days 1–4) included whole blood cell counts, coagulation parameters, and CFH levels (quantified by Western blot).

Results: Preincubation of CFH with FBG significantly reduces CFH-induced platelet activation (CFH alone; CD62P-PE = 3.49 ± 0.33 vs. CFH + FBG; CD62P-PE = 1.12 ± 0.06, $p < 0.001$), which was comparable to untreated controls (UT; CD62P-PE = 0.88 ± 0.08). Similar results were obtained using platelet aggregometry (CFH alone; 23.2 ± 1.7% vs. CFH + FBG; 11.1 ± 5.7%, $p < 0.05$).

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To determine the clinical threshold of CFH:FBG, we performed receiver operating curve analysis. CFH:FBG on ICU admission was used to predict the development of moderate-to-severe thrombocytopaenia (PLT < 100×10^9/L) during the study duration (AUC = 0.72). Sensitivity-specificity analysis and Youden's index calculation demonstrated an optimal CFH:FBG cut-off of 6.

29.5% patients (82/278) had CFH:FBG ≥ 6 and 70.5% (196/278) a CFH:FBG < 6. 59.8% (49/82) of those with CFH:FBG ≥ 6 developed moderate-to-severe thrombocytopaenia, compared to 22.2% (43/196) of those with CFH:FBG < 6 ($p < 0.001$). Nadir platelet count in the CFH:FBG ≥ 6 cohort was significantly lower compared to CFH:FBG < 6 [median = 140 (Q1–Q3 = 80–264) × 10^9/L vs. median = 237 (Q1–Q3 = 151–) × 10^9/L, $p < 0.001$].

Conclusion: FBG protects against CFH-induced thrombocytopaenia. CFH:FBG ≥ 6 may help predict development of clinically significant thrombocytopaenia.

BSH23-PO134 | Single-Centre experience on long-term safety and efficacy of Emicizumab in Paediatric patients

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Emicizumab, is bispecific humanised monoclonal antibody that promotes effective haemostasis in persons with haemophilia A (PwHA) with and without inhibitors. Given its easy administration as a subcutaneous injection, it became the preferred option for prophylaxis in both children and adults PwHA. Primary analyses of clinical trials and real-world data have shown emicizumab safety and efficacy. However, long-term data are limited.

We conducted a retrospective study to assess real-world long-term outcomes of paediatric patients on emicizumab in a tertiary paediatric haemophilia care centre. Only patients who had emicizumab prophylaxis for 1.5 years or more were included. Out 65 patients who started on emicizumab prophylaxis, 5 participants discontinued treatment early: 2 because of adverse events, 1 because of lack of efficacy, and 2 lost to follow-up. Moreover, 15 patients were either transitioned to adult services and follow-up data were not available or did not complete the 1.5 years of prophylaxis. Therefore, a total 45 patients (3 with inhibitors) were included.

Across a median efficacy period of 142.9 weeks (IQR, 130.2–159.7), ABR (annualised treated bleed rate) and percentage of treated bleeds were analysed in total and in 24-week intervals. The overall ABR was 0.24 (95% confidence interval [CI], 0.08–0.4). ABRs was stable at <1 in analysis of 24-week treatment intervals; with lowest ABRs recorded 0.1 at both 0–24 and 73–96 weeks intervals. During the total duration of follow-up 68.8% participants had 0 treated bleeds, 24.4% had ≤3 treated bleed. The lowest percentage of treated bleeds was at 0–24 weeks where 93.3% had zero treated bleeds. However, the percentage of 1–3 treated bleeds was the highest at the interval of 121–144 weeks (20%), and the percentage of >3 treated bleeds was the highest at 97–120 weeks interval (2.56%).

Emicizumab was well-tolerated, and no thrombotic events have been reported. Beyond the 5 patients previously discontinued emicizumab, on long-term follow-up one more participant discontinued treatment due to lack of efficacy. Also, one patient had recurrent joints bleeding with an ABR of 2, and therefore, he continued to receive recombinant FVIII prophylaxis for a period of 6 months concomitantly with emicizumab prophylaxis. In conclusion, emicizumab prophylaxis maintained low bleed rates in paediatric PwHAs with and without FVIII inhibitors and remained well-tolerated, with no new safety concerns.

BSH23-PO136 | Pregnancy outcomes following bone marrow transplant +/− total body irradiation: Retrospective data linkage

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Objective: Female childhood cancer survivors have increased risk of preterm birth (PTB), with relatively little known about pregnancy outcomes for women treated with bone marrow transplant (BMT) +/− total body irradiation (TBI), and management is likely variable. Preconception counselling and early referral to high-risk PTB prevention clinics (PTBPC) may improve outcomes. The British Society of Blood and Marrow Transplantation and Cellular Therapies (BSBMTCT) Data Registry holds detailed records for those who have received BMT in the UK and some units in the Republic of Ireland since 1974. The British Childhood Cancer Survivor Study (BCCSS; n = 17 980) and Teenage and Young Adult Cancer Survivor Study (TYACSS; n = 200 945) are large population-based cohort studies of five-year survivors of cancers, which have been linked to Hospital Episode Statistics for England (HES).

We are jointly funded by Action Medical Research/Borne to determine:

1. Current care offered to women who conceive following BMT +/− TBI.
2. Incidence of adverse pregnancy outcomes for these women.

These aims will be achieved using the following:

- National clinician survey
- Prospective UK Obstetric Surveillance Survey (UKOSS)
- Retrospective linkage of BSBMTCT and BCCSS/TYACSS datasets

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We will then work with stakeholders to develop national recommendations. Here we report on the retrospective data linkage.

Design: Retrospective data linkage.

Method: Anonymised linkage between the BSBMTCT and BCCSS/TYACSS datasets was performed, linking according to encrypted date of birth, diagnosis, and year of transplant. Strict adherence to GDPR was maintained throughout. Statistical analysis was then performed to provide information regarding pregnancy outcomes following BMT.

Results:

- Data for 34 pregnancies among 24 women with previous BMT +/− TBI were successfully matched. We continue to perform analysis and will present our data regarding the following pregnancy outcomes: underlying diagnosis, cancer management and treatment, decade of cancer treatment, gestational age at birth, mode of labour and birth, and birthweight.
- The UKOSS study has now completed data collection and will be analysed shortly.

Conclusions: We will present discussion of our experience of data linkage in this setting and will present our findings providing information regarding pregnancy outcomes for women who have previously had a bone marrow transplant. We will also present our initial prospective UKOSS findings.

BSH23-PO137 | Development of a functional assay to assess activation and proliferation in peripheral blood T cells

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Immunotherapies, either with engineered T cells or bispecific antibodies, are showing transformational activity in Diffuse Large B-Cell Lymphoma (DLBCL). Rational application of T-cell engaging therapies requires an understanding of T-cell function and phenotype. However, the functional status of the immune system in DLBCL, particularly T-cell fitness, remains poorly characterised. This highlights the need for reliable functional assays to determine T-cell fitness from the peripheral blood. In this study, an in vitro T-cell activation and proliferation assay was developed in healthy volunteers, which may enable the assessment of peripheral T-cell function in DLBCL and, therefore, inform treatment decisions for patients who are eligible for immunotherapy.

Peripheral blood mononuclear cells (PBMCs) were obtained from healthy donors and stimulated for 24–120 h with either anti-CD3/CD28 monoclonal antibodies or phorbol-12-myristate-13-acetate (PMA) plus ionomycin. Resultant T-cell activation (CD69 and CD25 expression), degranulation (CD107a expression), and proliferation (CellTrace Yellow [CTY] fluorescence) were assessed in CD4+ and CD8+ subsets using flow cytometry. The optimum antibody concentrations, protocol for stimulation and incubation time for proliferation were determined.

Optimum flow cytometry antibody concentrations ranged from 2 to 5 μg/mL. For optimal T-cell activation and degranulation after 24 h, PBMCs should be stimulated with 2 μg/mL plate-bound anti-CD3 and 2 μg/mL soluble anti-CD28. Sequential stimulation with anti-CD3 then anti-CD28 had no significant impact on T-cell activation compared to combined anti-CD3/CD28 delivery, but did result in inferior degranulation. Although PMA/Ionomycin stimulation produced superior activation and degranulation compared to anti-CD3/CD28, its downregulation of surface CD4 made it unsuitable for analysis of CD4+ T cells. Whilst assessing proliferative capacity at 72 h, additional phenotypic changes were observed, including increased forward and side scatter, and upregulation of CD4 and CD8 expression. Extending the duration of CD3/CD28 stimulation beyond 72 h did not change the proportion of T cells, which had divided, but did result in reduced CTY fluorescence, suggesting a greater number of divisions. Further optimisation is required to improve CTY peak resolution, which will enable more sophisticated analysis (including proliferation modelling) to be performed.

This study provides a reliable method of assessing T-cell activation and degranulation in CD4+ and CD8+ T cells. Once the feasibility of using cryopreserved patient samples has been confirmed, these functional assays will be used in translational studies to characterise the functional status of peripheral blood T cells in patients with DLBCL with the ultimate goal of guiding precision immunotherapy.

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BSH23-PO138 | MajesTEC-4: Phase 3 trial of Teclistamab+Lenalidomide versus Lenalidomide in patients with newly diagnosed multiple myeloma

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Background: Lenalidomide is standard-of-care in newly diagnosed multiple myeloma (NDMM); however, median progression-free survival (PFS) on Lenalidomide maintenance therapy is 52.8 months, long-term tolerability is poor, and most patients will relapse. Teclistamab is B-cell maturation antigen-(BCMA)×CD3 bispecific antibody that redirects CD3+ T-cells to induce cytotoxicity towards BCMA-expressing MM cells. In MajesTEC-1, Teclistamab monotherapy was well-tolerated and showed encouraging efficacy with deep and durable responses in heavily pretreated relapsed/refractory MM patients. Teclistamab+Lenalidomide is being investigated in MajesTEC-2 (NCT04722146) in MM patients who have received ≥2 prior line of therapy (LOT), including proteasome inhibitor (PI), immunomodulatory drug (IMiD), and anti-CD38 monoclonal antibody. Here we described MajesTEC-4, a randomised, open-label, multicentre, phase-3 study (NCT05243797) comparing efficacy of Teclistamab+Lenalidomide versus Lenalidomide alone as maintenance therapy in NDMM patients who have completed induction treatment followed by autologous stem cell transplant (ASCT), with/without consolidation. Teclistamab+Lenalidomide combination in maintenance setting is anticipated to provide greater efficacy than Lenalidomide alone.

Study Design and Methods: Eligible patients will be aged ≥18 years with NDMM (per International Myeloma Working Group-[IMWG] criteria) who received 4–6 cycles of 3 or 4 drug induction treatments (1 LOT, including PI and/or ImiD with/without anti-CD38 monoclonal antibody and achieved ≥partial response [PR]), had single or tandem ASCT, and had Eastern Cooperative Oncology Group performance status score 0–2. Patients with prior BCMA-directed treatment or progressive disease will be excluded. As combination of Teclistamab+Lenalidomide has not been previously evaluated in NDMM patients, a safety run-in will be performed. If no safety signal is observed, randomisation will begin. Approximately 1000 patients will be randomised 1:1 to receive 28-day cycles of Teclistamab+Lenalidomide or Lenalidomide. Study will continue until ~380 deaths or maximum 8-years after the last patient is randomised. Primary endpoint is PFS. Secondary endpoints include complete response (CR) conversion from PR/very good PR, minimal residual disease (MRD) negative conversion, ≥CR, MRD negative CR, sustained MRD negativity, OS, PFS2, time to next treatment, patient-reported outcomes, and adverse events (Aes). Responses will be assessed using 2016 IMWG criteria. Aes will be graded by Common Terminology Criteria v5.0, except for cytokine release syndrome and immune effector cell-associated neurotoxicity syndrome, which will be graded by American Society for Transplantation and Cellular Therapy guidelines. Patient enrolment began in May 2022. Results from MajesTEC-4 trial will provide insights into a possible new synergistic combination that could improve response and prolong survival versus current maintenance regimens for patients with NDMM.

BSH23-PO139 | Burden of respiratory coinfections in haematology patients with COVID-19

Dr Roshni Mitra[1], Dr Emily Hobbs[1], Mr Tim Smith[1], Dr Roochi Trikha[1], Dr Pramila Krishnamurthy[1], Dr Malur Sudhanva[1], Dr Austin Kulasekararaj[1]

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Introduction: Following the lifting of generalised restrictions and universal masking, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)-infected patients, especially the clinically extremely vulnerable (CEV) haematology patients, are at an increased risk for other respiratory viral coinfections; therefore, physicians need to be cognizant about excluding other treatable respiratory pathogens. Here, we report coinfection with SARS-CoV-2 and other respiratory pathogens in patients with haematological cancers presenting to a large tertiary care hospital.

Methods: From July 2022–December 2022, patients with haematological disorders were screened for SARS-CoV-2 and other 10 common respiratory pathogens by PCR. We

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performed a retrospective analysis of patients with concurrent respiratory viruses and will prospectively evaluate the same from Jan 2023 to March 2023.

Results: During this period a total of 322 inpatients had routine screening and additional 6213 swabs were done in the outpatient/ambulatory setting, of which 294 were positive in 221 patients. We excluded all patients who had a single positive PCR swab result and specifically analysed only patients with coinfections. We identified 30 patients (14%) who had respiratory coinfections with 73 viral infections/reactivations over 6 months period, which represented 25% of all positive swabs: 25 inpatients (19 symptomatic/6 asymptomatic) and 48 in outpatients (32 symptomatic/16 asymptomatic). The median age of the cohort was 47.3 years (21–77). Patients were post allograft (n = 15), autograft (n = 7), post CART (n = 5) and postchemotherapy (n = 4).

Of the 30 cases, 13 patients had concurrent infections: 5 SARS-CoV2, 10 Respiratory syncytial virus (RSV), 7 Rhino and 4 Influenza A, with all patients having dual viral infection. The remaining 17 patients had multiple viral infections but separated by a median of 54 days (range 27–137 days): 16 SARS-CoV2, 5 RSV, 6 Rhino, 2 Parainfluenza, 2 Adeno and one each of Influenza A, Influenza B, and metapneumovirus. Of the treatable infections (n = 46), 22% were detected on routine asymptomatic swabbing, with 50% of SARS-CoV2 detected on routine swabs. All 8 patients with Influenza were treated with oseltamivir, of 16 RSV cases one was treated with oral ribavirin and of the 22 SARS-CoV2 patients, 5 were treated (4 Paxlovid and 1 Remdesivir).

No patients needed intensive care support and no deaths were reported.

Conclusion: The burden of respiratory coinfections in CEV cohort has a significant impact on respiratory isolation and management, including appropriate & timely initiation of therapy for treatable viral infections. Although mortality was not increased secondary to respiratory coinfections and none needed intensive care, larger prospective cohorts are needed to assess the exact impact.

BSH23-PO14 | Developing an advanced clinical practitioner-led haematology ambulatory service (HAS)

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Introduction: Significant challenges continue to add pressure on the National Health Service (NHS) and its workforce. Haematology has seen its number of patients, lines of treatments with ever-improving safety profiles, and workload increase in recent years, which is only expected to increase further. The National Institute for Health and Care Excellence (NICE) places mounting emphasis on ambulatory care for haematology patients since it has been shown to improve patient outcomes and experience. As an advanced clinician capable of complex consultations, assessments and management planning, the Advanced Clinical Practitioner (ACP) can have an exciting role to develop and lead the Haematology Ambulatory Service (HAS).

At CRH the ACP-led HAS was developed initially for AML patients who have received intensive treatment and were predicted to have an extended period of neutropenia ($\leq 0.5 \times 10^9$) for ≥7 days. Initial process mapping showed that historically, patients would remain as an inpatient in hospital where they would receive low-intensity interventions whilst waiting for count recovery, with an expected timeframe of 4 to 6-weeks. An initial retrospective data collection study showed that there was an opportunity to save an average of 20.6 bed days per episode by allowing patients to be discharged home as part of the proposed HAS.

Using the Plan-Do-Study-Act principles, the HAS was developed and introduced in May 2022. A service audit was completed in November 2022, which was assessed against local and national clinical standards. Patient feedback was obtained from a patient questionnaire.

Results: From May 2022 to November 2022, a total of 86 bed days were saved over 5 HAS episodes (3 patients) with an average of 17.2 days per HAS episode. Patients graded the ACP-led HAS 9.5 out of 10 overall with the care received by the ACP scored 10 out of 10. 100% hospital readmissions for neutropenic sepsis were correctly managed and all patients were reviewed within 24 h of admission to hospital by the ACP or consultant.

Conclusion: The HAS at CRH stands as a successful example of how the Haematology ACP role can be optimised and evolved to lead an outpatient ambulatory service, which improves patient experience, clinical outcomes and departmental resourcefulness. It is prudent to expect that the ambulatory service and role of ACP may be expanded further to cover a wider range of patient groups of different haematology diagnoses.

BSH23-PO140 | *Stenotrophomonas maltophilia* infection within a UK teaching hospital haemato-oncology unit

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Stenotrophomonas maltophilia is an environmental bacteria demonstrating intrinsic resistance to most commonly used antimicrobials, including carbapenems. *S. maltophilia* typically causes intravascular device-related blood stream infections in the healthcare setting, but in the immunocompromised may also cause soft tissue, pulmonary and disseminated infection with substantial morbidity and mortality. The mainstay of treatment is removal of infected devices and

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treatment with co-trimoxazole (trimethoprim/sulfamethoxazole). Despite most isolates of *S. maltophilia* remaining co-trimoxazole susceptible, mortality remains 21%–69%.

Methods: We performed a retrospective analysis of all patients whose cultures yielded *S. maltophilia* between 2007 and 2022. Baseline patient characteristics were collected including: age, diagnosis, duration of neutropenia, CRP at day 0 and at nadir, site of infection, number of positive cultures, antimicrobial susceptibility, and antimicrobial therapy. Clinical infection was defined as positive blood culture or *S. maltophilia* grown from 2 sites. Hazard ratios were calculated by Cox's method. Statistical analysis was performed using the computing environment R.

Results: *S. maltophilia* was recovered from cultures taken from 116 patients; 112 (96%) of these were considered infections. 36 patients received allografts & 14 autografts & 48 patients had recently received chemotherapy. Median number of days of admission was 17 (range –3 to 123), mean number of prior admissions was 1.3 (*r* 0–12), 53% of these admissions were in the previous 90 days prior to *S. maltophilia* infection. 51% of patients were neutropenic prior to infection, median days of neutropenia was 14 (*r* 1–239). 31% patients with *S. maltophilia* infection required ITU admission.

Of 112 *S. maltophilia* infections; 85% were isolated from a central venous catheter (CVC), 26% from peripheral cultures, 10% from sputum or broncho-alveolar lavage, 3% were identified from other source. 6 (5%) of *S. maltophilia* isolates were resistant to co-trimoxazole, 8 (7%) were resistant to colistin, with one resistant to both. 41% of patients had other bacteraemia prior to *S. maltophilia*.

28 day all-cause mortality was 31%, Neutropenia at time of *S. maltophilia* infection was significantly associated with death (Hazard ratio: 1.99, p = 0.046, 95% CI 1.01–3.92), underlying diagnosis, timing of line removal, antibiotic resistance or whether the patient received co-trimoxazole treatment were not significant.

Conclusion: To our knowledge, this is the largest cohort of haematology patients with *S. maltophilia* infection described. Infection remains a significant burden in terms of mortality, and within our cohort optimum antimicrobial therapy (Colistin/co-trimoxazole & central line removal) did not alter outcome. Preventing colonisation and subsequent infection is, therefore, of paramount importance.

BSH23-PO142 | Re-vaccination against SARS-COV2 in allogeneic HSCT patients: Repeated primary vaccines doses increases seroconversion rates

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Patients undergoing treatment for haematological malignancies have been shown to have reduced antibody responses to vaccination against SARS-COV2. This is particularly important in patients who have undergone allogeneic haemopoietic stem cell transplantation (HSCT), in whom there is limited data about vaccine efficacy. In this retrospective single-centre analysis, we present data on serologic responses following one, two, three or four doses of either Pfizer-BioNTech (PB), AstraZeneca (AZ) or Moderna (MU) SARS-CoV-2 vaccines from a series of 75 patients who have undergone allogeneic HSCT within 2 years from the time they were revaccinated.

The seroconversion rates following post-HSCT vaccination were found to be 50.7%, 78%, 79% and 83% following the first, second, third and fourth primary post -HSCT vaccine doses, respectively. The median time from allograft to first revaccination was 145 days (range 79–700). Our findings suggest that failure to respond to the first SARS-CoV-2 vaccine post-HSCT was associated with the presence of acute GVHD (p = 0.042) and treatment with rituximab within 12 months of vaccination (p = 0.019). A statistical trend was observed with the presence of chronic GVHD and failure to seroconvert following the second (p = 0.07) and third (p = 0.09) post-HSCT vaccine doses. Patients who had received one or more SARS-CoV-2 vaccines prior to having an allogeneic stem cell transplant were more likely to demonstrate a positive antibody response following the first dose of revaccination against Sars-CoV-2 (p = 0.019) and retained this seropositivity following subsequent doses. The incidence of confirmed COVID-19 diagnosis among this cohort at the time of analysis was 16%. 17% of these were hospitalised and there was one recorded death (8%) secondary to COVID-19 in a patient who was 15.7 months post allogeneic transplant.

In summary, this study suggests that despite the initial low seroconversion rates observed postallogeneic transplant, increasing levels of antibody response are seen post the second primary vaccine dose. In addition, there seems to be lower risk of mortality secondary to COVID-19 in this vaccinated population, compared to what was reported in the earlier phases of the pandemic prior to use of SARS-COV2 vaccination. This adds support to the widely adopted policy of early full revaccination with repeat of primary vaccine doses and boosters post-HSCT to reduce mortality in this population. Finally, we have identified rituximab use and active GVHD as potential risk factors influencing serological responses to SARS-COV2 vaccination and further work should focus on further characterising this risk and optimum dosing schedule both pre- and post-transplant.

BSH23-PO143 | Haploidentical transplantation for patients with fanconi anaemia

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Background: Fanconi anaemia (FA) is a heterogeneous bone marrow failure disorder characterised by progressive

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marrow failure and predisposition to leukaemia and epithelial malignancies. Allogeneic stem cell transplantation (alloHSCT) is the only therapy that can correct the haematological manifestations of FA patients. Haploidentical transplantation has increased the pool of available donors.

Methods: Retrospective analysis of FA patients who underwent haploidentical allogeneic HSCT at King Hussein Cancer Center (KHCC) from Jan 2005 until Jun 2022.

Results: We are reporting the outcome of haploidentical alloHSCT for 8 FA patients, with overall survival of 75% (95% CI: 31.5–93.1). Five patients underwent upfront T-cell repleted transplant with post-transplant cyclophosphamide (PT-CY). The preparative regimen included Fludarabine and single fraction total body irradiation (TBI), one patient received cyclophosphamide, which was omitted later due to toxicity, and rabbit anti-thymocyte globulin was added to the last two patients to decrease the graft vs. host disease (GVHD). All patients received calcineurin inhibitors and Mycophenolate mofetil starting from day five post-transplant. The timing of immunosuppressants was adjusted for the last patient from day zero and one post-transplant, respectively. Three patients were rescued after primary graft failure, one after a related donor transplant using PT-CY, and two after unrelated cord blood transplantation. Alemtuzumab in the stem cells bag was used for T-cell depletion. The median time for neutrophil engraftment was 13 days (range, 9–18) and 14 days (range, 11–93) for platelet engraftment. Seven (90%) patients developed acute GVHD, six (75%) developed CMV reactivation (one had CMV colitis), six (75%) patients developed chronic GVHD, and three patients are still receiving immunosuppressants more than 4 years post-transplant. We report three deaths, one due to chemotherapy toxicity before engraftment, the second patient died 8 months post-transplant due to extensive chronic GVHD and viral reactivation, and the third death was in the rescue group due to squamous cell carcinoma 5 years post-transplant.

Conclusions: T-cell repleted haploidentical donor transplantation with PT-CY is feasible for FA patients without a matched related, with increased risk of post-transplant complications, and requires extensive supportive care. Modifying the timing of r-ATG and the GVHD prophylaxis may decrease the toxicity.

BSH23-PO144 | BTKi-treated CLL patients have Favourable survival outcomes but concerning levels of second malignancies and COVID-related mortality

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Introduction: Bruton's tyrosine kinase inhibitors (BTKi) are used extensively within the NHS to treat specific B-cell malignancies with patients stopping BTKi usually due to adverse events or disease progression. The objective of this study was to analyse effectiveness of BTKi therapy for chronic lymphocytic leukaemia (CLL) at our centre compared to previously published real-world data from the UKCLL Forum (Follows et al, Blood 2019). In addition, we investigated treatment-related adverse events (AE) and second malignancies.

Methods: This is a single-centre retrospective study of 112 CLL patients treated with a BTKi for a minimum of 4 weeks between 2014 and 2022 (ibrutinib n = 71, acalabrutinib n = 38, zanubrutinib n = 3). Treatment was first line (n = 39), second line (n = 44) and 3+ line (n = 29). Patient demographics, duration of BTKi therapy, Aes, discontinuation reasons and second malignancies were collected. Aes were compared with a parallel cohort of 53 non-CLL BTKi-treated patients.

Results: Median age starting treatment was 73 years, and 71% were male. Primary outcomes were discontinuation-free survival (DFS) and overall survival (OS). With a median follow-up of 3.90 years, the median DFS was 4.18 years (95% CI: 3.52–4.91) with a median OS of 6.35 years (95% CI: 5.52–NA). These compare favourably with previous UKCLL forum data (median DFS = 2.79 years; median OS = 4.66 years), although our patients were more likely to receive BTKi earlier in treatment (3rd line or beyond: 26% of our patients vs. 78% in the UKCLL Forum). The most common Aes included bleeding, cytopenia, infection, cardiac events and mouth ulcers, with 21% stopping BTKi for CLL due to Aes whilst 15% of non-CLL BTKi patients stopped due to an AE. Second malignancies were reported in 49% of CLL patients, yet only 34% of non-CLL patients. Among patients with a confirmed cause of death, infection was the most common cause (39%), followed by CLL (33%), then second malignancy (18%). Of the 31 deaths in 2020 and 2021, 7/31 (23%) were due to, or in association with COVID-19 infection. No COVID-19 deaths were associated with BTKi in non-CLL patients.

Conclusions: We demonstrate a favourable real-world DFS and OS for BTKi-treated CLL patients although a high number of patients still stop BTKi due to Aes. The very high incidence of second malignancies for all BTKi-treated patients and COVID-19 and infection-related deaths for CLL patients is concerning. As CLL is known to associate with high levels of second cancers, it remains unclear whether BTKi use increases this risk further.

BSH23-PO147 | A real-world study of belantamab mafodotin in a heavily pretreated population of multiple myeloma patients

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The antibody–drug conjugate belantamab mafadotin (belamaf) has been approved for the treatment of relapsed/refractory multiple myeloma after its evaluation in the DREAMM-2

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trial. Here we assess the efficacy and safety of belamaf in a cohort of patients treated at a UK myeloma centre.

After local ethical approval we retrospectively assessed all patients commenced on belamaf at The Royal Marsden Hospital before 21/06/2022 (follow-up until 01/08/2022). Belamaf was provided on a compassionate use basis and was initially given at 2.5 mg/kg every 3 weeks. Response was assessed using International Myeloma Working Group criteria. Progression-free survival (PFS) and overall survival (OS) were calculated from cycle1 day1 and medians compared using the log-rank test. Cytogenetic high-risk (HR) was defined as the presence of t(4;14), t(14;16), t(14;20), del(17p) or gain(1q) (most recent analysis prior to starting belamaf).

Twenty patients received belamaf, with a median age of 64 years (range 45–84) and a median 5 prior lines of therapy (range 3–11). Most patients had HR disease (89%, 16/18 available) and were International Staging System stage 3 (55%, 11/20). Of the 20 patients, 19 were assessable for response (1 patient had asecretory disease). The overall response rate (ORR), ≥ partial response (PR), was 47% (9/19) with 32% (6/19) achieving at least a very good partial response. The median PFS was 4.0 m (median follow-up 6.9 m) and the median OS was 8.4 m (median follow-up 9.4 m). Achievement of ≥PR was associated with a significantly longer PFS and OS (median PFS 6.8 m vs. 1.4 m, $p < 0.01$; OS not yet reached vs. 5.7 m, $p < 0.05$).

Ten patients (50%) had received ≥4 cycles at the time of data cut-off; 5 remain on treatment. The reason for stopping therapy was progression/lack of response in 93% (14/15) and CNS VZV infection in 1 patient. Eye toxicity was assessed in 18 patients (2 had no follow-up data). Any grade keratopathy occurred in 72% (13/18) and ≥grade 3 in 5.6% (1/18). Change in best-corrected visual acuity occurred in 56% (10/18) and ≥grade 3 in 33% (6/18). Haematological toxicities were also observed with 50% (10/20) requiring a transfusion of platelets and/or red cells during treatment.

In this heavily pretreated population, belamaf was associated with an ORR of 47% and the PFS and OS were 4.0 and 8.4 months, respectively. Our results are in keeping with data from the DREAMM-2 trial and real-world studies and support belamaf as an option in the relapsed/refractory setting with careful management of ocular toxicity.

BSH23-PO148 | Response rates to second line treatment with Daratumumab Bortezomib dexamethasone (DVD) in relapsed/refractory AL amyloidosis

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Background: Bortezomib based regimes are the most common first-line treatments in patients with systemic AL amyloidosis. The impact of 2nd line DVD in patients who have relapsed after or are refractory to frontline bortezomib has not been reported.

Objective: The primary objective was to assess the impact of second line DVD in patients who received Bortezomib therapy upfront analysing outcomes based on depth and duration of haematological response to frontline therapy.

Methods: All patients from a prospective observational study of newly diagnosed AL amyloidosis (ALCHEMY) who received frontline Bortezomib based therapy and second line DVD were analysed. The best haematological response achieved following DVD was compared to the response to frontline.

Results: 116 patients were included, 75 (64.7%) male and median age at diagnosis 64 years. 40 (34.5%) patients were Mayo stage 3a/3b. 85 (73.3%) patients had λ AL-type and 30 (25.9%) were κ AL-type.

Median number of cycles of frontline therapy received was 6 and median time to next line of treatment was 17 months (range 1–86) with 44 (37.9%) requiring second line treatment within 12 months from diagnosis. 41 patients (35.3%) achieved a complete haematological response (CR), 43 (37.1%) very good partial response (VGPR) and 32 (27.6%) achieved a partial response (PR) or no response (NR).

All patients received second line DVD. The best haematological response following DVD was; CR 42 (36.2%), VGPR 36 (31.1%) and PR/NR 35 (30.2%). There was no significant difference in proportion of patients achieving a CR/VGPR to DVD based on time of relapse. However, of patients achieving PR/NR to frontline treatment, only 44% achieved a CR/VGPR to DVD compared with 80% achieving a VGPR/CR at second line in those who had previously achieved VGPR/CR to 1st line.

19 patients required third line therapy with median time to next treatment of 13 months (range 0–34 months) and 21 (19%) patients died. Proportion of patients surviving 24 months after DVD was 96%/73%/52% in those achieving CR/VGPR/PR to second line DVD.

Conclusion: Patients with proteasome inhibitor sensitive disease (i.e. CR/VGPR to 1st line treatment) show excellent response to 2nd line treatment with DVD, irrespective of the duration of response to 1st line treatment. However, those achieving less than VGPR to initial bortezomib based regime have poorer responses to DVD. DVD remains a treatment option for relapsed proteasome inhibitor sensitive patients with AL amyloidosis.

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BSH23-PO149 | A new isoelectric focusing (IEF) based technique to diagnose and monitor non-secretory multiple myeloma

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In most patients with myeloma, the tumour cell produces a unique monoclonal immunoglobulin (Mig), which serves as a quantitative surrogate biomarker for tumour burden. Patients with non-secretory myeloma (NSM) represent 1%–3% of patients and do not produce a Mig suitable for standard monitoring. Therefore, clinicians rely on bone marrow biopsies and CT-PET/WB MRI scans, which are challenging for patients/clinicians, especially when required regularly for monitoring. Consequently, there is an unmet need for a blood-based biomarker in this patient group to determine remission induction and disease relapse.

We have developed a new method to detect monoclonal light chains in this series of patients with NSM, (negative on immunofixation and no measurable disease marker (sFLC <100 mg/L)). Isoelectric focusing (IEF) is well-established for detecting IgG oligoclonal banding in cerebrospinal fluid to diagnose multiple sclerosis. Using IEF, any monoclonal immunoglobulin displays a highly reproducible unique spectrotype of bands. We have developed serum IEF to identify monoclonal sFLCs using highly specific and sensitive monoclonal antibodies.

Study patients were identified from the MRC Myeloma IX, CRUK Myeloma XI or NIHR TEAMM trials. Overall, serum IEF detected monoclonal FLC in 28/32 patients (88%): 14 patients with κ FLC and 10 λ FLC. One patient had both κ and λ FLC bands detected, likely reflecting bi-clonality with both active myeloma clone and secondary MGUS clone picked up by the sensitive IEF assay.

Follow-up samples were available in 27/32 patients. Serum IEF identified response to therapy (disappearance of band) and relapse (reappearance of band). The method also allows serial sample dilution to determine percentage reduction in monoclonal protein and provide semi-quantitative results in response to therapy.

The IEF method identified monoclonal FLC in 86% of NSM patients studied, indicating that most are low-level secretors rather than true non-producers. Although their Mig cannot be monitored using standard techniques, our highly sensitive IEF assay allows NSM patients to be diagnosed and monitored using blood samples. This approach could reduce reliance on bone marrow biopsy and imaging in this patient cohort. Mass spectrometry (MS) is emerging as another exciting blood-based option for Mig detection. However, the limit of detection for MS is <10 mg/L compared to IEF sensitivity <1 mg/L. Further, the IEF assay utilises a low-cost electrophoresis platform available in many laboratories that conduct myeloma testing, enabling easier integration into clinical practice. Future clinical studies are required to validate IEF and should include comparison to other technologies, including MS.

BSH23-PO15 | The journey to e-Referrals in Haematology: A local experience of implementation at a tertiary Centre

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Background: Like many specialities, haematology departments are dealing with an increasing and under-recognised ad hoc workload relating to liaison referrals (mostly inpatient), which historically have involved phone calls to the on-call specialist trainees (STs). This makes it difficult to capture activity/plan workforce and also to ensure robust governance. We introduced non-urgent in-hours E-referral service for haematology using an advice and guidance format, at a large university hospital (1700 beds, 2 sites). Potential benefits include better clinical governance, ease of auditing, monitoring workload for operational planning, and freeing up of the on-call phone for urgent queries.

Method: Audit of consecutive samples of phone calls to the on-call STs before and after the introduction of the E-referral service. Data was collected on the subjective clinical appropriateness of the referral and whether it could have been handled instead as an E-referral. We also looked at the volume of E-referrals over the first year, a more detailed review of response times and outcomes for 89 consecutive E-referrals, and a consultant quality review of haematology STs responses to 20 random E-referrals in August–October 2022.

Results: Among 325 phone calls in the baseline survey, participating haematology STs reported that 28% of calls could have been managed as E-referrals instead. This remained a significant proportion 36% on reaudit 3 months after E-referrals went live. Only 59% of the referrals made were felt to be clinically appropriate queries in baseline audit, and similarly 62% in the follow-up sample.

From October 2021–September 2022, we received mean (range) 200 (135–238) E-referrals per month. Focused review of 89 E-referrals found that the target of response within 24 h was achieved in 85% of cases (98% by 7 days) with no adverse clinical incidents reported. Educational and constructive consultant feedback was provided to haematology STs for all the E-referrals selected for quality review.

Conclusion: The E-referral system enabled timely and safe responses to non-urgent haematology enquiries. We are still receiving a significant number of phone referrals that could be managed by e-referral. Moreover, the data highlighted

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that around 40% of phone enquiries were considered clinically inappropriate and could be more easily redirected if e-referrals became the standard for non-urgent requests, further freeing up the on-call ST to deal with urgent referrals. Recognition of the scale of referral activity should facilitate operational and workforce planning, particularly for haematology STs. Finally, the e-referral system may provide greater opportunities for clinical audit, education and training.

BSH23-PO150 | Preliminary safety of Bcl-2 inhibitor BGB-11417 in relapsed/refractory multiple myeloma Harbouring t(11;14): Phase 1b/2 study

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Background: B-cell lymphoma-2 (Bcl-2) proteins play an important role in multiple myeloma (MM) cell survival and represent an attractive therapeutic target. In prior trials, a subgroup analysis of patients with t(11;14)-positive relapsed/refractory (R/R) MM showed the combination of a Bcl-2 inhibitor, a proteasome inhibitor, and dexamethasone improved progression-free survival with no increased mortality. BGB-11417, a Bcl-2 inhibitor, is more potent and selective than venetoclax. BGB-11417-105 (NCT04973605) is a phase 1b/2 study assessing the safety and efficacy of BGB-11417 monotherapy, in combination with dexamethasone, or with dexamethasone+carfilzomib in patients with t(11;14)-positive R/R MM. Preliminary safety results for the combination of BGB-11417 + dexamethasone are presented.
Methods: Eligible patients had t(11;14)-positive R/R MM and had been exposed to a proteasome inhibitor, immunomodulatory agent, and anti-CD38 therapy. Patients received 80-, 160-, 320-, or 640-mg BGB-11417 daily with 40-mg dexamethasone weekly until death, intolerability, or disease progression. Dose escalation was guided by a mTPI-2 design and overall review by a safety monitoring committee. Pharmacokinetics (PK) were also assessed.
Results: As of 1 July 2022, 10 patients were enrolled in the 80-, 160-, and 320-mg (3 patients each) and 640-mg (1 patient) dose-escalation cohorts of BGB-11417 + dexamethasone. The median age was 69 years (range, 52–81) and median prior lines of therapy was 3 (range, 1–5). The median treatment duration was 3.2 months (range, 0.5–6.5). No patients experienced dose-limiting toxicity at any dose level. Three patients died whilst on study: 1 due to COVID-19 complications 157 days after treatment discontinuation (day 208), 1 due to progressive disease 50 days after treatment discontinuation (day 89), and 1 due to COVID-19 whilst on study treatment (day 78). No deaths were associated with study treatment. Two patients experienced Grade ≥ 3 treatment-emergent adverse events (TEAEs). One patient in the 160-mg cohort experienced Grade 3 increase in liver enzymes and lymphopenia. One patient in the 320-mg cohort experienced Grade 3 lymphopenia. The most common TEAEs were insomnia (50%), fatigue (30%), arthralgia (20%), back pain (20%), lymphopenia (20%), and nausea (20%). BGB-11417 exposure increased dose-dependently from 80 mg to 320 mg with high interpatient PK variability. BGB-11417 exposures after single and multiple doses appeared similar, indicating limited accumulation.
Conclusion: BGB-11417 plus dexamethasone was generally well-tolerated in patients with R/R MM harbouring t(11;14) at doses ≤640 mg. Efficacy data are forthcoming. Recruitment is ongoing in the US, Australia, and New Zealand; the BGB-11417, dexamethasone, and carfilzomib combination arm will open in the future.

BSH23-PO151 | Evaluation of novel IKK-alpha inhibitors for the treatment of chronic lymphocytic leukaemia (CLL)

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Introduction: Despite inhibitors to Bruton's tyrosine kinase, phosphatidylinositol-3 kinase and B-cell lymphoma-2 revolutionising chronic lymphocytic leukaemia (CLL) treatment, not all patients respond, and these treatments are not curative. Nuclear factor kappa B (NF-κB) is a transcription factor that is commonly dysregulated in CLL, and recurrent NF-κB-associated mutations such as NFKBIE, NOTCH1 and BIRC3 confer a poor prognosis. NF-κB signalling can be triggered by a canonical pathway (IKKα/β/γ signalling) and a non-canonical pathway (two IKKα homodimers). Although therapeutic targeting of NF-κB holds great promise, IKKβ inhibition is highly toxic. In contrast, the therapeutic potential of targeting IKKα in CLL has not been fully explored.
Methods: Five novel IKKα inhibitors were evaluated in vitro (SU1261, SU1349, SU1644, SU1680 and SU1699). MEC-1 cells and primary CLL cells (n = 14) were plated (1×10^6/mL) in 24-well plates. All inhibitors were compared to a structurally similar compound with no activity against IKKα (SU1715) as a control. Cells were incubated with serial dilutions of each compound for 48-h, followed by Annexin V-APC and 7-AAD labelling. The percentage apoptosis was quantified using flow cytometry and the LC_{50} values (concentration required to kill 50% of cells) were interpolated from sigmoidal dose–response curves. In primary samples, cells were stained with CD5, CD19 and Annexin V-APC. The percentage apoptosis of CLL cells (CD19+CD5+) and autologous T cells (CD19-CD5+) were determined using flow cytometry. Drug responses were compared in prognostic subsets including immunoglobulin gene (IGHV) mutational status, CD38 and CD49d expression.

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Results: In kinase screening assays, SU1261 and SU1349 showed 68-fold and 209-fold selectivity for IKKα over IKKβ, respectively, and in MEC-1 cells, SU1261 was identified as the most potent inhibitor. In 14 patient samples, SU1349 and SU1261 were most effective at inducing apoptosis in CLL cells compared to the control drug SU1715 (SU1349: 85%, $p < 0.0001$; SU1261: 63%, $p < 0.0001$) and had the lowest LC_{50} values (SU1349 = 2.42 μM, SU1261 = 7.49 μM). Both SU1349 and SU1261 were effective irrespective of CLL IGHV mutational status or CD38 and CD49d expression. Importantly, both compounds preferentially killed CLL cells when compared with autologous T cells (SU1349: $p < 0.0001$, SU1261: $p < 0.0007$), indicating a tumour selective response.
Conclusion: We have demonstrated that two IKKα selective compounds have low micro-molar toxicity in MEC-1 and primary CLL cells, with no loss of potency in CLL cells derived from poor prognostic subsets, and minimal cytotoxicity on autologous T cells. These data highlight the potential for selective IKKα inhibitors for the treatment of CLL.

BSH23-PO152 | Insertional diversification of the immunoglobulin heavy chain in chronic lymphoproliferative disorders

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Insertional diversification of the immunoglobulin heavy chain (IgH) switch and variable regions (also known as templated mutagenesis) occurs when nucleic acid sequences from non-IgH sites are integrated into these regions. Insertion of DNA is hypothesised to occur during B-cell maturation when the B-cell receptor genes undergo variable, diversity and junctional (VDJ) gene recombination, somatic hypermutation and class-switch recombination. Proposed mechanisms for the development of these insertions include R-loop formation. Insertional diversification events have been described in normal donors, but not in lymphoproliferative disorders.
We screened 819 gel electrophoresis samples from routine immunoglobulin heavy chain variable region (IGHV) sequencing in Chronic Lymphoproliferative Disorder (CLPD) patients for the presence of primary or secondary bands that correlated with longer than expected polymerase chain reaction (PCR) amplicons. These bands could represent insertional diversification events or duplicate VDJ gene sequences. 64 samples had secondary bands, and on review of the consensus sequences 9 had duplicate V or VDJ sequences and 2 had inserted sequences, one in a chronic lymphocytic leukaemia sample, and one in a systemic marginal zone lymphoma sample. The inserts were 209 and 222 bp in length and aligned to the same intergenic region on chromosome 6 that was characterised as a 'Dnase hypersensitive' site.
To uncover whether insertional diversification events also occurred in the switch region of clonal samples a targeted PCR assay for next-generation sequencing (NGS) of the switch region was developed. Loci for insertion events were identified using a publicly available datasets and mapped onto the switch region where they clustered to form 'hotspots' in both the switch Mu and Gamma-1 sections. In-silico primer design and PCR modelling was used to successfully target the hotspots within this highly repetitive intergenic region of DNA. Samples from 21 CLPD patients, and control samples from peripheral blood of a normal donor (providing polyclonal controls for switched and unswitched B cells) and a non-B-cell line (HeLa) underwent PCR and NGS of the hotspot targets. We optimised a bioinformatic pipeline that is designed to detect insertional diversification events and found Insertion events in polyclonal samples ($n = 2$), but not in clonal samples. Interestingly, like the variable region, the control insertions also correlated with Dnase hypersensitivity sites.
To our knowledge the VDJ region insert results are the first demonstration of insertional diversification events in clonal samples. We have also shown that both the switch and variable region insertions correlate with Dnase hypersensitive sites.

BSH23-PO153 | Monoclonal Gammopathy of renal significance: Are we investigating possible cases appropriately? A quality improvement project

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Monoclonal gammopathy of renal significance (MGRS) differs from the well-known monoclonal gammopathy of unknown significance (MGUS) in that the monoclonal protein produced is nephrotoxic. Early diagnosis of MGRS is paramount to initiating treatment to prevent devastating decline in renal function. Renal biopsy is critical for MGRS diagnosis. The International Kidney and Monoclonal Gammopathy Research Group (IKMG) produced an algorithm to guide clinicians, suggesting referral for renal biopsy if urine albumin creatinine ratio (ACR) is >30 mg/mmol.
The primary aim of cycle one was to quantify urine ACR measurements in all newly diagnosed MGUS patients at a large tertiary centre. The baseline audit included all patients diagnosed between January 2014–December 2020 ($n = 503$). Results were presented at the Myeloma multidisciplinary-team (MDT) to improve adherence to measurement of urine ACR as a screening investigation for MGRS and improve awareness of IKMG guidance for MGRS diagnosis. A second audit was performed assessing for improvement in rates of urine ACR measurement in patients diagnosed with MGUS between April–October 2022 ($n = 18$). Additionally, patients from either cycle who met the urine ACR threshold for biopsy (urine ACR >30 mg/mmol) were identified. The frequency of referral of these patients for renal biopsy and any subsequent biopsy results were analysed. Results of the second cycle were presented at the joint Renal and Myeloma

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MDT to identify further necessary interventions to improve appropriate renal biopsy referral.

There was improvement in measurement of urine ACR in cycle two following educational interventions, with urine ACR measured at diagnosis in 61.1% (n = 11) versus 33.0% (n = 166) in cycle one. Forty-five patients across both cycles met the IKMG threshold for recommendation of renal biopsy, of those 24.4% (n = 11) had a renal biopsy performed and only one had biopsy findings diagnostic of MGRS.

This quality improvement project has led to increased awareness of the role of urine ACR measurement in MGRS diagnosis. This is demonstrated by increased rates of measurement following educational interventions. Adherence to IKMG guidance for referral for renal biopsy was low. However, it was felt it would not have been practically possible or clinically appropriate to have performed this invasive investigation on all patients who met urine ACR criteria. Further work is required to develop a centre-based algorithm for which patients are put forward for discussion at Renal and Myeloma MDT as potential candidates for biopsy. This will help guide appropriate referral for renal biopsy to maximise diagnosis of MGRS whilst minimising unnecessary invasive procedures.

BSH23-PO154 | BGB-11417 (Bcl-2 inhibitor) monotherapy or combination with Zanubrutinib in CLL/SLL patients: Preliminary phase 1 data

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Background: Bcl-2 inhibitors are effective for treating chronic lymphocytic leukaemia (CLL) and small lymphocytic lymphoma (SLL); however, the emergence of resistance limits their utility. The combination of Bcl-2 and Bruton tyrosine kinase (BTK) inhibitors is tolerable with synergistic activity in CLL. BGB-11417 is a novel Bcl-2 inhibitor that is more potent and selective than venetoclax. Zanubrutinib, a next-generation BTK inhibitor, has favourable activity and safety in CLL/SLL patients. BGB-11417-101 is an ongoing phase 1/1b dose-escalation/expansion study (NCT04277637) in patients with B-cell malignancies; data from CLL/SLL cohorts are presented.

Methods: In separate Monotherapy/Combination cohorts, patients received escalating doses of BGB-11417 (40, 80, 160, 320, or 640 mg QD) with a dose ramp-up to minimise tumour lysis syndrome (TLS) risk. Combination cohorts: zanubrutinib (320 mg QD or 160 mg BID) beginning 8-12 weeks before BGB-11417. Dose-limiting toxicity for each cohort was evaluated during ramp-up through intended dose day 21. Adverse events (Aes) reported per Common Terminology Criteria for Aes v5.0. Minimal residual disease (MRD) was assessed by a European Research Initiative on CLL flow cytometry assay.

Results: As of 15 May 2022, 50 CLL patients received treatment: 6 monotherapy (all relapsed/refractory [R/R]) and 44 combination (22 R/R; 22 treatment-naïve [TN]). BGB-11417 doses: Monotherapy, up to 160 mg; combination R/R CLL, up to 640 mg; TN CLL, up to 320 mg (includes 8 patients in zanubrutinib pretreatment). Maximum tolerated dose has not been reached; dose escalation is ongoing. Median follow-up: Monotherapy, 11.5 months (range, 8.5–18.3); Combination, 5.8 months (range, 0.2–10.5). Most common treatment-emergent Aes (TEAEs): Monotherapy, thrombocytopenia and neutropenia (both ≥50%; grade ≥3, 33%); Combination, contusion, neutropenia (grade ≥3, 11.4%), and low-grade gastrointestinal toxicity (all ≥22.7%). One discontinuation (Combination: disease progression; Richter transformation). Laboratory TLS (Monotherapy: 1 high-risk patient; resolved with no intervention). No clinical TLS reported. Diarrhoea was mostly grade 1; none were grade ≥3. Early efficacy data: notable reductions in absolute lymphocyte count (ALC); responses seen at 1 mg dose. Four responses (66%), partial response [PR] or better, were observed with Monotherapy and 32 responses (72.7%), PR with lymphocytosis or better, with combination. Early MRD data (4 MRD-evaluable patients at 160 mg); 3 (2 Monotherapy; 1 Combination) had a peripheral blood CLL count <10−4 at 24 weeks after BGB-11417 initiation.

Conclusion: Preliminary data show BGB-11417 (monotherapy and combination with zanubrutinib) is well-tolerated. Grade ≥3 neutropenia was uncommon and manageable. TLS rates are low. Early MRD response data are promising; mature MRD data are forthcoming. Venetoclax-treated CLL/SLL cohorts will soon be open for enrolment.

BSH23-PO155 | Impact of baseline ocular conditions on belantamab mafodotin-related corneal events in relapsed/refractory multiple myeloma

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Introduction: Belantamab mafodotin (belamaf), an antibody–drug conjugate targeting B-cell maturation antigen, has demonstrated deep and durable responses in heavily pretreated patients with relapsed or refractory multiple myeloma (RRMM). Belamaf is associated with reversible corneal changes (keratopathy), which can lead to decreased/blurred vision and dry eye. Patients with RRMM may have impaired ocular health (baseline ocular conditions [BOCs]) due to comorbidities or effects of prior myeloma therapies. We report an analysis of pooled data from DREAMM-2 to examine the

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association between BOCs, belamaf treatment duration and resulting ocular symptoms.

Methods: DREAMM-2 evaluated belamaf (2.5 or 3.4 mg/kg every 3 weeks) in patients with RRMM following ≥3 lines of therapy. Exclusion for BOCs was limited to corneal epithelial disease, except mild punctate keratopathy. Baseline and subsequent eye examinations were conducted prior to each dose. Dose modifications were allowed for toxicity. In this analysis, ocular symptoms (patient-reported) were analysed via the presence/absence of the most common BOCs for all patients, and according to belamaf treatment duration (≤3 or >3 months), to assess the impact of drug exposure. Differences were defined as a ≥20% variance in ocular symptoms between patients with/without each BOC.

Results: Most (198/218) patients who received belamaf had BOCs. Cataracts (66%) and cornea-related BOCs, including keratopathy (52%) and dry eye (22%), were the most common. Among patients with BOCs, blurred vision and dry eye were the most common ocular symptoms following belamaf treatment. There were no differences (≥20% variance) in ocular symptoms in patients with/without BOCs associated with the lens (cataract), optic nerve (glaucoma) or eyelids (blepharitis). Blurred vision was more common in patients with dry eye (44% vs. 21%) and, although confounded by small numbers, in patients with age-related macular degeneration at baseline (38% vs. 14%). Differences in ocular symptoms, by the presence or absence of BOCs, were not observed in patients exposed to belamaf for <3 months (N = 123). However, with longer treatment (>3 months; N = 95), a difference in blurred vision was seen in patients with dry eye (65% vs. 40%) and keratopathy (56% vs. 36%) at baseline.

Conclusions: BOCs are common in patients with RRMM; however, only patients with cornea-related BOCs were likely to have increased ocular symptoms with belamaf, predominantly blurred vision. There was little evidence for an effect of other BOCs on treatment-emergent ocular symptoms. These findings inform risk/benefit decision-making with belamaf and indicate that belamaf can be used to treat patients with RRMM despite BOCs.

BSH23-PO156 | Preliminary results from phase 2 of CC-92480-MM-001: Mezigdomide plus dexamethasone in relapsed/refractory multiple myeloma (RRMM)

Dr Rakesh Popat[1], Paul G. Richardson[2], Suzanne Trudel[3], Hang Quach[4], Sagar Lonial[5], Robert Z, Orlowski[6], Kihyun Kim[7], María-Victoria Mateos[8], Charlotte Pawlyn[9,10], Karthik Ramasamy[11], Joaquín Martinez-Lopez[12], Ignacio Casas-Avilés[13], Alessia Spirli[14], Jing Gong[15], Michael Amatangelo[15], Jessica Katz[15], Paulo Maciag[15], Teresa Peluso[14], Nizar J. Bahlis[16]

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In phase 1 of CC-92480-MM-001 (NCT03374085), the recommended phase 2 dose (RP2D) of mezigdomide plus dexamethasone (MEZI-d) was selected at 1 mg once daily for 21/28 days. Here we report preliminary results from the MEZI-d dose-expansion cohort in patients with heavily pretreated RRMM.

Key eligibility criteria were: RRMM; ≥3 prior lines of therapy; disease progression ≤60 days of last myeloma therapy; refractoriness to lenalidomide/pomalidomide, a proteasome inhibitor, a glucocorticoid, and an anti-CD38 monoclonal antibody. Oral mezigdomide 1 mg was given on days 1–21 of each 28-day cycle, plus weekly dexamethasone (40 mg; 20 mg if >75 years of age). Primary objective was to evaluate efficacy (overall response rate [ORR]); secondary objectives included safety/tolerability and additional efficacy assessments. Pharmacodynamics was an exploratory objective.

As of 16/Sep/2022, 101 patients had received MEZI-d at the RP2D. Median age was 67 (range 42–85) years, median time since initial diagnosis was 7.4 (1.1–37.0) years; 39.6% of patients had plasmacytomas and 37/101 patients had high-risk cytogenetics (56/101 not evaluable). Median number of prior

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regimens was 6 (3–15); prior therapies included stem cell transplantation (77.2%) and anti-BCMA therapy (29.7%). All patients were refractory to last myeloma regimen and triple-class refractory. Median follow-up was 7.5 (0.5–21.9) months, with a median of 4 (1–20) cycles; 10.0% of patients continued treatment; progressive disease was the main reason for discontinuation (60.4%).

ORR was 40.6% for all patients. Whilst data are not mature yet, median PFS was 4.4 (95% CI 3.0–5.5) months and median duration of response was 7.6 (95% CI 5.4–9.5) months. ORR was 30.0% in patients with plasmacytomas ($N = 40$) and 50.0% in patients with prior anti-BCMA therapy ($N = 30$).

Ninety-one (91.1%) patients experienced a grade 3/4 treatment-emergent adverse event (TEAE). Most frequent hematologic grade 3/4 TEAEs were neutropenia (75.2%), anaemia (35.6%), and thrombocytopenia (27.7%); 34.7% of patients had grade 3/4 infections, including grade 3/4 pneumonia (15.8%) and COVID-19 (7.0%). Occurrence of other grade 3/4 non-hematologic TEAEs was generally low. Due to TEAEs, 76.2% and 29.7% of patients had mezigdomide dose interruptions and reductions, respectively; 90.1% of patients discontinued mezigdomide.

Mezigdomide induced substrate degradation and increases in activated and proliferating T cells in patients, including those directly refractory to pomalidomide-based therapies. MEZI-d had a manageable safety profile with encouraging efficacy in patients with triple-class refractory RRMM, including patients with prior BCMA-targeted therapies. These results strongly support the continued development of mezigdomide in MM, and especially in combination.

BSH23-PO157 | Challenges of providing first-line imaging in suspected myeloma: A user perspective and national UK survey

Dr Sandra Quinn[1], Prof. Vicky Goh[2,3], Dr Olwen Westerland[2,3], Dr Matthew Streetly[2,3], Prof. Guy Pratt[4]

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Access to cancer services may depend on where a patient lives in the UK and these inequalities may affect time to diagnosis, the available treatment options and the quality of life of patients. Whilst we know that key diagnostics, such as recommended first-line imaging in myeloma with techniques such as whole-body MRI and PET-CT are difficult for some clinical teams to access and provide, the extent of this problem from the patient perspective across the whole of the UK remains unknown.

This service evaluation aimed to examine the provision of first-line imaging from the user perspective (i.e. myeloma patients and their carers) and whether provision matched the National Institute for Health and Care Excellence (NICE) guidelines (NG35, 2016) in first-line imaging practice in the UK.

Myeloma UK invited patients and carers to complete an online survey to examine the types of imaging myeloma patients received at first-line, identify how imaging practice had changed over time and look at geographical variation across the UK.

895 patients and carers participated in the survey between the 4 and 14 March 2022. Skeletal survey (SS, whole-body X-rays) remained a common first-line investigation and most patients had more than one test. Magnetic resonance imaging (MRI) was used at first-line in 48.7% of cases and 18F-fluorodexyglucose positron emission tomography/computed tomography (18F-FDG PET/CT) was used least often (23.1%). There was no improvement in first-line imaging practice since the introduction of the guidance, $X^2(8, N = 895) = 29.91$, $p < 0.001$, with a small effect size according to Cramér's $V = 0.064$. However, SS was used more often in East England (57.9%) and Scotland (61.2%) than in South East England (36.3%), $X^2(36, N = 827) = 70.615$, $p < 0.001$, Cramér's $V = 0.048$, post-hoc tests $p < 0.05$. Overall, little has improved in first-line imaging practice since the introduction of the guidance.

Despite NICE recommendations regarding first-line MRI in suspected myeloma, both SS and multiple imaging investigations are common at diagnosis with some geographical variability in imaging practice. To challenge this issue, we must emphasise to healthcare professionals the superiority of first-line MRI compared to SS, the need to avoid unnecessary multiple imaging requests and drive for improvements in care across the UK.

BSH23-PO158 | Retrospective multicentre study comparing survival outcomes in chronic lymphocytic leukaemia (CLL) with genetic risk stratification

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Purpose: This retrospective West Midlands multicentre study of CLL patients was conducted to review results of genetic testing in CLL patients and impact on survival.

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Methods: 349 patients across the West Midlands were included in this retrospective study collected between December 2018 and March 2022. Clinical centres were asked to obtain data relating to the type and number of lines of treatment, overall response and reported genetic abnormalities. Treatment and response were categorised as per international workshop on chronic lymphocytic leukaemia (iwCLL) criteria.[1] Genetic testing comprised fluorescence in situ hybridisation (FISH) for copy number abnormalities of 17p &11q, IgHV mutation status and Oxford Gene Technology CLL Next-Generation Sequencing (NGS) Panel to detect sequence variants in key genes associated with CLL (including NOTCH1, SF3B1, BIRC3, ATM and TP53). Patients were risk stratified into good, poor or not poor risk categories as per Rodríguez-Vicente et al.[2]

Statistical Analysis and Results: For 349 patients analysed, 143 (41%) patients were under active surveillance, and 206 (59%) patients received 1st line treatment. Out of the 206 patients that received 1st line treatment, 92 (44%) patients proceeded to 2nd line treatment, 31 (15%) patients proceeded to 3rd line treatment and 21 (10%) of patients required treatment beyond 3rd line Within the poor risk category, 173 patients were identified with 38 had TP53 deletion or mutation and 135 patients had other poor risk mutations such as NOTCH1, SF3B1, ATM. Statistical analysis did not show a difference in survival either from the diagnosis or from the date of NGS sample in the two groups of high-risk patients.

Conclusion: The data identifies a cohort of patients with poor outcome that are negative for TP53 mutation, highlighting the importance of NGS in CLL patients at the point of treatment.

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BSH23-PO159 | Single-agent belantamab mafodotin in patients with relapsed/refractory multiple myeloma: Final analysis of the DREAMM-2 trial

Dr Karthik Ramasamy[1], Prof Ajay Nooka[2], Prof Adam Cohen[3], Prof Hans Lee[4], Prof Ashraf Badros[5], Prof Attaya Suvannasankha[6], Prof Natalie Callander[7], Prof Al-Ola Abdallah[8], Dr Suzanne Trudel[9], Prof Ajai Chari[10], Prof Edward Libby[11], Prof Maria Chaudhry[12], Dr Malin Hultcrantz[13], Dr K. Martin Kortüm[14], Dr Paul Richardson[15], Prof Rakesh Popat[16], Prof Douglas Sborov[17], Dr Shawn Hakim[18], Dr Eric Lewis[19], Mr Bharat Bhushan[20], Dr Boris Gorsh[18], Dr Ira Gupta[18], Dr Joanna Opalinska[18], Dr Sagar Lonial[2]

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Introduction: Belantamab mafodotin (belamaf), an antibody–drug conjugate targeting B-cell maturation antigen, demonstrated deep and durable responses, with a manageable safety profile in previous analyses of the DREAMM-2 study, in patients with relapsed or refractory multiple myeloma (RRMM). Here, we present final efficacy and safety data from DREAMM-2.

Methods: DREAMM-2 (NCT03525678) is a Phase 2, open-label study of single-agent belamaf (2.5 or 3.4 mg/kg every 3 weeks) in patients with RRMM who had ≥3 prior therapies and were refractory to an immunomodulatory agent and a proteasome inhibitor, and refractory/intolerant to an anti-CD38 monoclonal antibody. The primary endpoint was overall response rate (ORR); secondary endpoints included progression-free survival (PFS), overall survival (OS), safety and health-related quality of life (HRQoL).

Results: As of 31 March 2022, median follow-up was 12.48 and 13.77 months for patients randomised to belamaf 2.5 mg/kg (N = 97) and 3.4 mg/kg (N = 99), respectively. ORR was 32% and 35%; 19% and 24% of patients (58% and 69% of responders) achieved very good partial response (VGPR) or

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better. Median time to response was 1.5 and 1.4 months, and median duration of response was 12.5 and 6.2 months, in the 2.5 and 3.4 mg/kg cohorts, respectively. Median PFS was 2.8 and 3.9 months; median OS was 15.3 and 14.0 months. In patients with ≥VGPR, median PFS was 14.0 and 16.8 months, and OS was 30.7 and 35.5 months, in the 2.5 and 3.4 mg/kg cohorts, respectively. The most common Grade ≥ 3 adverse events (Aes) were keratopathy, anaemia and thrombocytopenia. Incidences of AE-related dose reductions, delays and permanent discontinuations were 36% and 44%, 54% and 62% and 9% and 5% in the 2.5 and 3.4 mg/kg cohorts, respectively. Discontinuation due to ocular events was 3% in both cohorts. The most common ocular events in the 2.5 mg/kg cohort included keratopathy (71%), blurred vision (23%), best-corrected visual acuity (BCVA) reduced to 20/50 (21%) and dry eye (15%). BCVA reductions and blurred vision resolved by end of follow-up in ≤86% of patients; no permanent complete vision loss occurred. EORTC-QLQ-C30 data suggest that overall global health status/QoL, physical and role functioning, and overall disease symptoms were maintained or improved during treatment.

Conclusions: Single-agent belamaf (2.5 and 3.4 mg/kg) results in rapid, deep, durable, and clinically meaningful responses, with a manageable safety profile in patients with RRMM. No new safety signals were observed. Dose modification remained effective in resolving ocular events, and HRQoL was maintained or improved.

BSH23-PO16 | Demographics of haematology trial participants at the Christie. How can we widen participation?

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In the UK there are pronounced gradients in health, longevity and disability free life expectancy. Individuals who are part of minority groups or in lower socio-economic groups fare worse in these analyses. A key factor in health equality is equal access to health services. For this reason, we have undertaken a retrospective audit of referrals to the Christie Haematology Research department for consideration trial participation to evaluate to what extent our referrals are distributed evenly over the region in terms of the sex, ethnic group, geographic location and socio-economic groups. This analysis is intended both as a way of identifying gaps in our referral cohort and as a benchmark prior to implantation of interventions.

We found a profound lack of ethnic diversity in the referrals to the service as 95.4% of referred patients were Caucasian. Postcode analyses would suggest that from a socio-economic perspective our referral cohort is roughly representative of the northwest region; however, the referrals are not evenly distributed over the hospitals in the region.

We present potential solutions to these issues and will re-evaluate the effectiveness of these interventions over the coming year. It is anticipated that ensuring a representative patient cohort will increase the validity of the trial data obtained as well as helping to address health inequalities within the haematology trials population.

BSH23-PO160 | Patient-reported outcomes with daratumumab, lenalidomide, and dexamethasone in frail transplant-ineligible patients from MAIA

Dr Karthik Ramasamy[1], Prof Thierry Facon, Prof Torben Plesner, Prof Saad Usmani, Prof Shaji Kumar, Prof Nizar Bahlis, Dr Cyrille Hulin, Prof Robert Orlowski, Prof Hareth Nahi, Dr Peter Mollee, Dr Aurore Perrott, Dr Arnaud Jaccard, Prof Michael Delforge, Dr Lionel Karlin, Dr Bertrand Arnulf, Prof Ajai Chari, Dr Huiling Pei, Ms Niodita Gupta, Ms Shuchita Kaila, Ms Kathryn Matt, Ms Katharine Gries, Mr Robin Carson, Mr Fredrik Borgsten, Prof Katja Weisel
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Introduction: In the phase 3 MAIA trial, the addition of daratumumab to lenalidomide and dexamethasone (D-Rd) improved progression-free survival and overall survival in transplant-ineligible (TIE) patients with newly diagnosed multiple myeloma vs. Rd. Clinically meaningful improvements in patient-reported outcomes (PROs) with D-Rd have also been demonstrated in MAIA. Frail patients are particularly vulnerable and there may be concerns around the impact on health-related quality of life (HRQoL) with a triplet regimen. Here, we assess PROs in frail TIE patients from MAIA (median follow-up, 64.5 months).

Methods: Patients were randomised 1:1 to D-Rd or Rd until disease progression (PD) or unacceptable toxicity. Frailty was assessed using the simplified frailty score. PROs were assessed using the European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire Core 30-item. Questionnaires were completed at baseline, on day 1 of cycles 3, 6, 9, and 12 for year 1, and every 6 months thereafter until PD. Analyses were conducted on intent-to-treat patients. Treatment effect was assessed for change from baseline by a mixed-effects model for repeated measures. Meaningful thresholds for worsening were defined a priori based on published literature.

Results: Overall, 737 patients were randomised; 341 patients were classified as frail (D-Rd, $n = 172$; Rd, $n = 169$). Over time, frail patients treated with D-Rd showed large reductions in pain from baseline (≥20-point change) and were greater with D-Rd than Rd. Fatigue symptoms were moderately reduced with both treatments. Increases in global health status (GHS) were consistent over time for both treatment groups. Patients treated with D-Rd showed improvements from baseline in physical functioning that were greater than Rd until cycle 42 and increases in emotional and social functioning were seen with both treatments that were numerically greater with D-Rd at several time points. Improvements in role functioning were seen with both treatments with no difference

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between treatments. No meaningful changes from baseline were observed in nausea and vomiting with either treatment. Median time to first worsening was longer with D-Rd than Rd for GHS. Median time to first worsening for the other functional (emotional, social, role) and symptom (nausea and vomiting) scales numerically favoured D-Rd vs. Rd, except cognitive functioning and fatigue. Median time to worsening of pain symptoms was not reached for D-Rd.
Conclusions: Frail TIE patients treated with D-Rd reported improvements in GHS (an overall HRQoL measure) and physical functioning, with a notable reduction in pain from baseline throughout the duration of therapy.

BSH23-PO161 | Management of cardiovascular complications of BTKis: Are we complying with the good practice guidelines?

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Bruton tyrosine kinase inhibitors (BTKis) were approved for use at the end of 2013 and have since been used for indications including chronic lymphocytic leukaemia (CLL), Waldenstrom's macroglobulinaemia and mantle cell lymphoma. The use of BTKis has increased significantly in the UK since they achieved NICE (National Institute for Health and Care Excellence) approval for frontline treatment of CLL in 2021. However, they are associated with significant adverse cardiovascular events. In September 2021 the British Journal of Haematology published good practice guidelines for the management of cardiovascular complications of BTKis.
Our aim was to see whether these guidelines had been adhered to for patients taking BTKis.
Method: Data was collected for all patients being prescribed BTKis (ibrutinib and acalabrutinib) in the South Tees NHS Trust in July 2022. Patients' medical records were used to assess whether their management adhered to the good practice guidelines. Data was collated for 67 patients in total.
Results: The data showed that although all patients were consented for the risk of atrial fibrillation only 6% were consented for hypertension and only 1.5% for ventricular arrhythmias and sudden cardiac death. The guidelines recommend a baseline ECG (electrocardiogram) on commencement of treatment; however, only 7% had this completed and 0% had the minimum monitoring recommendation of 6-monthly ECGs. Thirty patients (45%) had an indication for a baseline echocardiogram; however, only one had this completed. For patients reporting symptoms of syncope, dizziness or palpitations only 50% had an ECG completed. Three patients developed worsening heart failure. The recommendations suggest referral to a cardio-oncologist; however, due to lack of availability of this service the referrals were instead made to the usual cardiologist.
Conclusion: Although there was a lack of compliance with guideline recommendations, it should be considered that most usual checks were affected by COVID-19 outbreaks and a drop in face-to-face clinics, which were replaced by phone clinics and home delivery of medications. However, the premade consent forms for BTKis need to be updated to include consent for ventricular arrhythmias and sudden cardiac death. There also needs to be routine procedures in place to ensure that regular blood pressure testing and ECG monitoring occurs and that there is prompt recognition of cardiovascular complications.
Action and implementation: To ensure improved compliance with these guidelines we plan to update our consent forms and create a proforma for clinic use to ensure that clinicians are aware of the various monitoring criteria required.

BSH23-PO162 | Evaluating outcomes of myeloma diagnosed via emergency presentations in Cardiff and Vale health board

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Myeloma's often non-specific presentation contributes to patients' delayed presentation to healthcare services. This can result in its diagnosis via emergency presentations, often at more advanced stages and through various complications, including spinal cord compression or acute renal failure, with resulting poorer prognosis.
This project aimed to assess incidence of myeloma diagnosis via emergency presentations in Cardiff and Vale Health Board (CVHB), identifying patient outcomes and ways to increase earlier detection via non-emergency presentations. Using a dataset compiled from Cancer Network Information System Cymru (CANISC) and Welsh Clinical Portal (WCP), a retrospective review of 89 patients referred to CVHB between April 2018 and 2022 was undertaken. Focus was on a number of variables including, date and mode of presentation (as emergency or other) leading to haematology referral and myeloma diagnosis, the presenting complaint, any documented prior presentations not leading to diagnosis, date of referral receipt, disease staging and date of death. Chi-squared statistical testing was performed to evaluate the significance of any relationships between particular variables, with survival rates assessed using SPSS (Statistical Product and Service Solutions) software.
The study identified that 37% of myeloma cases in CVHB were diagnosed through emergency presentations. The implications of this finding were shown through a statistically significant relationship identified between emergency presentations and advanced disease stage at diagnosis, with 64% of those diagnosed via this route at ISS (International Staging System) stage 3 at diagnosis (compared to 29% of non-emergency presentations) (p-value 0.006 (<0.05)). Furthermore, reduced 1-year survival rates were identified, with a 1-year survival of 56.3% for emergency presentations compared to 83.8% for non-emergency presentations—a

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difference deemed statistically significant (p-value 0.023 (<0.05) on log rank (Mantel-Cox testing)). No apparent difference in diagnostic delay was identified to account for worse outcomes for emergency presentations (11.5-week average symptom duration prior to presentation compared to 12.8 weeks for non-emergencies). However, the study found a greater proportion of emergency presentations had prior healthcare service presentations, which did not lead to diagnosis (45%, most commonly to general practice) compared to non-emergency presentations (29%), though this was not deemed a statistically significant association (p-value 0.106 (>0.05)). Despite this, the study found there remains scope to reduce the number of prior presentations occurring before myeloma diagnosis in CVHB, thereby aiding earlier diagnosis and preventing emergency presentations and associated worse outcomes.

BSH23-PO163 | Patient-reported symptoms and diagnostic journey in multiple myeloma

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Background: Late presentation of myeloma is associated with greater mortality and increased risk of complications. In the UK, current standards state that multiple myeloma should be diagnosed within 28 days of referral. However, the variability in initial symptoms can lead to a delay to the referral being made.

Objective/purpose: To assess the diagnostic journey of patients with multiple myeloma from initial symptom(s), other diagnoses, and healthcare professionals seen.

Methods: We used the online UK-based RUDY research database (www.rudystudy.org), where patients self-report their diagnosis of multiple myeloma and provide information about their diagnostic journey. Symptoms and other diagnoses were grouped where appropriate and logistic regression was performed with the primary outcome of diagnosis within 3 months. Analysis was performed using R version 4.2.2.

Results: 300 adults self-reporting multiple myeloma were included in the analysis. 41% were female, and the mean age was 66.2 (±9.74) years. The median time from symptom onset to final diagnosis was 4 months (IQR 2–10 months, range 0–172). Older age at symptom onset was a significant independent predictor for being diagnosed <3 months after symptom onset (OR = 1.05, 95% CI [1.02, 1.08]).

Back pain/vertebral fractures was the most commonly reported initial symptom (47%). This was followed by chest/shoulder/rib pain (20%) and fatigue/tiredness (19.7%). Chest/shoulder/rib pain (OR = 0.41, 95% CI [0.19, 0.85]), and infection (OR = 0.32, 95% CI [0.11, 0.83]) were less likely to be diagnosed <3 months of symptom onset.

52% reported other diagnoses before myeloma was diagnosed. 31% received a musculoskeletal diagnosis such as osteoporosis, costochondritis, or muscle strains. The healthcare professional most commonly seen before haematology were allied healthcare professionals (AHP) (i.e. physiotherapist, osteopath or chiropractor) (24%) and orthopaedic surgery (14%). Individuals seeing an allied health professional during the diagnostic journey were also less likely to be diagnosed <3 months of symptom onset (OR = 0.23, 95% CI [0.10, 0.47]).

Conclusion: Patients diagnosed with myeloma tended to present with musculoskeletal symptoms, and chest/shoulder/rib pain and seeing an AHP predicted a longer diagnostic interval. Developing resources for allied health professionals might be considered to improve early recognition of myeloma.

BSH23-PO164 | Belantamab Mafodotin for relapsed/refractory myeloma: A retrospective real-world, multicentre study from the UK and Ireland

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Introduction: Belantamab mafodotin (belamaf) is a first-in-class antibody–drug conjugate targeting B-cell maturation antigen (BCMA), licensed in Europe as monotherapy in relapsed-refractory multiple myeloma (RRMM) after four prior treatment lines. For patients in the UK, fourth line therapies are typically pomalidomide-based whilst BCMA-targeting agents represent a fifth line option. Here we report the first analysis of belamaf in a real-world cohort from the UK and Ireland, with the key aims of assessing efficacy and tolerability.

Methods: Thirteen myeloma centres across the UK and Ireland contributed patient cases to this retrospective study. Data were entered pseudonymously on a secure online data collection form, regarding: patient baselines (including eye health); belamaf administration and response to therapy; toxicity profile and ophthalmological assessment whilst on treatment; and survival statistics.

Results: 74 patients were submitted for analysis. The median age of the cohort was 65 years old (interquartile range [IQR]

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of 57–72), with 58% male. Patients had received a median 5 lines of therapy (IQR 4–6); 85% were triple-class refractory and 82% were pomalidomide-exposed.

Median follow-up by inverse Kaplan–Meier was 6.6 months (IQR 4–12.4). The median cycles belamaf received was 4 (IQR 2–6). ORR for the total cohort was 57%, with 32% obtaining very good partial response or better, and 24% a partial response; 18% saw a best response of stable disease and 19% progressive disease (PD) (7% unknown). Median overall survival (OS, $n = 70$) was 17.8 months (IQR 3.2–20.4) and event-free survival 9.3 months (IQR 2.7–19.6).

62 of the 68 patients for whom toxicity data were available experienced an adverse event (91%). Ocular toxicity occurred in 39 patients (53%), median grade 2 and caused a dose delay in 20/39 (51%) and a dose reduction in 12/39 (31%).

At the cut-off point, 38/74 (51%) had discontinued treatment. Reasons for discontinuation were death in 9/38 and PD in 26/38; only one patient actively stopped their treatment because of toxicity (grade 2 keratopathy). Discontinuation reason unknown in 2/38.

Conclusions: In a real-world cohort of 74 patients, belantamab mafodotin was associated with an ORR of 57%. This study is inevitably limited by its retrospective design and consequent limitations in data entry, as well as brief follow-up period. Nonetheless, the ORR data are better than in the registration DREAMM-2 study (31%) and further analysis is ongoing to explain this difference.

BSH23-PO165 | TRIM24 expression and regulation in chronic lymphocytic leukaemia upon BTK inhibition

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Studies on the epigenetic landscape of chronic lymphocytic leukaemia (CLL) have provided valuable insights into disease pathogenesis. B-cell-receptor signalling inhibitors are the state-of-the-art for CLL treatment, including Ibrutinib (IB), which targets the Bruton Tyrosine Kinase (BTK). Ibrutinib was shown to alter the epigenomic and transcriptomic dynamics of CLL. We explored the effects of BTK inhibition (BTKi) on the CLL epigenome in a global and gene-specific level.

Among the gene candidates screened, TRIM24 was found to follow IB-related expression patterns. The E3 ubiquitin ligase mRNA and protein levels were upregulated upon Ibrutinib treatment in CLL primary cells and cell lines. The same trend was identified in the mantle cell lymphoma (MCL) cell line REC-1, suggesting that the reported effects are not disease-specific but rather BTKi-specific. In vitro blockade of BTK pathway with IB (irreversible) or ARQ-500 (reversible) inhibitor did not alter the protein levels of TRIM24 in a REC-1 BTK KO cell line. This indicates that BTK blockade may drive TRIM24 upregulation. Immunofluorescence (IF) staining showed that TRIM24 was located mostly in the cytoplasmic and perinuclear area at pre-IB state and a shift to the nucleus was observed post-IB treatment. Screening for TRIM24 promoter area at matched promoter capture HiC sequenced CLL primary samples (pre/post BTKi), uncovered a significant long-range interaction gained at the post-IB state, which could represent an IB-responsive regulatory element responsible for the TRIM24 upregulation upon treatment.

Functional downregulation of TRIM24 by shRNA in HG3 cells led to activation of markers responsible for the DNA Damage response (DDR) pathway after 48 h. By 53BP1 foci staining through IF in HG3 wildtype and TRIM24 knock out (KO) cell line, we observed significant upregulation of the marker at 4 h and 24 h for the cells lacking TRIM24. Interestingly, the accumulation of DNA damage by irradiation (2 Gy) in the same context, led to an enhanced 53BP1 signal and increased protein levels of markers involved in DDR including the phosphorylated form of BRCA1, ATR and CHK1 at the TRIM24 KO cells. This could not be restored even after 24 h when compared to the wildtype cells. This finding highlights the critical role of TRIM24 to maintain the functionality of DDR machinery in CLL.

Altogether, our work provides first evidence for a role of TRIM24 and its epigenetic regulation in CLL in the context of BTKi treatment. Shedding light on its regulatory properties can provide valuable knowledge on the epigenetic phenomena taking place upon BTKi.

BSH23-PO166 | STATIC—A randomised phase III trial comparing intermittent with continuous treatment strategies in CLL

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Trial Design: STATIC is designed with multiple pathways, the 'Randomisation Pathway' and the 'Clinical Need Cohort', which route a participant enters will be determined by their eligibility.

Randomisation Trial: A prospective, national, multicentre, open-label, randomised, controlled, 2-arm, parallel-group, non-inferiority, phase III trial to assess whether patients with CLL on long-term treatment with a BTK inhibitor, (including ibrutinib) have similar disease control with an intermittent treatment strategy (experimental arm) compared with standard continuous treatment (control arm).

Clinical Need Cohort: A prospective, national, multicentre, open-label, single-arm, non-randomised cohort to assess the

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safety and overall survival of patients with CLL receiving long-term continuous treatment with ibrutinib.

Aims and Objectives: The primary aim was to assess whether patients with CLL on long-term treatment with BTK inhibitor, (including ibrutinib) have similar disease control with an intermittent treatment strategy, known as the 'Pausing Ibrutinib' arm compared with standard continuous treatment, the 'Continuous Ibrutinib' arm.

The primary objective is to assess non-inferiority (NI) of the intermittent treatment strategy in terms of time to treatment strategy failure, defined as the first documented instance of active disease (as defined by the 2018 iwCLL criteria) that does not respond to treatment, or death.

The secondary objectives are to assess Overall survival, Toxicity and tolerability, cost effectiveness, Quality of life, Summative treatment-free interval, Response to retreatment in intermittent treatment arm, Time to next treatment for CLL, Response to next treatment for CLL, Rate of resistance mutations and Evolution of resistant sub-clones.

Hypothesis: We hypothesise that an intermittent treatment strategy using ibrutinib, will reduce treatment-emergent resistance and thus be at least non-inferior to continuous treatment with regards to time to treatment strategy failure whilst reducing resource impact for the NHS and improving quality of life.

Number of participants: 800 participants will be randomised into the randomisation trial over a 6 year period with ~45% coming from ibrutinib-based treatment arms of the frontline CLL trial FLAIR and ~55% receiving ibrutinib off-trial (via NHS access) for previously treated for CLL. An additional cohort of ~30 patients from the ibrutinib-based treatment arms of FLAIR will be enrolled to a non-randomised Clinical Need Cohort to receive continuous ibrutinib.

STATIC trial is now open to recruitment in UK running from University of Leeds.

BSH23-PO167 | Residual disease kinetics among patients treated with first-line fixed-duration ibrutinib-venetoclax versus chlorambucil-obinutuzumab: The GLOW study

Dr Talha Munir[1], Dr Carsten Niemann, Prof Carol Moreno, Dr Carolyn Owen, Dr George Follows, Dr Ohad Benjamini, Prof Ann Janssens, Dr Mark-David Levin, Prof Anders Osterborg, Prof Tadeusz Robak, Dr Martin Simkovic, Dr Sergey Voloshin, Dr Vladimir Vorobyev, Prof Munci Yagci, Dr Loic Ysebaert, Dr Keqin Qi, Dr Qianyaya Qi, Dr Lori Parisi, Dr Srimathi Srinivasan, Dr Natasha Schuier, Dr Kurt Baeten, Dr Angela Howes, Dr Donne Bennett Caces, Prof Arnon Kater

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Background: In the GLOW study (NCT03462719) fixed-duration ibrutinib plus venetoclax (I+V) demonstrated superior progression-free survival (PFS) and undetectable minimal residual disease (uMRD) responses versus Clb+O in older pts with previously untreated chronic lymphocytic leukaemia (CLL; Kater et al., 2022). Unmutated IGHV (uIGHV) and TP53 mutations confer worse outcomes with both chemoimmunotherapy and venetoclax plus anti-CD20 therapies for CLL. Here we investigate MRD kinetics and outcomes in GLOW with longer follow-up, including these risk factors.

Methods: Pts aged ≥65 years or 18–64 years with cumulative illness rating scale score >6 or creatinine clearance <70 mL/min were randomised 1:1 to I+V (3 cycles ibrutinib lead-in, 12 cycles I+V) or 6 cycles of Clb+O (28 days/cycle). Pts with del(17p) or known TP53 mutations at screening were excluded. The primary end point was IRC-assessed PFS. MRD endpoints were evaluated in peripheral blood (PB) during and post-treatment.

Results: 106 pts were randomised to I+V and 105 to Clb+O. uMRD rates increased from 46.2% after 6 cycles of the I+V combination to 54.7% at 3 months after end of treatment (EOT+3), showing the majority of disease clearance from PB occurred within the first 6 months of combination treatment. 77.6% (45/58) of pts in the I+V arm had sustained uMRD post-treatment (from EOT+3 to EOT+18) versus 12.2% (5/41) in the Clb+O arm.

uMRD rates after 6 cycles I+V and at EOT+3, respectively, were 52.2% and 59.7% in pts with uIGHV and 31.3% and 40.6% in pts with mIGHV. uMRD was sustained post-treatment in 80.0% (32/40) of pts with uIGHV and 76.9% (10/13) of pts with mIGHV. Five of 7 pts with TP53-mutated CLL achieved uMRD at EOT+3 in the I+V arm, which was maintained through EOT+18. Among pts with detectable MRD (≥10–4) at EOT+3, 2/16 with uIGHV and 0/14 with mIGHV had clinically progressed by EOT+18 in the I+V arm; corresponding rates were 83.3% (25/30) and 42.9% (6/14) in the Clb+O arm.

At median follow-up of 34.1 months, PFS was sustained post-treatment regardless of IGHV status in the I+V arm, whilst uIGHV pts in the Clb+O arm relapsed more quickly. PFS, overall survival, and subgroup analyses with longer (46 month) follow-up will be presented.

Conclusion: Fixed-duration I+V achieved better-sustained uMRD responses than Clb+O during the first 18 months post-treatment irrespective of IGHV status and clinical progressions were uncommon. MRD kinetics and sustained responses demonstrate strong efficacy of I+V in older pts including those with high-risk features.

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BSH23-PO168 | Teclistamab in combination with daratumumab and lenalidomide in multiple myeloma (MajesTEC-2 study)

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Introduction: Combination of daratumumab (Dara) and lenalidomide (Len) may enhance the function of teclistamab (Tec), potentially resulting in improved antimyeloma activity in a broader population. We present initial safety and efficacy data of Tec-Dara-Len combination in patients with multiple myeloma (MM) in a phase 1b study (MajesTEC-2; NCT04722146).
Methods: Eligible patients who received 1–3 prior lines of therapy (LOT), including a proteasome inhibitor and immune-modulatory drug, were given weekly doses of Tec (0.72-or-1.5 mg/kg with step-up dosing) + Dara 1800 mg + Len 25 mg. Responses per International Myeloma Working Group criteria, adverse events (Aes) per CTCAE v5.0, and for cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS) per ASTCT guidelines, were assessed.
Results: 32 patients received Tec-Dara-Len (0.72 mg/kg, n = 13; 1.5 mg/kg, n = 19). At data cut-off (11 July 2022), median follow-up (range) was 5.78 months (1.0–10.4) and median treatment duration was 4.98 months (0.10–10.35). Median age was 62 years (38–75); 87.5% were male. Median prior LOT was 2 (1–3), 18.8% were refractory to Dara and 28.1% refractory to Len. CRS was most frequent AE (81.3% [n = 26], all grade 1/2), 95% occurred during cycle1. Median time to onset was 2 days (1–8), median duration was 2 days (1–22). No ICANS were reported. Frequent Aes (≥25.0% across both dose levels) were neutropenia (75.0% [n = 24]; grade 3/4: 68.8% [n = 22]), fatigue (43.8% [n = 14]; grade 3/4: 6.3% [n = 2]), diarrhoea (37.5% [n = 12]; all grade 1/2), insomnia (31.3% [n = 10]; grade 3/4: 3.1% [n = 1]), cough (28.1% [n = 9]; all grade 1/2), hypophosphatemia (25.0% [n = 8]; all grade 1/2), and pyrexia (25% [n = 8]; grade 3/4: 6.3% [n = 2]). Febrile neutropenia frequency was 12.5% (n = 4). Infections occurred in 24 patients (75.0%; grade 3/4: 28.1% [n = 9]). Most common were upper respiratory infection (21.9% [n = 7]), COVID-19 (21.9% [n = 7]), and pneumonia (21.9% [n = 7]). Three (9.4%) had COVID-19 pneumonia. One (3.1%) discontinued due to COVID-19 infection and this patient subsequently died of this infection. Overall response rate (ORR, median follow-up) was 13/13 (8.61 months) at 0.72 mg/kg and 13/16 evaluable patients (less mature at 4.17 months) at 1.5 mg/kg. 12 patients attained very good/better partial response at 0.72 mg/kg dose, and response was not mature for 1.5 mg/kg group. Median time to first response was 1.0 month (0.7–2.0). Preliminary pharmacokinetic concentrations of Tec-Dara-Len were similar as seen with Tec monotherapy. Tec-Dara-Len-treatment led to proinflammatory cytokine production and T-cell activation.
Conclusions: The combination of Tec-Dara-Len has no new safety signals beyond those seen with Tec or Dara-Len individually. Promising ORR supports the potential for this combination to have enhanced early disease control through the addition of Tec. These data warrant further investigation.

BSH23-PO169 | Daratumumab clinical trial analysis: Outcomes in patients with lenalidomide-refractory relapsed/refractory multiple myeloma, 1–3 prior LOT

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Introduction: Survival outcomes for patients with multiple myeloma (MM) have improved following new treatment

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combinations with proteasome inhibitors (Pis), immunomodulatory drugs (ImiDs), and anti-CD38 monoclonal antibodies. However, for patients who have received multiple lines of therapy (LOT), especially those who are PI-exposed and lenalidomide (len) refractory, selecting the next regimen can be challenging. Data characterising treatments and outcomes in this difficult-to-treat population are limited.

Methods: Individual patient-level data were derived from a pooled analysis of daratumumab clinical trials. The analysis included the initial study, long-term follow-up, and all treatment arms and regimens with or without daratumumab: APOLLO, CASTOR, CANDOUR, EQUULEUS, ALCYONE, MAIA, GRIFFIN, POLLUX, and CASSIOPEIA. Len-refractory MM patients with 1–3 prior LOTs (including a PI and ImiD) and ECOG <2 were identified. Time zero (T0) was defined as the time when subsequent LOT started after patient met inclusion criteria for each eligible index line. Patients with multiple therapies subsequent to meeting eligibility criteria contributed multiple observations. Treatment patterns and outcomes were assessed by number of prior LOT. For all index lines, descriptive statistics for patient characteristics were assessed at T0. Kaplan–Meier method (starting at T0) was used to evaluate progression-free survival (PFS), time to next treatment (TTNT), and overall survival (OS).

Results: Of 4764 patients, 915 with 1230 index lines (prior LOT, 1 [n = 114]; 2 [n = 516]; 3 [n = 600]) met inclusion criteria (median age 65.5 years; male 57.5%; White 76.4%; Black 3.2%). Median time from diagnosis was 3.6 years. At T0, 5.4% had baseline plasmacytoma, 16.1% had ISS stage III disease, and 14.3% had high-risk cytogenetics. 13.8% of all eligible index lines were triple-class refractory; 58.9% received stem cell transplant prior to T0. The most common treatment regimens subsequent to meeting eligibility criteria were DPd (13.3%), DKd (13.2%), Pd (11.4%), Kd (8.9%). For all patient indices, the overall response rate was 50.1% (very good partial response or better 29.0%; complete response or better 12.0%); median OS was 23.9 months, median PFS was 12.2 months, and median TTNT was 8.34 months. Response rates and PFS decreased as number of prior LOT increased.

Conclusions: This analysis of patients enrolled in daratumumab randomised clinical trials shows that response rates decrease with each additional prior LOT for PI-exposed, len-refractory patients with 1–3 prior LOTs. There is a need for a new effective and safe regimen for this patient population, which has poor PFS and moves quickly through available therapies.

BSH23-PO17 | Should COVID-19 pandemic mitigations in the myeloma clinic continue? A summary of patient opinions

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Introduction: The COVID-19 pandemic necessitated multiple changes to the format of myeloma clinics to minimise the risk of infection among patients and staff. These included changing in-person clinic appointments to telephone appointments when there was no medical need for face-to-face review and instituting a courier service for delivery of oral or self-administered medications. As COVID-19 restrictions relaxed, we sought to investigate the acceptability of these changes to our patients and to determine which, if any, of the new arrangements should continue.

Methods: Patients who attended the Myeloma Clinic at The Royal Marsden Hospital, both in-person and by telephone, on four separate dates in August and September 2022 were asked to complete a questionnaire to provide their opinions using a combination of multiple-choice, Likert scale and free-text questions. These covered the main domains of change outlined above along with questions about blood test location and attendance with family and friends.

Results: Questionnaires were returned by 59 patients, 11 relating to in-person appointments and 48 to telephone appointments. 86.0% of patients were in favour of continuing the option of telephone appointments, with many highlighting their convenience and the avoidance of long travel and waiting times, with some also mentioning their COVID-19 security. However, a number of patients expressed concerns including communication difficulties, the inability to effectively assess physical health with an examination and a lack of reassurance. Furthermore, those who attended in-person appointments felt they were very COVID-secure, assigning them a mean of score of 4.5, where 1 was very insecure and 5 very secure. Several suggested that the optimum schedule would include regular telephone appointments with occasional in-person meetings. Interestingly, only 25.5% of patients wanted a video calling option.

Patients were also very positive about receiving medications by courier, with 94.1% of patients receiving their medications within two working days of their clinic appointment. 81.8% of patients expressed a wish for this option to continue, highlighting the increased convenience and reduction in waiting times.

Conclusions: These results suggest that changes made to the Myeloma Clinic in response to the COVID-19 pandemic have improved the patient experience. A mixture of telephone and in-person appointments may be preferable for this cohort of patients, many of whom require regular appointments for chemotherapy approval but are medically

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stable, and whose frailty makes long travel and waiting times challenging. These findings have implications for the planning of myeloma clinics across the UK.

BSH23-PO18 | Modernisation of the clinical haematology workforce

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Clinical Haematology departments have been significantly impacted by difficulties in recruiting medically qualified consultant haematologists and meeting the demands of increasing workloads. This has been evidenced in recent professional body reports from the Royal College of Pathologists and British Society for Haematology. These workforce survey reports cited the effect of increased cancer-related workload, staff recruitment and retention, and demands on staff beyond their working capacity. One respondent observed that medically qualified consultants are increasingly absent from the laboratory and that Consultant Clinical Scientists are ideally placed to fill the gap between laboratory and clinical services at the diagnostic-end of the patient's journey.

Lancashire Haematology Centre (Blackpool Teaching Hospitals, NHS Foundation Trust) adopted a blended multiprofessional workforce model to address these issues. Outpatient patient referrals were seen in a Consultant Clinical Scientist-led Haematology Diagnostic Clinic. Audit showed that 80% have non-haematological causes of abnormal laboratory tests or have haematological diagnoses that are suitable for management in primary care. Only 20% required transfer of care to a medically qualified consultant for a management decision. Once a management plan is in place, patients may then be suitable for monitoring and maintenance under the care of an Advanced Nurse Practitioner. The Clinical Nurse Specialist supports each stage.

The service has also benefitted from the Consultant Clinical Scientist also fulfilling leadership roles that have traditionally been delivered by medical consultants such as Laboratory Director and Consultant with responsibility for Blood Transfusion. Other specialist leadership roles are also possible.

The National School of Healthcare Science (NSHCS) Higher Specialist Scientist Training (HSST) programme and associated Fellow of the royal College of Pathologists (FRCPath) curriculum has been central in achieving this workforce model, alongside a specific work-based training programme. The Haematology HSST programme has three specialist tracks: general/haem-oncology; haemostasis and thrombosis; and transfusion. These are essential to fulfilling specific specialist roles in the department as well as contributing to general haematology. Stage 1 HSST/part 1 FRCPath are common to each specialist track, whilst stage 2 HSST/part 2 FRCPath focus on the trainee's specialist track. The Professional Doctorate programme (Doctor of Clinical Science; DclinSci) also includes doctoral-level research, and leadership and management training.

We present this new workforce model as a proven and progressive solution to the existential issues currently facing Clinical Haematology services in the National Health Service.

BSH23-PO19 | The nomadic life of the district general hospital haematology registrar: Reviewing how far we travel

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As a district general hospital (DGH) registrar, clinical duties are varied. The geographical location of these clinical duties can vary in different hospital settings. In our trust, the inpatient Haematology ward was reallocated to become the designated 'COVID ward' at the start of the pandemic, due to a lack of availability of side rooms in the hospital. Haematology inpatients are now managed on general medical wards, with no specific Haematology ward available. This has further increased the geographical spread of registrars' clinical duties.

In order to quantify the impact of this geographical spread, we undertook an audit of physical activity over a four-week period. We included the physical activity during working hours recorded by the attending haematology registrar and an on call medical registrar for comparison. We collected data using smart devices on steps walked, distance travelled, time spent walking and calories burnt whilst walking. We collected data for all day shifts worked from 09:00–17:00 for all days of the week.

Overall, the attending haematology registrar walked an average of 10 241 steps a day, covering 7.87 km over a period of 107 min and burning 410 calories whilst active. The medical registrar walked an average of 7498 steps a day, covering 5.76 km over a period of 79 min and burning 300 calories whilst active. By comparison the attending haematology registrar covered 37% more steps per day than the medical registrar, a statistically significant difference (p-value 0.002, students unpaired t-test).

During a 7.5 h working day (taking into account a contractual 30-min unpaid lunch break), our DGH haematology registrars spend 107 min walking, which is 24% of their working hours. Our results highlight the time pressures on DGH haematology registrars. Time pressures on registrars in London have become more important as a result of Health Education England's (HEE) medical specialty redistribution programme, which will see the number of specialty trainees in London reduced by 46% over 5 years. Addressing the geographical spread of clinical duties could help to reduce the time pressures on registrars allowing them to spend more time on providing a clinical service. We have presented our data to trust management as evidence to lobby for the

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creation of a new haematology ward, which has now been included in the plans for a new hospital at our trust.

BSH23-PO21 | Sharing the knowledge and expertise of BSH members throughout the global Haematology community

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'BSH Global Speakers' was established in 2015 as a core project of the BSH Global Haematology Special Interest Group (SIG). As the project enters its eighth year, we present an update and reflection on the successes and challenges encountered.

Initially known as the 'Plenary Speaker Scheme', the project was developed following a stakeholder meeting in 2015 at the inception of the SIG. Haematology colleagues from the UK and low- and middle-income countries (LMIC) came together to discuss how the BSH may be best placed to support haematologists practicing in LMICs. Sharing of expertise and building collaborative networks were identified as key priorities. The 'Plenary Speaker Project' was conceived; BSH haematologists would be supported in delivering plenaries at the meetings of colleagues in LMICs, with the aim that each visit could act as a catalyst for creating networks and developing collaborative projects in education, research, and capacity building.

We established a yearly cycle of inviting applications from LMIC societies for a funded speaker at their scientific meetings, selecting the most impactful meetings, then recruiting appropriate UK-based speakers. We place emphasis on the likelihood of ongoing collaborative working or other impacts, for example engagement with local haematology trainees. To date, ten speakers have represented BSH at the meetings of LMIC societies, presenting on diverse topics, from molecularly guided interventions to prevent relapse in AML, to adapting lymphoma treatment strategies for low resource settings. Recently we have opened applications to nurse specialists and scientists, with our first scientific speaker presenting in Thailand May 2023.

The COVID-19 pandemic created significant challenges for the project due to the disruption in international travel and the cancellation of many haematology meetings around the globe. We were, however, able to adapt the project to support virtual speakers at meetings in South Africa, Vietnam, and Ghana. Although virtual meetings do not naturally lend themselves to collaborative working, we were pleased that a longer term joint educational program in haemoglobinopathy care has been established with the Vietnamese Society of Haematology as a result of BSH support.

The impact of BSH Global Speakers is significant. Even at smaller meetings, speakers will have the ear of the majority of practicing haematologists in a country. From the relationships built between societies and speakers we have seen the development of fellowship programmes, online education programmes, laboratory support, and numerous networks for informal advice in clinical care, research, and more.

BSH23-PO22 | Can patients meaningfully contribute to genomics research?

Miss Hira Mayet[1], Dr Catriona Gilmour Hamilton[2,3], Mr Adam Holloway[2], Prof Anne Goriely[4], Dr Christian Babbs[5], Mr John Ansell[2], Ms Penny Clarke[2], Ms Sally Jeans[2], Prof Thomas Milne[6], Dr Noemi BA Roy[3,7]

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Background: Genomics research, including inherited and acquired conditions, is becoming increasingly clinically relevant to patient groups. In the UK, samples sent for whole genome sequencing are made available for research with patient consent. Despite this, Patient and Public Involvement (PPI) projects that tackle genomics are few, possibly due to a perception that explaining basic science to a lay audience is too complex an undertaking for their meaningful involvement. We present our findings following a series of workshops explaining genomics to haematology patients and members of the public affiliated with our haematology PPI network.

Methods: We initially recruited 8 participants for this genomics education programme through our established haematology PPI network. Participants received one introductory session on genomics definitions by a consultant haematologist and three collaborative workshops from basic science researchers who utilise genomics in their research. This culminated in a final session where participants and presenters discussed what had been learned about genomics through the project, and participants gave feedback on the most effective use of presentation and style to deliver key messages.

Participants and researchers were surveyed at the start and following the workshops on their experiences. Participants' demographics data was collected.

Results: Participants self-assessed their knowledge about genomics and epigenetics before and after the project. A

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rating of 0 denoted no knowledge and 10 denoted expert knowledge.
Mean rating at the start of the project was 2.33, and 5.33 at the end; $p = 0.0011$.
Presenters self-assessed their confidence about presenting genomics and epigenetics concepts to patients and members of the public before and after the workshop. Interestingly, whilst there was no statistical improvement in this, presenters recommended engaging in PPI activities to other researchers. Common obstacles noted to involving patients in genomic and epigenetic research included lack of time, lack of funding to support PPI activities, and a fear that patients will not understand the science. All presenters felt that there is room for provision to better support researchers in doing more PPI.
Conclusions: Interventions to allow patients and members of the public to gain a better understanding of genomics, especially as pertaining to haematology, are possible. PPI contributors felt their knowledge on the subject benefitted significantly from the series of workshops. By contrast, the benefit to the presenters was not as clear. Work is ongoing in understanding the best methods to explain genomics concepts to patients, and how presenters' confidence in discussing the subject can be increased.

BSH23-PO23 | Ensuring quality and consistency of on-call haematology advice using a NEQAS-Style framework

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[1]*Wythenshawe Hospital, Manchester, UK,* [2]*Manchester Royal Infirmary, Manchester, UK*

Several centres in our region have received outcomes from the UK Accreditation Service (UKAS) highlighting a lack of process for standardising the quality and consistency of haematology advice given to clinicians in other specialities, particularly in the context of abnormal laboratory Results. Most clinicians agree that there is, anecdotally at least, a lack of consistency in the 'on-call' haematology advice given by both trainees and consultants.
We aimed to improve the consistency and quality of advice given by haematologists to clinicians in other specialities by developing a model of feedback and assessment. By repeating the exercise periodically we aimed to form a regular quality improvement exercise in our region.
We opted for a workflow analogous to that used by UK NEQAS for quality assurance in laboratory morphology. A survey was circulated to trainees and consultants expected to give general haematology advice whilst on-call. The survey consisted of ten questions reflective of the breadth of clinical and laboratory queries received 'on-call', covering transfusion, haemostasis, complications of sickle cell disease, and abnormal blood counts. Each question contained a small clinical vignette and allowed the responder to select from a list of investigation and management options.
Responses were summarised and results presented back anonymously to respondents with a commentary and evidence summary suggesting reasons for variations in practice and highlighting focus areas where practice could be improved, particularly where advice was not aligned with national guidelines or recent data.
20 clinicians responded, 14 trainees and 6 consultants. Variation was observed in results for all ten questions often with no distinct consensus observed. Important practice variations were observed in the setting of severe gastrointestinal bleeding; 25% advising the use of prothrombic complex concentrate and 75% suggesting andexanet alfa, with 20% respondents also advising the use of tranexamic acid. Wide practice variations were observed in the management of suspected heparin-induced thrombocytopenia, with 75% opting to stop heparin, 25% advising to transfuse platelets, and an even split between switching to fondaparinux, danaparoid or argatroban.
Several clinicians have found this exercise to be practice changing and we feel that it will continue to improve the quality and consistency of haematology advice given in our region. We expect that the exercise will both satisfy UKAS requirements and support the revalidation needs of individual clinicians. We envision that haematologists in other regions will benefit from incorporating our quality improvement model.

BSH23-PO24 | Are pharmacist-led clinics the way forward?

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Introduction: Bruton Tyrosine Kinase Inhibitors (BTKi) have revolutionised the treatment of chronic lymphocytic leukaemia (CLL), mantle cell lymphoma (MCL) and Waldenstroms' Macroglobulinaemia (WM).[1] Ibrutinib is licensed for frontline treatment of CLL, and relapsed/refractory CLL and MCL. Acalabrutinib is licensed for both frontline and relapsed CLL.
Extended outcome data have revealed late toxicities, such as hypertension.[2,3] Current BSH guidelines suggest BP monitoring at every clinic attendance.[3] Compliance to this recommendation has been poor, given the number of patients on BTKi's and overstretched physicians.[4] Pharmacist-led clinics are a possible solution.
Aim: To implement a pharmacist-led BTKi clinic at The Christie@Macclesfield and evaluate its outcomes.
Objectives: To introduce and review a pharmacist-led BTKi clinic, specifically looking at:

- Blood pressure (BP) monitoring
- Medication shortages
- Knowledge of next appointment
- Clinician time saved
- Patient experience

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Methods: All patients on a BTKi were included. Patient records were accessed to record, age, sex, medication type and dose. BP recording, medication shortages, appointment knowledge, clinician time saved, and patient experience were captured before and after implementation, via a questionnaire.

The data were subjected to descriptive statistical analysis, including grouping and frequency counting.

Results: A total of 24 patients (7 female, 17 male), on BTKi's were included in the study. Seventy-nine percent (19 patients) were treated for CLL, 13% (3 patients) for WM and 8% (2 patients) for MCL. The mean cycle number was 25 (range 4–57). Twenty-five percent of patients were on acalabrutinib, and 75% on ibrutinib.

There were improvements in BP recordings (58% to 65%), medication shortages (17% to 0%) and patient appointment knowledge (58% to 100%) before and after pharmacist-led clinic intervention, respectively.

Patient experience remained high. Several patients commented on the timeliness of appointments, and the benefits of collecting several cycles of medication, which minimised shortages with the pharmacist-led clinic.

Discussion: A pharmacist-led BTKi clinic has led to an average timesaving of two physician hours per week. BP monitoring has improved; however, it was not as high as expected, and is not yet in line with the British Society of Haematology good practice paper, which recommends measurement at every visit.[3] This is potentially due to an increased number of telephone consultations. Other benefits included: improvements in medication supply, appointment knowledge and patient experience, suggesting overall clinic implementation success.

The main limitation was the small sample size. Future work could address this with a larger scale study.

BSH23-PO25 | Machine learning to distinguish causes of abnormal red cell indices: Garbage in garbage out?

Dr Sobia Sharif[1], Dr Sara Trompeter[2], Dr Wai Keong[2]

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Introduction: Much of medicine concerns pattern recognition. Machine Learning holds much promise in detecting patterns influencing outcomes in large datasets. There is a saying within data science circles: 'garbage in, garbage out', that is, data quality provided to the model directly affects the quality of its predictions. Healthcare generates vast amounts of data. Automation can be applied to assist audit and research. How then to ensure 'quality in, quality out' instead?

Objectives: To correctly ascertain the underlying diagnosis pertaining to abnormal red cell indices in a cohort of women presenting at booking (~12 weeks pregnancy) using full blood count (FBC) parameters. Hence, FBC could be used as an accurate prediction tool rather than a filter for further testing, saving on cost and patient anxiety.

Methods: Data was stored in EPIC in two databases: Clarity and Caboodle. Booking appointment data was obtained via Clarity and full blood count and haemoglobinopathy results from Caboodle in two datasets. 12 586 patients were included. Haemoglobinopathy conclusion (via capillary electrophoresis) was used as the 'ground truth' or diagnosis.

This dataset was curated extensively in Pandas and FastAI package used to create a machine learning model utilising 90% of data to train and 10% to test predictions.

Results: The model favoured predicting patients as 'normal' despite controlling for class imbalance (i.e. the differing frequencies of certain conditions vs. 'normal') with 96% accuracy. The other 4% incorrectly were diagnoses such as iron deficiency and alpha/beta thalassaemia trait. Interestingly, sickle cell disease was predicted with 98.7% accuracy notwithstanding small patient numbers.

Conclusion: Ground truth established from Capillary electrophoresis was not ideal as: (1) data was entered manually (requiring relabelling) and (2) it is not able to distinguish alpha thalassaemia with intercurrent iron deficiency. The ideal test for ground truth is genetics; however, this was entered into EPIC via PDF uploads, making data extraction challenging.

As a prediction tool, it was thus limited. Nevertheless, model accuracy was 96%, 98.7% and 59% for 'normal', sickle cell and alpha thalassaemia/iron deficiency, respectively.

Routinely collected data in healthcare for machine learning has its pitfalls. Data recording is optimised for clinical use and not geared towards analysis. This wealth of information is thus being rendered less useful by way of organisation and storage. Machine learning models have the ability to crunch large amounts of data. We must, therefore, focus on methods of data entry that allow us to take better advantage of this innovation.

BSH23-PO27 | The psychological and emotional burden of blood cancer for children and their caregivers

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Introduction: Being diagnosed and living with blood cancer poses a complex psychosocial/emotional challenge for children. This complexity stems from factors unique to children, such as their maturity level, intellectual capacity, previous

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healthcare experiences, disruption of normality (e.g. school and social activities), and the role of peer acceptance in their lives, especially for teenage patients.

Aim: This systematic literature review identified and examined psychological stressors, unmet needs, and impacts on daily life due to blood cancers from the point of view of children and their parent caregivers across the patient journey.

Methods: Two databases, Embase and PsycINFO were searched and after removing duplicates, 1523 titles and abstracts were screened by two independent researchers. The twelve eligible studies included in this review reported primary qualitative findings on psychological/emotional experiences and/or outcomes of children with blood cancer aged between 0 to 19 years and their caregivers, and were published in English between January 2002 and July 2022.

Results: Common themes specific to the child patient throughout their blood cancer journey were identified. Fear was a major psychological stressor throughout the entire journey, starting as a fear of the hospital, needles, and death at diagnosis and early treatment. This progressed to fear of experiencing physical changes from illness, late effects, and negative impacts on their health if they participated in physical and social activities. Children reported that their need for resources and information was unmet throughout their journey, from having a lack of access to information about their condition and procedures, to information on how to cope with anxiety about disease recurrence and navigate relationships with peers/teachers. Notable impacts on their daily life included changes from their normal routine, physical scarring, inability to participate in sports, fatigue, and a reduction in school attendance/time with peers, as well as postponing their social lives. This impacted children's maturity development. Parents' observations further supported child patients' reports of blood cancers impacts on their maturity, emotions, social life, and schooling.

Conclusion: The findings of the psychological and emotional impacts on child patients who are living with or have been previously treated for blood cancer indicate an effect on children both whilst in the hospital/clinic and far beyond the hospital's/clinic's walls. There is a need to amplify child patient voices and those of their caregivers in the blood cancer literature and a need for healthcare providers to learn and act on those identified gaps to better support child patients.

BSH23-PO28 | The challenges of embedding advanced care planning into a Haematology service

Miss charlotte Bloodworth[1], Mrs Ellie Cooke[1], Mrs Faye Blackborow[1], Miss Charis Hillberg[1], Mrs Rebecca Bowen[1], Mrs Sherron Davies[1], Mrs Cheryl Reed[1], Mrs Nichola Coles[1], Mrs Annette Blackmore[1], Mrs Kayleigh Griffiths[1], Mrs Bethan Hellier[1], Miss Emma Dimond[1]
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Following an increased need for individual patient escalation plans during the COVID-19 pandemic we recently created a working group to embed Advance Care Planning (ACP) into our service. Modern ACP is not only about end-of-life planning, it involves meaningful conversations and supporting patients to make decisions throughout all stages of their disease and treatment. With early conversations and the opportunity to pre plan, the stress and anxiety attached to the difficult decisions at a time when someone may be acutely unwell should become easier (from the perspective of both staff and patients).

We carried out a preproject audit to ascertain patient opinions on how we could embed ACP into our service. 50 patients were offered a questionnaire, 38 chose to partake.

Results: 100% (38) of patients had never been approached by a nurse to discuss ACP.

82% (31 out of 38) said they would not want to be approached about ACP.

18% (7 pts) would like to be approached but 13% (5pts) noted only if end of life. On asking when the best time to be approached: three patients said 'at diagnosis', two said 'anytime', one said 'never', five said 'when ready' and seven said 'end of life only'.

Suggestions on the best way to raise ACP issues, five said poster displays (one noting the need for this poster to be positive and also available in Welsh), four suggested routine discussions in the current appointment and five people suggested a separate appointment. Other comments included, "I only want to discuss these issues with my solicitor", "I would like my family to be involved", "make sure the nurses can answer questions".

Average age was 68.5 years (range 49–85).

Discussion: This was a limited audit on an outpatient population, but it raised a number of important issues.

The actual carrying out of an audit on an emotive subject highlighted one the difficulties of embedding ACP initiatives into a service with a quarter of patients choosing not to answer the questionnaire, and of those who chose to answer 83% said they did not necessarily want to be approached to discuss ACP. In addition, the terminology 'ACP' appeared confusing, many people linking it to end-of-life discussions or to legal aspects such as writing of Wills.

The barriers this audit has highlighted, has helped to shape the future direction of our working group and highlighted the need for increased training.

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BSH23-PO29 | Upskilling haematological services for WGS and germline assessments through a regional skin biopsy training programme

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NHS England Genomics introduced whole genome sequencing (WGS) with standard-of-care (SoC) genetic testing for haemato-oncology patients who meet eligibility criteria, including patients with acute leukaemia across all ages, and exhausted SoC testing. Alongside, the role of germline mutations in haematological cancers is becoming increasingly recognised. DNA samples are required from the malignant cells (somatic sample) via a bone marrow aspirate, and from non-malignant cells (germline sample) for comparator analysis. Skin biopsy is considered the gold-standard tissue to provide a source of fibroblast DNA for germline analysis.

Performing skin punch biopsies is not within the traditional skillset for haematology teams and upskilling is necessary to deliver WGS/germline testing safely, independently and sustainably. A teaching programme was designed and piloted by the dermatology and haematology teams in Sheffield and delivered throughout the NHS trusts in North East & Yorkshire Genomic Laboratory Hub.

The training programme consisted of a 90-min session, slides, video and practical biopsy on pork belly or synthetic skin, designed to teach up to six students at one time. To disseminate best practice, the standard operating procedure and patient information used routinely in Sheffield were shared, to be adapted for local service delivery.

From January 2021 to December 2022, 136 haematology staff from 11 hospitals, including 34 consultants, 41 registrars, 34 nurses and 8 physician associates, across the NEY GLH region completed the skin biopsy training programme. Feedback from the course was outstanding, with consistently high scores in all categories. Practical components of the course were especially valued; 98.6% (71/72) trainees scored the practical element of the programme a top score of 5 out of 5, highlighting that despite the challenges of delivering face-to-face teaching due to COVID-19, teaching of practical skills was highly valued; training in this way could not have been replicated virtually.

Costs of the programme have been approximately £16 000, including consultant input and teaching/educational materials. Recent support has been provided by a separately funded Genomic Nurse Practitioner (GNP), with succession planning for the GNP to take over leadership from the consultant dermatologist. Plans are in place to use the remaining budget to disseminate the programme nationally.

Our training programme has shown that skin biopsy can be formally embedded into training for haematology consultants, trainees, nursing team, and physician associates. Delivery of training can be effective and affordable across regional GLHs with appropriate leadership and interspeciality coordination, and ultimately sustainable with specialist nursing staff, including GNPs.

BSH23-PO30 | Improving blood cancer awareness and increasing confidence in managing patient care through online education

Miss Khadijha Sundus[1]

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The primary aim of this project was to develop an e-learning resource for new starters on haematology wards to increase their knowledge of blood cancer and improve confidence in managing patient care. Our secondary aim was to reduce burden on senior nurses from regularly having to teach basic haemato-oncology nursing concepts to junior nurses, consequently increasing time available for patients and clinical work.

The 'Introduction to blood cancer' e-learning resource consists of 11 chapters and has been accredited by the Royal College of Nursing since April 2017. Upon completion, the course contributes 4 h 45 min towards continuous professional development revalidation requirements.

The learning outcomes are to:

1. Understand basic haemato-oncology concepts to support a foundation of nursing practice for caring of blood cancer patients.
2. Identify common complications and haematological emergencies and know how to respond.
3. Understand the importance of providing emotional support and signposting.

To date, 7209 participants have signed up to the course, of which 49.2% have completed it, a higher than average signup to completion rate for many e-learning courses.[1] The five-year survey results are positive, and participants reported increased knowledge and confidence in all three learning outcomes. The professionals who completed the course are predominantly from the intended audience with the majority below Band 7 and over half working in blood cancer for less than one year.

Application of knowledge was established by the question "Please tell us a specific work situation where you will apply what you have learned". We received over 2000 examples of specific situations where participants have, or were going to, apply the knowledge gained from the course. 85.5% of participants agreed or strongly agreed that because of taking the

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course and applying what they have learnt, their results at work have been positively impacted.

Overall enjoyment was rated at 10/10 by 41.6% of participants and 48.1% rated it 8–9/10. Qualitative evidence from participants suggest that the resource empowers nurses, particularly those at a junior level, improves their confidence in their roles and enables them to support patients more effectively. We conclude that this e-learning resource is an effective tool for new starters working in haematology and it should be integrated into all new nurse inductions.

Reference:

[1] Beynon, K (Macmillan), personal communication, April 2017.

BSH23-PO31 | Unusual presentation of BCR-ABL1 mutation with isolated thrombocytosis: Case series

Dr Laura Aiken[1], Dr Tanya Freeman[1], Dr Shamzah Araf[1], Dr Matthew Smith[1], Dr Tom Butler[1], Dr Fredrick Chen[1], Ms Marianne Grantham[1]

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Chronic myeloid leukaemia (CML) traditionally presents with a leucocytosis, thrombocytosis and splenomegaly, with the classic feature being that of a leucocytosis. A BCR::ABL1 clone presenting with isolated thrombocytosis is a recognised but rare presentation. The recent BSH Good Practice Paper on genetic testing in myeloproliferative neoplasms (MPN) recommends testing for BCR::ABL1, but only if the initial screen for JAK2 V617F, CALR and MPL is negative. We present four interesting cases from our institution of thrombocytosis with BCR::ABL1 fusion.

Patient 1, a 57-year-old, presented with asymptomatic isolated thrombocytosis (1579×10^9/L). As per BSH guidelines he was screened for JAK2 V617F, MPL and CALR mutations in peripheral blood. He was found to have a CALR mutation by allele-specific PCR and was given a diagnosis of Essential Thrombocytosis (ET). He went on to have a bone marrow examination, which was consistent with the diagnosis of ET and also showed a t(9;11)(q34;q13) translocation. FISH studies for BCR::ABL1 were undertaken, but were negative with no rearrangement of ABL1 (9q34). Eleven months later he went on to develop Precursor-B-cell acute lymphoblastic leukaemia (B-ALL). BCR::ABL1 was negative again by FISH, but positive by standard PCR and by next-generation sequencing (NGS) RNA fusion gene panel.

Patient 2, a 68 year-old, presented with extreme thrombocytosis (1812×10^9/L), and a mild leucocytosis (WBC 12.9×10^9/L). JAK2 V617F, CALR and MPL were all negative, the patient was diagnosed with 'triple-negative' ET. BCR::ABL1 was not tested at diagnosis, and the patient was commenced on hydroxycarbamide. Four years later she presented with Philadelphia positive (Ph+) B-ALL. Patient 3, a 25-year-old, presented with thrombocytosis (2839×10^9/L) without leucocytosis. JAK2 V617F, MPL, CALR were negative but BCR::ABL1 positive by PCR. Patient 4, a 59 year-old, also presented with isolated extreme thrombocytosis (2359×10^9/L). MPN screen was negative, but BCR::ABL1 positive by PCR.

All four patients had extreme thrombocytosis and were positive for BCR::ABL1. Only one had a mild leucocytosis. These highlight the importance of BSH guidelines in checking BCR::ABL1 in cases of an isolated thrombocytosis with a negative initial MPN screen. Patient 1 had a CALR mutation alongside a cryptic (negative by FISH) BCR::ABL1 fusion. Coexistence of CALR mutation and BCR::ABL1 is rare and has not been not reported in the context of Ph+ B-ALL. Our institution now recommends screening for BCR::ABL1 by PCR and FISH in all cases of thrombocytosis even if a conventional MPN mutation is found.

BSH23-PO33 | Impact of momelotinib on patient-reported quality of life for patients with myelofibrosis from MOMENTUM

Prof Ciro Rinaldi[1], Dr Ruben Mesa[2], Prof Claire Harrison[3], Dr Jeanne Palmer[4], Prof Vikas Gupta[5], Prof Donal McLornan[3], Prof Mary Francis McMullin[6], Prof Jean-Jacques Kiladjian[7], Prof Lynda Foltz[8], Prof Uwe Platzbecker[9], Prof Maria Laura Fox[10], Prof Adam Mead[11], Prof David M Ross[12], Dr Stephen Oh[13], Prof Andrew Perkins[14], Prof Michael Francis Leahy[15], Dr Jun Kawashima[16], Sunhee Ro[16], Rafe Donahue[16], Samineh Deheshi[17], Dr Srdan Verstovsek[18]

[1]*University of Lincoln, Lincoln, UK,* [2]*Mays Cancer Center at UT Health San Antonio MD Anderson Cancer Center, San Antonio, USA,* [3]*Guy's and St Thomas' NHS Foundation Trust, London, UK,* [4]*Mayo Clinic, Phoenix, USA,* [5]*Princess Margaret Cancer Centre, University of Toronto, Toronto, Canada,* [6]*Queen's University, Belfast, UK,* [7]*Universite de Paris, AP-HP, Hopital Saint Louis, Centre d'Investigations Cliniques, Paris, France,* [8]*University of British Columbia, Vancouver, Canada,* [9]*University of Leipzig Medical Center, Leipzig, Germany,* [10]*Vall d'Hebron University Hospital, Barcelona, Spain,* [11]*Oxford University Hospitals NHS Foundation Trus, Oxford, UK,* [12]*Royal Adelaide Hospital and SA Pathology, Adelaide, Australia,* [13]*Washington University School of Medicine, St Louis, USA,* [14]*Australian Centre for Blood Diseases, Monash University, Melbourne, Australia,* [15]*East Metropolitan Health Service PathWest Laboratory Medicine, Perth, Australia,* [16]*Sierra Oncology Inc, San Mateo, USA,* [17]*Sierra Oncology Inc, Vancouver, Canada,* [18]*The University of Texas MD Anderson Cancer Center, Houston, USA*

Disease-related inflammation, anaemia, and splenomegaly contribute to burdensome symptoms in patients with myelofibrosis (MF), decreasing health-related quality of life (HRQOL). Momelotinib (MMB) is an ACVR1/AK1/JAK2 inhibitor with demonstrated symptom, spleen, and anaemia

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benefits in patients with MF. In MOMENTUM, MMB significantly improved symptoms vs. danazol (DAN) (24.6% vs. 9.2%) by Total Symptom Score reduction ≥50% (TSS50) at week 24 (W24) vs. baseline in symptomatic, anaemic patients previously treated with a JAK inhibitor. We report patient-reported outcomes (PROs) from MOMENTUM.

PRO assessments included the MFSAF v4.0, EORTC QLQ-C30, and PROMIS Physical Function Short Form. Responder analyses were summarised by corresponding meaningful change thresholds. Longitudinal change from baseline scores were analysed using mixed model for repeated measures, and longitudinal analyses of response were performed by generalised estimating equation after multiple imputation for missing scores. Time to first response was analysed using the Kaplan–Meier method. A stratified log-rank test was performed to compare treatment arms; hazard ratios (HRs) were estimated from a stratified Cox regression model.

Proportionally greater improvements in symptoms, physical function, and global QOL were observed with MMB vs. DAN at W24 by MFSAF, PROMIS, and EORTC QLQ-C30, respectively. Improvements in all MFSAF individual item scores from baseline at W24 were greater with MMB vs. DAN, with the greatest treatment differences noted for night sweats, abdominal discomfort, bone pain, and rib pain. Longitudinal MFSAF TSS response analysis indicated a higher likelihood of response in the MMB arm (overall odds ratio, 2.50 throughout the 24-week randomised treatment period). Response HRs were >1 (range, 1.47–3.82) in the time to first response analysis for MFSAF TSS and individual items, excluding itching (0.77), indicating a faster symptom response with MMB. Responder analyses showed greater improvement for fatigue with MMB based on MFSAF (18.5% vs. 9.2%) and EORTC QLQ-C30 (41.5% vs. 21.5%). Similarly, responder analysis using the PROMIS assessments related to physical function also favoured MMB (17.7% vs. 4.6%), which was supported by shorter times to first response for physical function (HR, 1.93; 95% CI, 0.95–3.91). Improvement in global QOL by EORTC QLQ-C30 was also greater in the MMB arm: proportion difference of 14.6% between MMB and DAN (95% CI, 1.5%–27.7%; $p = 0.04$).

Consistent with the positive primary endpoint (TSS50) result of MOMENTUM, these responder, longitudinal, and time-to-event analyses demonstrate that MMB improves disease-related symptoms with associated improvement in fatigue, physical function, and overall HRQOL vs. DAN in patients with MF.

BSH23-PO34 | Ruxolitinib and Magrolimab significantly increases Calreticulin expression in Myelofibrosis. Proof of concept for combination therapy

Dr Kristian Boasman[1], Dr Matthew J Simmonds[1], Prof Ciro Rinaldi[1]

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Background: Primary myelofibrosis (MF), is a myeloid cancer characterised by bone marrow fibrosis and either overproduction or depletion of blood cells, and a high tendency to evolve into acute myeloid leukaemia. In solid tumours, calreticulin (CALR) overexpression produces a pro-phagocytic signal and is counteracted by anti-phagocytic CD47, reflecting an apoptosis vs. survival mechanism. We have previously found that CD47, not CALR, is overexpressed on the membrane of patients with MPN. Anti-CD47 monoclonal antibody Magrolimab, is currently being evaluated in clinical trials in AML, MDS but have yet to be tested in MF.

Aims: To investigate the alterations in expression of CALR and CD47 in MF patients when treated with standard therapies and/or CD47 inhibitors, in comparison with healthy controls.

Methods: Mononuclear cells were collected by Ficoll separation from bone marrow of 6 MF patients and from 2 controls. CD34+ cells and monocytes were purified using Dynabeads kits from Thermo Fisher. Cells expression of CALR and CD47 was measured by flow cytometry, before and after incubation with ruxolitinib alone and in combination with magrolimab.

Results: CD47 membrane expression was 57% in untreated MF cells compared to 12% in the control samples and CALR membrane expression was 51.8% in untreated MF cell compared to 12.5% in the control samples. After incubation with ruxolitinib, MF CD34+ cell survival decreased to 98.2%. Following Incubation with magrolimab alone, MF CD34+ cells show no changes in survival (100% alive). The combination of ruxolitinib and magrolimab reduced slightly the MF CD34+ cell survival to 92%. A substantial membrane overexpression of CALR was observed in MF CD34+ cells when incubated with Ruxo and Magrolimab, 74.1%, vs. Ruxo alone 52.6%, Magrolimab alone 53.4% and in untreated (51.8%). Interestingly, in the healthy controls, the exposure to combination therapy did not cause the same degree of CALR overexpression (20.7% vs. 12.5%), indicating a much stronger pro-phagocytic effect following magrolimab/ruxolitinib combo in MF cancer cells.

Conclusion: In vitro, treatment with ruxolitinib or ruxolitinib plus magrolimab, slightly reduce the overall cell survival rate in CD34+ MF samples. However, the combination of Ruxolitinib and Magrolimab significantly increases CALR membrane expression comparing with Ruxo or Magrolimab alone, signalling a much stronger pro-phagocytic message in MF cells. The induced CALR overexpression, could stimulate a stronger immune response in vivo, attracting macrophages towards MF clone. These data support the rational of combining Ruxolitinib with Magrolimab in the next clinical trial stage in myelofibrosis.

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BSH23-PO35 | Impact of momelotinib on patient-reported quality of life for patients with myelofibrosis from MOMENTUM

Prof Ciro Rinaldi[1], Dr Ruben Mesa[2], Prof Claire Harrison[3], Dr Jeanne Palmer[4], Prof Vikas Gupta[5], Prof Donal McLornan[3], Prof Mary Francis McMullin[6], Prof Jean-Jacques Kiladjian[7], Prof Lynda Foltz[8], Prof Uwe Platzbecker[9], Prof Maria Laura Fox[10], Prof Adam Mead[11], Prof David M Ross[12], Dr Stephen Oh[13], Prof Andrew Perkins[14], Prof Michael Francis Leahy[15], Dr Jun Kawashima[16], Sunhee Ro[16], Rafe Donahue[16], Samineh Deheshi[17], Dr Srdan Verstovsek[18]

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Disease-related inflammation, anaemia, and splenomegaly contribute to burdensome symptoms in patients with myelofibrosis (MF), decreasing health-related quality of life (HRQOL). Momelotinib (MMB) is an ACVR1, JAK1, and JAK2 inhibitor with demonstrated symptom, spleen, and anaemia benefits in patients with MF. In MOMENTUM, MMB significantly improved symptoms vs. danazol (DAN) (24.6% vs. 9.2%) by Total Symptom Score reduction ≥50% (TSS50) at week 24 vs. baseline in symptomatic, anaemic patients previously treated with a JAKi. We report patient-reported outcomes (PROs) from MOMENTUM.

PRO assessments included the MFSAF v4.0, EORTC QLQ-C30, and PROMIS Physical Function Short Form. Responder analyses were summarised by corresponding meaningful change thresholds. Longitudinal change from baseline scores were analysed using mixed model for repeated measures, and longitudinal analyses of response were performed by generalised estimating equation after multiple imputation for missing scores. Time to first response was analysed using the Kaplan–Meier method. A stratified log-rank test was performed to compare treatment arms; hazard ratios (HRs) were estimated from a stratified Cox regression model.

Proportionally greater improvements in symptoms, physical function, and global QOL were observed with MMB vs. DAN at week 24 by MFSAF, PROMIS, and EORTC QLQ-C30, respectively. Improvements in all MFSAF individual item scores from baseline at week 24 were greater with MMB vs. DAN, with the greatest treatment differences noted for night sweats, abdominal discomfort, bone pain, and rib pain. Longitudinal MFSAF TSS response analysis indicated a higher likelihood of response in the MMB arm (overall OR, 2.50 throughout the 24-week randomised treatment period). Response HRs were >1 (range, 1.47–3.82) in the time to first response analysis for MFSAF TSS and individual items, excluding itching (0.77), indicating a faster symptom response with MMB. Responder analyses showed greater improvement for fatigue with MMB based on MFSAF (18.5% vs. 9.2%) and EORTC QLQ-C30 (41.5% vs. 21.5%). Similarly, responder analysis using the PROMIS assessments related to physical function also favoured MMB (17.7% vs. 4.6%), which was supported by shorter times to first response for physical function (HR, 1.93; 95% CI, 0.95–3.91). Improvement in global QOL by EORTC QLQ-C30 was also greater in the MMB arm: proportion difference of 14.6% between MMB and DAN (95% CI, 1.5%–27.7%; $p = 0.04$).

Consistent with the positive primary endpoint (TSS50) result of MOMENTUM, these responder, longitudinal, and time-to-event analyses demonstrate that MMB improves disease-related symptoms with associated improvement in fatigue, physical function, and overall HRQOL vs. DAN in patients with MF.

BSH23-PO36 | Zanubrutinib in acalabrutinib-intolerant patients with B-cell malignancies

Dr Rocco Crescenzo[1], Dr. Mazyar Shadman[2], Dr. Ian W. Flinn[3], Dr. Edwin C. Kingsley[4], Dr. Benjamin Freeman[5], Dr. Moshe Y. Levy[6], Dr. Houston Holmes[6], Dr. Charles M. Farber[7], Dr. Arvind Chaudhry[8], Dr. Adam Idoine[1], Dr. Xiaoping Zhang[1], Dr. Aileen Cohen[1], Dr. Kunthel By[1], Dr. Jeff P. Sharman[9]

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Bruton tyrosine kinase inhibitors (BTKi) are effective treatments for various B-cell malignancies. However, treatment-related adverse events (Aes), potentially due to off-target kinase binding, limit their use. Zanubrutinib, a potent next-generation BTKi, is designed to selectively bind to

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BTK and maximise tolerability by minimising off-target kinase binding. Results from the BGB-3111-215 phase 2 study (NCT04116437) showed that zanubrutinib is well-tolerated in patients with intolerance to ibrutinib and/or acalabrutinib (Blood 2021;138[suppl 1]:1410). Here, we report updated results of the tolerability and efficacy of zanubrutinib in patients intolerant to acalabrutinib (cohort 2).

Eligible patients with chronic lymphocytic leukaemia/small lymphocytic lymphoma (CLL/SLL), Waldenström macroglobulinemia (WM), mantle cell lymphoma (MCL), or marginal zone lymphoma (MZL) who met protocol-defined criteria for acalabrutinib intolerance received zanubrutinib 160 mg twice daily or 320 mg once daily. Patients who progressed on prior BTKi therapy were excluded. Safety and efficacy were evaluated. Investigators used parameters at study entry as baseline and assessed responses every 3 cycles based on indication-specific standard response criteria.

As of 6 June 2022, 17 patients in cohort 2 received zanubrutinib (12 CLL/SLL; 3 WM; 1 MCL; 1 MZL). Median age was 74 years (range, 51–87); median treatment duration was 9.2 months (range, 0.5–20.9), median follow-up was 10.4 months (range, 1.1–20.9); median number of prior therapies was 2 (range, 1–6); 9 (53%) patients received prior ibrutinib and acalabrutinib. Five patients discontinued treatment (Aes $n = 2$, withdrawal $n = 2$, progressive disease $n = 1$). A total of 28 acalabrutinib-intolerance events were reported in 17 patients, most commonly arthralgia, myalgia, headache (4 each), haemorrhage, and fatigue (2 each). Twenty-one (75%) acalabrutinib-intolerance events did not recur on zanubrutinib, corresponding to 11 (65%) patients not experiencing any intolerance recurrence. Seven (25%) events recurred (1 lower grade, 6 same grade); 2 patients discontinued owing to recurrence (myalgia and diarrhoea same grade). Two patients who experienced the same intolerance events (pain in extremity and atrial fibrillation) on ibrutinib and acalabrutinib did not have a recurrence of either event on zanubrutinib. Among 14 efficacy-evaluable patients on zanubrutinib, 13 (93%) achieved at least stable disease and 9 (64%) achieved a deepening of response.

With a longer median zanubrutinib exposure (9.2 months) compared with acalabrutinib exposure (3.8 months), acalabrutinib intolerances were unlikely to recur, and disease was controlled in 13 (93%) efficacy-evaluable patients treated with zanubrutinib. Our results suggest that switching to zanubrutinib may yield clinical benefit in patients intolerant to other BTKi. Enrolment is ongoing.

BSH23-PO37 | Evaluation of a smartphone app (my CML) to support patients with chronic myeloid Leukaemia

Mr Nick Duncan[1,2], Mr John Choi[5], Ms Cassandra Cooper[5], Dr Zahraa Jalal[2], Ms Jennifer Jupp[5], Mr Sam Spurgeon[4], Ms Andrea Preston[3]

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Introduction: There is strong evidence that suboptimal adherence has a negative effect on clinical endpoints in patients with Chronic Myeloid Leukaemia (CML). Optimising treatment adherence is, therefore, a key strategy to improve patient outcomes. Given the increasing focus on providing patients with greater autonomy and responsibility for managing their health, alongside the ubiquitousness of smartphones and associated technological innovations, we created a smartphone application (app) to assist patients. Functionalities were designed based on our findings from an earlier international survey of CML patients.

In this study, we aimed to evaluate the app's usability from patient and healthcare professional perspectives.

Methodology: Using a mixed methods approach we evaluated the app via:

- usage data from Google's Firebase platform (https://firebase.google.com/)
- healthcare professional benchmarking, using the Mobile App Rating Scale (MARS) questionnaire
- patient questionnaires, using the mHealth App Usability Questionnaire (MAUQ)
- patient focus groups
- one-to-one patient interviews

Patient recruitment was undertaken via CML online support groups. Eligible participants were ≥18 years with a diagnosis of CML and previous experience (minimum of 4 weeks) with using the My CML app. Ethical approval was granted by University of Birmingham ethics committee.

Results: There were 1100 downloads in 79 Countries within 9 months of launching. The majority (50%) were from the UK, followed by USA (20%) and India (7%). The average engagement time per session was 1 min 26 s and the most popular app functions were the My Progress and Drug Interaction Checker functions. Of 44 patients surveyed, 82% were still using the app and the majority (53%) used the app every day. The overall mean usability score (on a scale of 1–7) among respondents was 5.8 (95% confidence interval (CI) 5.5–6.1). Usability were scores were significantly higher among patients who were still using the app [6 0.0 vs. 4.7, $p = 0.0007$].

When assessing app functionality, practicality, and quality, the My CML app achieved the highest score of the four CML

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patient apps currently on the market and the overall MARS score for the app (4.24) compared favourably with other mobile apps for haematological conditions.
Focus groups and interviews demonstrated high levels of patient satisfaction, with all participants happy to recommend to other CML patients.
Conclusions: The My CML app has generated positive feedback from CML patients and achieved a global reach. This evaluation has generated a wealth of patient-driven data to inform further development of the app to maximise patient benefit/outcomes.

BSH23-PO38 | Transfusion independence: Potential surrogate for overall survival in JAK inhibitor-experienced patients with myelofibrosis from MOMENTUM

Prof. Joanne Ewing[1], Prof Donal McLornan[2], Dr Srdan Verstovsek[3], Dr Stephen Oh[4], Prof Jean-Jacques Kiladjian[5], Prof Uwe Platzbecker[6], Prof Francesco Passamonti[7], Dr Aaron Gerds[8], Prof Alessandro Vannucchi[9], Prof Vikas Gupta[10], Prof Sebastian Grosicki[11], Prof Haifa Kathrin Al-Ali[12], Dr David Lavie[13], Dr Andrew Kuykendall[14], Prof Sung-Eun Lee[15], Prof Alessandra Iurlo[16], Prof Luminita Ocroteala[17], Dr Jun Kawashima[18], Prof Rafe Donahue[18], Bryan Strouse[19], Dr Ruben Mesa[20]
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Momelotinib (MMB), an oral ACVR1, JAK1, and JAK2 inhibitor, has previously demonstrated clinical activity on symptoms, anaemia, and spleen volume in JAK inhibitor (JAKi)-naive and -experienced patients with myelofibrosis (MF). In the SIMPLIFY-1 study, JAKi-naive patients with MF who received MMB and achieved transfusion independence response (TI-R; no transfusions and no haemoglobin [Hb] <8 g/dL during the 12 weeks preceding week 24 [W24]) experienced prolonged OS vs. those who were non-TI-R. We applied similar analyses to the phase 3 MOMENTUM study of MMB vs. danazol (DAN) in anaemic, symptomatic patients with MF who were previously treated with a JAKi.
Study design was as follows: randomisation (2:1) to MMB 200 mg daily (n = 130) or DAN 600 mg daily (n = 65) for 24 weeks, followed by open-label (OL) MMB. Eligibility criteria included primary or post-ET/PV; DIPSS high risk, Int-2 or Int-1 MF; Hb <10 g/dL; prior JAKi for ≥90 days (≥28 days if RBC transfusions ≥4 units in 8 weeks; grade 3/4 thrombocytopenia/anaemia/hematoma); and palpable spleen ≥5 cm. Anaemia benefit was evaluated by TI-R, Kaplan–Meier (KM) estimates, HR, and OR; survival was evaluated using KM analysis. Transfusion dependence (TD) was defined as ≥4 units of transfusions in the 8 weeks immediately prior (each in response to Hb ≤9.5 g/dL), and transfusion requiring (TR) as neither TD nor TI.
In the MMB and DAN arms, mean Hb were 8.1 and 7.9 g/dL, 13% and 15% of patients were TI, and 49% and 52% were TD at baseline. W24 TI-R rates were 31% and 20% for the MMB and DAN arms (noninferiority, p = 0.0064). Over the 24-week period, 35% of MMB patients had zero RBC units transfused vs. 17% of DAN patients (OR, 2.7; p = 0.0107). Median days to first, third, and fifth RBC transfusion were 40, 106, and not reached on MMB, and 15, 52, and 109 on DAN. Of the 63 MMB patients with baseline TD, 14% were TI and 30% were TR (end of W24). Of the 34 DAN patients with baseline TD, 9% were TI and 9% were TR (end of W24). Interim OS analysis of both arms shows that patients who achieve W24 TI-R have prolonged OS vs. those who were non-TI-R.
These analyses demonstrate that MMB increased the likelihood of TI and lower transfusion burden vs. DAN, consistent with previously reported data from the SIMPLIFY studies, and suggest that W24 TI-R is a potential surrogate for improved OS.

BSH23-PO39 | Evusheld prophylaxis increases social interactions and improves anxiety, depression, agoraphobia and quality of life scores

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Evusheld is a combination injection of tixagevimab/cilgavimab for pre-exposure COVID-19 prophylaxis and was made available to UK private clinics from October 2022. NICE review is ongoing. Whilst efficacy analysis of Evusheld has focused on the risk-reduction of contracting COVID-19,

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anecdotal reports suggest additional psychological benefits from Evusheld, although supportive objective data are lacking. In this study, we used 4 well-established psychological health questionnaires to assess different psychological parameters (EQ5D-3 L quality of life (QoL) score, DSM5 Agoraphobia score, Duke's Social Support Index (DSSI) and the hospital anxiety and depression score (HADS)) in blood cancer patients treated with Evusheld at the Genesis Care (GC) Clinic, Cambridge. Patient data (pre- and post-Evusheld) were compared with a control group of GC blood cancer patients who had not received Evusheld. The study was approved by GC and all patients had consented to email contact. Questionnaire replies were anonymised and free-text comments were invited.

Questionnaires were completed by 29/40 Evusheld and 54/100 control patients. With EQ5D, Evusheld did not impact mobility, self-care and pain/discomfort scores and patient/control groups scored at similar levels. EQ5D scores for 'usual activities' and 'anxiety/depression' improved post-Evusheld (patients reporting 'normal activities' increased from 52% to 76% (control = 78%); patients reporting 'no anxiety/depression' increased from 45% to 66% (control = 65%)). The mean global EQ5D QoL score improved post-Evusheld [69.4% to 72.9% (control = 75.7%)]. With the DSM5 agoraphobia score, Evusheld treatment improved agoraphobia parameters, reducing the mean score from 15.7 to 5.1 (control = 3.7; max = 40) with certain striking changes; 72% of pretreatment patients avoided crowded situations all of/most of the time, reducing to 14% post-Evusheld (control = 11%). The DSSI score assessed social/work interactions with external household contacts and post-Evusheld the mean number of interactions over 3 weeks increased from 1.48 to 3.37 (control = 3.77). Pre-Evusheld, 52% of patients had no interactions outside their household, dropping to 20% in the 3 weeks post-Evusheld (control = 17%). Using the HADS 14-point analysis of depression and anxiety revealed on average that each parameter was 25.3% 'significantly improved' and 25.4% 'a little improved' post-Evusheld.

Accepting limitations of a small study and potential biases associated with a self-funding patient cohort, Evusheld treatment broadly improved all psychological scores assessed. Free-text comments clearly indicate that Evusheld had a major positive impact on QoL/social mobility for specific patients. The Evusheld patients had higher baseline scores for social isolation, anxiety, depression and agoraphobia compared with control patients, yet Evusheld treatment appeared to improve these parameters to a level similar to control patients.

BSH23-PO40 | Incidental finding of JAK2 V617F mutations in patients with solid cancers: Implications for reporting variants

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Genomic testing is changing the landscape of the management of cancer. The National test directory in England mandates a standardised approach to genomic testing in cancer ensuring equitable access. At Barts Health NHS Trust, specimens with insufficient DNA for the centralised solid tumour large panel (RMH220) are run locally on a 50 gene pan-cancer panel (CHPv2) including TP53, NRAS, KRAS but also JAK2.

In the past two years we have identified four patients with incidental JAK2 V617F mutations in their solid tumours using our local 50 gene pan-cancer panel. Three patients had thrombocytosis, with one patient presenting with a normal blood count. One of the patients with thrombocytosis also had iron deficiency anaemia, which could have contributed to the thrombocytosis. The JAK2 V617F mutations were confirmed in the peripheral blood and were present at a higher variant allele frequency (VAF) compared to the tumour sample consistent with peripheral blood contamination of tumour.

The unexpected presence of the JAK2 V617F mutation in these cases indicated the coexistence of an unsuspected myeloproliferative neoplasm (MPN), in addition to their solid tumour. High platelet counts may be seen as a reactive feature in solid tumours and so oncologists may not expect or test for MPNs.

In terms of treatment, two patients were started on Aspirin and one was already on Rivaroxaban for a venous thromboembolism. One patient required cytoreductive therapy with hydroxycarbamide. One patient did not have further investigations as they were for palliative treatments only.

Conclusion: The growing use of large pan-cancer genomic panels will increase the chances of 'incidental findings'. The American College of Medical Genetics (ACMG) recommends reporting incidental findings in over 70 genes predisposing to medically actionable conditions. However, the JAK2 V617F is not listed in this document, or usually included in 'virtual' panels used for solid tumours. Similarly, the European Society of Human Genetics (ESHG) propose limiting disclosure to 'medically actionable variants' We argue that JAK2 V617F is a driver mutation and, therefore, meets the criteria as medically actionable and should be disclosed and investigated further. We have reviewed our local practice to ensure that all patients whether tested locally or using the centralised gene panel have JAK2 V617F mutations reported and acted on if clinically appropriate.

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BSH23-PO41 | Ruxolitinib versus best available therapy for PV intolerant or resistant to Hydroxycarbamide a randomised trial

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Polycythaemia vera (PV) is characterised by thrombosis, haemorrhage, systemic symptoms and disease transformation. In high-risk PV ruxolitinib controls blood counts, improves symptoms, but broader efficacy is uncertain. Clinical benefit of reduction of JAK2V617F variant allele fraction (VAF) is also unproven.

MAJIC-PV is a randomised phase II trial of ruxolitinib vs. Best Available Therapy (BAT) in patients resistant/intolerant to hydroxycarbamide (HC-INT/RES). Primary outcome was complete response (CR) (European LeukemiaNet criteria) within 1 year. Secondary outcomes included response duration, event-free survival (EFS), symptom/molecular response.

180 patients were evaluated. CR was achieved in 40 (43%) patients on ruxolitinib vs. 23 (26%) on BAT (OR 2.12, 90% CI 1.25, 3.60, $p = 0.02$) @1 year. Duration of CR was superior for ruxolitinib (HR 0.38, 95% CI 0.24, 0.61, $p < 0.001$). Symptom durably improved with ruxolitinib (mean 52 months). Thromboembolic-event-free survival significantly improved with ruxolitinib (HR 0.56, 95% CI 0.32, 1.00, $p = 0.05$) and correlated with average number of venesections per year (HR 1.20, 95% CI 1.08, 1.33, $p < 0.001$). EFS was superior for patients attaining CR @1 year (HR 0.41, 95% CI 0.21, 0.78, $p = 0.01$); and on ruxolitinib (HR 0.58, 95% 0.35, 0.94, $p = 0.03$). >50% JAK2V617FVAF reduction, @12 m occurred in in 14% ruxolitinib, 18% BAT patients, respectively, and more common with CR @1 year. Molecular responses occurred in 56% ruxolitinib and 25%, BAT ($p < 0.001$) at latest time point and was associated with progression-free survival $p = 0.001$, EFS $p = 0.001$, overall survival $p = 0.01$. To analyse clonal burden in stem/progenitor cells (HSPC), targeted single-cell genotyping was performed on three ruxolitinib treated patients confirming substantial (72%–100%) reduction in JAK2V617F+ HSPC. Survival analysis demonstrated impaired EFS in patients with additional driver mutations (treatment, age and sex-adjusted HR = 1.92, 95% CI 1.16, 3.19, $n = 167$, $p = 0.01$). Specifically, mutated-ASXL1 conferred impaired EFS (adjusted HR 3.02, 95% CI 1.47, 6.17, $n = 167$, $p = 0.003$) correcting for age, gender and the presence of mutations in TET2. The safety profile of ruxolitinib was as previously reported.

MAJIC-PV demonstrates novel benefits for ruxolitinib, which improved thrombosis-free and EFS in high-risk HC-int/res PV. Importantly, preplanned analyses within this study demonstrate for the first time that attainment of a 50% reduction in JAK2V617F VAF, which occurred more frequently with ruxolitinib, was associated with important clinical benefits and clearance of MPN stem cells. This data confirms and challenges the current therapeutic algorithm, supporting the benefit of targeted therapy and molecular monitoring in this disease.

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BSH23-PO42 | Updated results from MOMENTUM: Momelotinib vs. Danazol in JAK inhibitor-experienced, symptomatic, and anaemic myelofibrosis patients

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The phase 3 MOMENTUM study (NCT04173494) of the ACVR1/JAK1/JAK2 inhibitor momelotinib (MMB) vs. danazol (DAN) in patients with myelofibrosis (MF) previously treated with a JAK inhibitor (JAKi) met the primary endpoint and all key secondary endpoints at week 24 (W24). We provide updated results from week 48 assessments.

Eligible patients had primary or post-ET/PV MF; DIPSS high, Int-2, or Int-1 risk; Total Symptom Score (TSS) ≥10; haemoglobin (Hb) <10 g/dL; platelets ≥25×10^9/L; prior JAKi for ≥90 days (≥28 days if red blood cell [RBC] transfusions ≥4 units in 8 weeks or grade 3/4 thrombocytopenia/anaemia/hematoma); and palpable spleen ≥5 cm. Randomisation was 2:1 to MMB 200 mg/day or DAN 600 mg/day for 24 weeks, followed by open-label (OL) MMB. Week 48 endpoints included durations of response (TSS, transfusion independence [TI], splenic) and overall and leukaemia-free survival (OS, LFS).

As of 17 May 2022, 93/130 (72%) MMB → MMB and 41/65 (63%) DAN → MMB patients received OL MMB; mean MMB durations were 48 weeks and 24 weeks, respectively. Analyses for W24 responders showed the following: of TSS responders, 31/32 (97%) MMB → MMB and 6/6 DAN → MMB patients had TSS < baseline; of TI responders, 36/40 (90%) and 10/13 (77%) had no RBC transfusions or Hb <8 g/dL; and of spleen responders, all patients had splenic volume < baseline.

In the OL phase, the most common grade ≥3 treatment-emergent adverse events (TEAEs) were thrombocytopenia (MMB → MMB, 9%; DAN → MMB, 15%) and anaemia (MMB → MMB, 9%; DAN → MMB, 2%). Grade ≥3 infections occurred in 19% of MMB → MMB and 10% of DAN → MMB patients, including grade ≥3 (nonfatal) COVID-19. Peripheral neuropathy (PN) occurred in 2% of patients in each arm, and none discontinued MMB due to PN. TEAEs led to MMB discontinuation in 18% (MMB → MMB) vs. 10% (DAN → MMB).

A trend towards improved OS up to W24 was previously observed with MMB vs. DAN (hazard ratio [HR], 0.506; $p = 0.0719$); after all patients crossed over to OL MMB, OS and LFS curves for both arms converged (HR, 0.945, 95% CI, 0.528–1.693; HR, 0.830, 95% CI, 0.473–1.4555).

Sixty of 81 (74%) MMB → MMB and 29 of 43 (67%) DAN → MMB patients with baseline platelets ≤150×10^9/L entered the OL phase. Efficacy and safety results in thrombocytopenic subgroups in the OL period were consistent with the intent-to-treat (ITT) population.

OL MMB maintained symptom, TI, and spleen responses with continued good survival and safety in the ITT and low platelet populations. MMB may address an unmet need in anaemic patients with MF.

BSH23-PO43 | Audit of sepsis-six implementation in adult patients with neutropenic sepsis

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Background: Neutropenic sepsis is a common life-threatening presentation to haematology/oncology services. Completion of the Sepsis-Six protocol has been shown to reduce mortality in neutropenic sepsis (Daniels et al, 2011). A National Early Warning Score (NEWS) of ≥5 identifies patients who are likely to deteriorate.

Aim: To assess compliance with the Sepsis-Six pathway in adult patients with neutropenic sepsis and to evaluate the factors associated with adverse outcomes.

Methods: We retrospectively reviewed neutropenic sepsis cases in a UK district general hospital from 07/2020 to 11/2022. Data on demographics, clinical parameters (NEWS score, neutrophil count, urine output, blood culture, lactate, albumin), administration of antibiotics, oxygen and intravenous fluids, and outcomes (ITU admission, transfusion requirement, length of stay and mortality) were collected from medical records.

Results: Eighty-three cases (M:F 42:41, mean age 63) met the inclusion criteria.

The trust Sepsis-Six proforma was fully completed in 14% (12/83) of cases, although only 8% (8/83) had this actioned.

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The proforma was partially completed in 8% (7/83), not completed at all in 60% (50/83) and no documentation was available in 17% (14/83) of cases.
In 75% (62/83), there was incomplete actioning of Sepsis-Six, with failure to measure urine output the most common omission.
Overall mortality within 28 days of presentation was 14% (12/83). Mortality was 19% (5/26) among patients with a NEWS score ≥5 and 12% (7/57) in those with a NEWS <5 during the first 24 h of admission.
Deceased patients had a background of haematological malignancy in 50% (6/12), of solid malignancy in 33% (4/12) and no history of malignancy in 17% (2/12) of cases.
Mean length of stay was 10 days.
Discussion: Adherence to the Sepsis-Six protocol remains the gold standard in emergency sepsis management in the UK. There was poor compliance with completion of Sepsis-Six documentation or its actioning in this data set. Despite limited data, patients with actioned Sepsis-Six protocols did not have superior outcomes.
NEWS ≥5 at presentation was more common in those patients who subsequently died. The observed mortality rate was 14%, compared to expected in hospital mortality of approximately 10% recognised by the European Society for Medical Oncology. In order to improve our compliance, better training of staff and key parameters will be incorporated into the mandatory fields of electronic medical notes in the future. Cooperation with other trusts in the sector aimed to increase the sample size and obtain more robust data for future audits.

BSH23-PO44 | Assessing whether cardiovascular risk assessments are conducted for patients with CML taking tyrosine kinase inhibitors

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Background: Chronic myeloid leukaemia (CML) is a myeloproliferative neoplasm of clonal haematopoietic stem cells, commonly affecting myeloid cell lines. Tyrosine kinase inhibitors (TKIs) have revolutionised the treatment of CML, converting a life-limiting disease into a long-term condition. As patients are living longer with CML, it is important to consider the increased risk of cardiovascular disease associated with TKIs. British Society for Haematology (BSH) guidelines for CML recommend baseline and annual cardiovascular risk assessments via the QRISK3 screening tool.
Aim: This audit aimed to assess the degree of implementation of these guidelines in the haematology department at the Royal Wolverhampton NHS Trust.
Method: The Trust Clinical Portal database was utilised to identify patients diagnosed with CML who were prescribed TKIs over the last 12 months. The final list consisted of 55 patients, and data was collected regarding patient demographics, past medical history, medication history and any baseline assessments they had prior to commencing a TKI. For any patients who did not have a QRISK3 score calculated and were not currently on a statin, their estimated QRISK3 score was calculated from the patient records available and those who required a statin according to BSH guidelines were identified.
Results: The results of this audit suggest that only 5.45% of patients were managed according to the guidelines, with a cardiovascular risk assessment and documented QRISK3 score as advised. This means that 94.55% of patients did not have a documented QRISK score. Importantly, 23.64% of these patients did not have a documented QRISK score, were not on statins and had a 10% or higher risk of a cardiovascular event in the next 10 years. These are the individuals who are currently at risk of harm because their QRISK score was not calculated.
Conclusion: Overall, training and education regarding the guidelines to all healthcare professionals is key to ensure that patients are receiving appropriate care. A 1-h training session was delivered to all medical staff, including consultants, in the haematology department at the Trust, where the results of the audit were disseminated and the BSH guideline was re-emphasised.
Furthermore, the introduction of a pro forma to be completed annually in face-to-face consultations with specific reminders to conduct a cardiovascular risk assessment is recommended. A reaudit in 12 months will highlight any progress made towards the implementation of this guideline recommendation.
Key Words: Chronic myeloid leukaemia, tyrosine kinase inhibitors, cardiovascular disease, risk management, QRISK3 score.

BSH23-PO46 | UK retrospective observational study of romiplostim and other treatments in early immune thrombocytopenia

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Background: As of 2021, the thrombopoietin receptor agonist (TPO-RA) romiplostim is licensed for adults with immune thrombocytopenia (ITP) across all disease phases, including duration ≤1 year refractory to first-line treatments. Whilst aligned with treatment guidelines, real-world UK data on this indication are lacking.

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Methods: To assess the response to romiplostim, we performed a retrospective UK-wide study of real-world treatment patterns in adults with primary ITP ≤1 year from diagnosis between 2 July 2015 and 23 March 2020. Recruited patients were divided into those who either did (Group 1) or did not (Group 2) receive romiplostim during the first year from diagnosis. TPO-Ras, including eltrombopag, could be received in either group after the first year (excluding eltrombopag in the first year). All efficacy and safety analyses were descriptive.

Results: Seventy-eight patients enrolled at seven UK sites; Group 1 (n = 13) 62% female, median (min-max) platelets 4 (2–44) × 10^9/L and Group 2 (n = 65) 59% female, median (min–max) platelets 7 (0–108) × 10^9/L.

In Group 1, romiplostim was initiated (off-licence) at a median of 6 weeks with a median treatment duration of 63 weeks; 2/13 received romiplostim first-line and 11/13 as the second treatment or later. Romiplostim treatment continued in 8/13 patients at study end. Other treatments in Group 1 in more than one patient included: prednisolone (n = 11), mycophenolate mofetil (MMF, n = 5), IVIg (n = 4), and rituximab (n = 3). In Group 2, two patients (3%) received no treatment; 33 patients (51%) received corticosteroids ± IVIg ± tranexamic acid only without further escalation; 26 patients (40%) required MMF or rituximab. In the second year, six patients (9%) in Group 2 required a TPO-RA. The median treatment duration was shorter for Group 1 vs. 2 for prednisolone (3 vs. 10 weeks for those discontinuing prednisolone) and MMF (13 vs. 39 weeks).

By the end of the 2-year treatment period, median (interquartile range) platelet counts had increased to 157 (81–275) and 122 (81–202) × 10^9/L in Groups 1 and 2, respectively. During the observation period in Groups 1 and 2, respectively, bleeding episodes occurred in 54% and 63% of patients, and rescue therapies were required by 39% and 31% of patients. The mean number of rescue therapies/patient was 0.6 and 0.8 in Groups 1 and 2, respectively. Thromboembolic events occurred in one patient in each group (Group 1: subacute left lower limb arterial ischemia; Group 2: stroke).

Conclusion: These findings support further exploring the earlier use of romiplostim for adult patients with primary ITP.

BSH23-PO47 | Comparative effectiveness of Fostamatinib vs. rituximab in refractory chronic immune thrombocytopenia: A network meta-analysis

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Introduction: Fostamatinib, a spleen tyrosine kinase inhibitor, is recommended by the National Institute for Health and Care Excellence (NICE) as an option in refractory chronic ITP (cITP) in adults who previously had a thrombopoietin receptor agonist (TPO-RA), or when a TPO-RA is unsuitable. We performed a network meta-analysis (NMA) to estimate the relative efficacy of fostamatinib vs. standard treatments after TPO-Ras in refractory cITP.

Methods: A systematic literature review was conducted in 2019 (updated 2021). Six studies informed the NMA: 2 placebo-controlled trials of fostamatinib, 2 placebo-controlled trials of rituximab, 1 randomised trial of 3 rituximab regimens, and 1 non-randomised study of 2 rituximab regimens. No studies assessing additional comparators were found that could inform the NMA.

The outcome of interest was overall platelet response. Definitions varied across studies, so the following analyses were performed to investigate sensitivity in outcome variation: (1) using published definitions for each study; (2) using a standardised definition (platelet count >30 × 10^9/L by week 4) for the fostamatinib trials, and published definitions for each of the remaining studies; (3) using the standardised definition for all studies, and restricted to those with results available. No treatment effect modifiers were identified by UK experts at a Delphi panel.

Random-effect models were employed in the analyses. The Odds Ratio (OR) for outcome occurrence of each treatment compared to fostamatinib (Ors <1 indicating reduced response relative to fostamatinib), the median rank, and the probability of being the best treatment are reported. Analyses were performed following NICE Decision Support Unit guidance and implemented with WinBUGS and R.

Results: Analysis 1: fostamatinib was statistically significantly more effective than placebo and all rituximab regimens. Ors for comparators vs. fostamatinib ranged from 0.11 to 0.20.

Analysis 2: fostamatinib was associated with improved response vs. placebo and all rituximab regimens (OR range 0.14–0.25) but statistical significance was only reached vs. placebo and rituximab 4 × 100 mg/week.

Analysis 3: pairwise comparisons suggested increased efficacy of fostamatinib relative to placebo and rituximab 4 × 375 mg/m^2/week (Ors: 0.25 and 0.34, respectively). However, OR was only statistically significant vs. placebo.

Across analyses the probability of fostamatinib being most effective ranged 0.91–0.97 and median rank for fostamatinib was 1.

The between study standard deviations indicated moderate heterogeneity in treatment effects.

Conclusions: Fostamatinib was associated with greater likelihood of overall platelet response relative to rituximab at all regimens evaluated, and may be a better alternative than rituximab to increase platelet counts in refractory cITP.

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BSH23-PO48 | Efficacy and safety of erythropoiesis stimulating agents (ESAs) in Myelofibrosis—A single-centre experience

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Background: Anaemia is one of the disease features of myelofibrosis (MF) associated with poor prognosis and response to anaemia targeted therapies has been associated with improvement in quality of life in MF patients.
Aims: To assess the response to erythropoiesis stimulating agents (ESAs) and the development of complications in clinical practice.
Methods: An audit was conducted with retrospective data collection on patients with MF receiving ESAs at St. James University Hospital, Leeds. Analysed data included age, gender, duration from diagnosis to onset of treatment with ESAs, baseline erythropoietin levels (epo); response in symptoms, haemoglobin levels (Hb), transfusion requirements, complications of treatment and causes of discontinuation.
Results: 21 patients received ESAs from 2020 to 2022: 12 with primary MF, 8 with secondary MF, and one patient with 1 MDS/MPN with ring sideroblasts and thrombocytosis; median age 76 years (range, 39–92), 57% male. A median time from diagnosis to treatment of 1 year (range, 0–8). Median baseline Hb was 93 g/L (range, 66–108). Baseline erythropoietin levels were available for 18 patients with a median of 32.6 mIU/mL (range, 7.3–390). All patients received Epoetin alpha (Eprex), median starting dose was 30 000 IU weekly (range, 10 000–40 000).
Response was assessed with regards to symptoms, a rise in Hb of 15 g/L and red cell transfusion requirements according to the available data during two intervals: between (6–9) weeks and between (15–18) weeks; 15 (71.4%) and 6 patients (28.5%) achieved a symptomatic improvement and a Hb response, respectively. The second timescale showed 12 patients (57%) with symptomatic improvement and 8 (38%) with an adequate Hb response. Of those who has received red cell transfusion prior to starting ESAs, 10% became transfusion independent and 20% had a reduction in transfusion requirement at both time points.
One (4.7%) patient developed venous thromboembolism likely attributed to metastatic malignancy and 3 (14.2%) patients developed possible treatment-emergent or exacerbation of hypertension. Median treatment duration was 14 months (range 1–48), 5 patients (23.8%) remained on treatment. The causes of discontinuation: commencement on an end-of-life pathway in 6, no or loss of response in 5 patients, sustained Hb rise in 3, hypertension in 1 and allogeneic bone marrow transplant in 1.
Conclusions: ESAs are effective first-line therapies for the management of anaemia in patients with MF. Improvement in Hb may be observed as late as 15–18 weeks after initiation of ESAs. Hypertension is one of the commonest side effects, careful blood pressure monitoring is, therefore, warranted.

BSH23-PO49 | Generic imatinib: Data from the UK National Registry for chronic myeloid leukaemia patients

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The expiration of the patent for imatinib (IM; Gleevec®, Novartis) and introduction of generic(s) IM to the UK in early 2017 elicited different reactions. The expected savings are welcomed by the healthcare sector, whilst patients and physicians have had understandable concerns regarding efficacy and safety.
The UK National Registry for patients with chronic myeloid leukaemia (CML) is a web-based population based cancer registry, initially established as a regional registry in 2010, then expanded nationally in 2015.
Using the available registry data collected between 2009 and 2020, we analysed the differences in efficacy, safety, and overall survival (OS) probability in patients treated with generic(s) and branded IM. As generic(s) IM replaced branded IM across the whole of the UK in 2017; we hypothesised that patients diagnosed before 2017 received the branded formulation and those diagnosed after received the generic formulation(s).
Patients treated with IM as first line were included, up until the time of switching treatment for those who did so. As such, 132 patients who were diagnosed, and had their TKI switch or last follow-up, before 2017 were considered to have been treated with the branded formulation and 36 patients who were diagnosed after 2017, were considered to have been treated with the generic formulation(s).
To assess the efficacy and safety of generic(s) IM, the achievement of treatment responses as per the 2020 ELN guidelines, occurrence of treatment side effects, and OS probability were compared with those obtained on branded formulation.
Patients treated with generic(s) IM were older (median: 60.3 (range: 31.8–82) vs. 53.7 (18.6–88.7) years; $p = 0.025$) and had a shorter median follow-up duration (548 (range: 45–1093) vs. 1816 (range: 61–5347) days; $p < 0.01$). There were no statistically significant differences in gender or proportions in the different risk groups (Sokal, Euro, EUTOS, or ELTS) between both groups. Seventeen patients (12.9%) receiving branded IM and 3 (8.3%) receiving generic(s) IM were reported to have experienced side effects ($p = 0.572$). There were no statistically significant differences in response achievement at 3, 6, and 12 months from start of TKI therapy, or diagnosis, between both groups. OS probability was

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not different between patients treated with generic(s) and branded IM ($p = 0.831$).

Our results demonstrates at the least non-inferiority of generic(s) as compared to branded IM. Longer follow-up is crucial to ascertain our findings.

BSH23-PO50 | A single-centre retrospective analysis comparing the prognostic utility of the risk-stratification tools used in Myelofibrosis

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Myelofibrosis is a myeloproliferative neoplasm that carries an aggressive disease course with a median survival of 6 years, yet it has a highly variable rate of clinical progression between individuals. Current therapeutic interventions, such as allogeneic stem cell transplantation and JAK2 inhibitors, carry a risk of treatment-related morbidity and may be financially costly. Over the past decade, several prognostic risk-stratification tools have been developed to assist therapeutic decision-making in patients with myelofibrosis. In a single-centre retrospective study, we compared the prognostic utility of the International Prognostic Scoring System (IPSS), dynamic IPSS (DIPSS), revised DIPSS (DIPSS-plus), mutation-enhanced international prognostic scoring system for transplant-age patients (MIPSS70), revised MIPSS70 (MIPSS70+v2), genetically inspired prognostic scoring system (GIPSS), and the Mysec prognostic model (MYSEC-PM). Data were collected on patients with a confirmed diagnosis of either primary or secondary myelofibrosis. Each patient was stratified as per the seven prognostic models. For survival analysis, event-free survival (EFS) was chosen as the primary endpoint. EFS was defined as the occurrence of either leukaemic transformation or death during the follow-up period. Median event-free survival was calculated using the Kaplan–Meier method for each model, and differences between the subgroups were compared using the log-rank test. The Cox proportional-hazards model was used for univariate analysis. p-values <0.05 were considered significant. The corrected Akaike's information criterion (AICc) was used to compare the prognostic utility of the prognostic tools. Lower AICc values suggest superior predictive accuracy and quality within a model.

33 patients were identified. Median follow-up length was 47 months (range 1–180). The median EFS of the cohort was 10.5 years. Of the risk factors used within the prognostic models, transfusion dependency ($p < 0.001$), blasts in peripheral blood ($p < 0.001$), ASXL1 mutation presence ($p < 0.001$), and unfavourable karyotype ($p = 0.032$) were significantly associated with lower EFS. Furthermore, triple-negative mutation status was identified as an independent poor prognostic factor (OR 12.164, $p = 0.009$). Across the highest risk categories within the seven tools, only DIPSS-plus failed to demonstrate a statistically significant difference in EFS ($p = 0.068$). All seven tools showed a significant difference in EFS between individual risk categories. Using the AICc to identify the most predictive prognostic tools, the GIPSS and MIPSS+v2 demonstrated superior prognostic accuracy.

Current guidelines in the UK recommend using the DIPSS-plus model to risk-stratify patients with myelofibrosis. However, cytogenetic and mutational data have crucial prognostic significance, hence, newer risk-stratification tools such as the MIPSS+v2 and GIPSS may be preferable when these data parameters are available.

BSH23-PO51 | The role of next-generation sequencing in myeloproliferative neoplasms associated with splanchnic vein thromboses (MPN-SVT)

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Background and Aims: Splanchnic vein thromboses (SVT) represent a distinct complication of myeloproliferative neoplasms (MPN). Of note, SVT can be the initial presentation of the underlying haematological condition, prompting testing for the MPN-associated driver mutations (JAK2V617F, CALR exon 9 and MPL). Myeloid next-generation sequencing (NGS) has also identified other, lower-frequency mutations of diagnostic and prognostic significance. The aim of our study was to utilise NGS in our MPN-SVT cohort and consider its future usage in identifying these mutations; closer evaluation is given to their overall prevalence, and correlation with haematological or vascular disease severity.

Methods: MPN diagnosis was based on established diagnostic criteria (WHO, 2016). DNA samples from 41 patients with MPN-SVT (38 JAK2V617F, 2 CALR and 1 JAK2V617F and CALR) were analysed by NGS, using Illumina TruSight or ArcherDx VariantPlex myeloid panels of 54 and 75 genes, respectively (sensitivity 5% variant allele frequency; VAF). The cohort comprised 22 women, 19 men; diagnoses included 16 patients with polycythaemia vera (PV), 9 MPN-unclassifiable (MPN-U), 7 myelofibrosis (MF), 6 essential thrombocythaemia (ET) and 3 isolated JAK2V617F mutations. 9/41 had >1 NGS study.

Results: JAK2V617F alone was recorded by NGS in 23/41 patients (56.1%). NGS identified additional mutations in a further 5/41 (12.2%), comprising TET2, BCOR, DNMT3A, EZH2 and an ASXL1 variant of uncertain significance (VUS). In this group, 2 patients had MF, 2 MPN-U, 1 PV and all had JAK2V617F mutations; 2 showed disease progression (MF shift DIPSS 1 to 2; new VTE). Presence of an additional mutation also correlated with higher JAK2V617F VAF (median 33.7%, versus 12.5% in those without additional mutations). In a further 9/41 (21.9%), single nucleotide polymorphisms (SNPs) were identified; interestingly, a TP53 germline mutation was seen in a CALR-mutated patient.

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4/41 patients (9.75%) had no mutations recorded by NGS. Of those with repeat NGS, 3/9 (33.3%) showed increased VAF over time, correlating with disease progression in 2 (MF DIPSS 1; new SVT).

Summary and Conclusions: Additional mutations were detected in 12.2% of 41 MPN-SVT patients, lower than reported elsewhere. Explanations for this include methodologic differences: the lower sensitivity of older NGS technologies and different thresholds for reporting abnormal NGS. The presence of additional mutations by NGS was associated with higher JAK2V617F VAF, whilst rising VAF was seen alongside clinical disease progression; these findings concur with those seen elsewhere. Whilst myeloid NGS has a role in investigating MPN-SVT, the prognostic relevance of additional myeloid mutations remains unclear.

BSH23-PO52 | Effectiveness and safety of ponatinib in treatment resistant chronic myeloid leukaemia: A systematic review

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Background: Chronic myeloid leukaemia (CML) in patients that are resistant to multiple tyrosine kinase inhibitors (TKI) or harbour known resistant mutations pose a challenge to treatment. Ponatinib (AP24534), a third generation TKI, is a more recent treatment designed to overcome these limitations. However, there are limited evidence regarding stratifying patients most at risk from adverse side effects: most notably adverse vascular events. Additionally, there is limited evidence for the paediatric population. The purpose of this systematic review is to survey all the existing evidence to date and summarise it into a coherent whole.

Methods: A search was undertaken of the Cochrane database, and of PubMed for relevant published studies. The focus of the systematic review was formulated using the PICO framework. The study types and number included for adults were 1 open-label trial, 1 prospective cohort study, 4 retrospective cohort studies, and 3 case reports. The paediatric evidence included 1 retrospective study and 1 case report.

Results: The efficacy of ponatinib was uniformly demonstrated across the evaluated studies to result in high rate of treatment response in terms of overall survival, cytogenetic response, and molecular response. In the chronic-phase population the average overall survival rate at 36-months was 83.4%.

Multivariate analyses suggested that both younger and less heavily pretreated patients experience better response rates. This may be a function of this group's ability to tolerate higher dose regimens.

The strongest risk factors for vascular adverse events determined across the studies were ponatinib dosage and pre-existing vascular disease and risk factors.

Conclusion: The data for safety and efficacy in the paediatric population is limited but demonstrates good efficacy and safety in this treatment group, and as a good bridge to stem cell transplant.

Analysis of the studies of the pretreated adult population demonstrates the evidence is unanimously in agreement of the strong efficacy of ponatinib.

Further research in optimal dosing strategies is required and is currently being investigated by the ongoing OPTIC trial.

Limitations of the evidence include lack of randomised trials, and small study sizes. This limits the ability to more precisely identify factors that are associated with both positive and negative outcomes. With further studies we may better be able to direct individualisation of treatment.

BSH23-PO53 | An audit of paroxysmal nocturnal haemoglobinuria (PNH) requesting at University Hospitals of North Midlands

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Early identification of PNH, a rare life-threatening disease is essential to ensure appropriate management via the UK PNH service. Since testing for PNH is expensive (£75.97 per test), we set out to assess the suitability of PNH requests against guidelines with the aim to feedback to colleagues and reduce unnecessary testing.

To determine whether PNH requests at UHNM were in line with criteria in British Society for Haematology (BSH) guidelines.

All patients over 18 years of age who had PNH testing for the first time and those who had repeat testing for monitoring between 01/04/2019 and 31/03/2020 were included. Patients were selected from electronic records of PNH sample receipt to laboratories. Hospital records were reviewed for clinical details and investigation Results.

82 requests including 79 individual patients were audited. 57% were male and 43% were female. Median age was 56 years. 97.6% of PNH tests were requested by a haematologist whilst only 2.5% requests were done by non-haematology clinicians. 52.4% requests were in keeping with BSH recommendations, whilst 47.6% tests did not meet criteria for testing. All patients tested outside of guideline recommendations were negative. Of the reasonable requests, only 23.3% (10) were positive. Of the PNH positive patients, 8 patients were known to have a PNH clone with aplastic anaemia; one patient had a hypoplastic bone marrow and a known PNH clone whilst only one patient with cytopenia had a new positivity for PNH. The frequency monitoring for aplastic anaemia and a PNH clone was 100% concordant with BSH recommendations.

With appropriate testing, only one new patient was identified. Our audit has limitations. We have not been able to assess whether any patients outside of those monitored for PNH in aplastic anaemia have been overlooked for testing. Also, the time period includes the COVID-19 pandemic so our findings may not reflect usual practice.

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New BSH guidelines for thrombophilia testing were published in 2022 and recommend testing for PNH in patients with thrombosis at unusual sites and abnormal haematological parameters and for patients with arterial thrombosis and abnormal blood parameters. This will likely limit excess tests although the sensitivity and specificity of such an approach has not been formally evaluated.

In a finite health system, it is our responsibility to rationalise investigations. The cost of a year's testing was £6229.54 and that of inappropriate testing was £2962.83. As a department, we could, therefore, save £2962.83.

BSH23-PO54 | The efficacy of decision aids to support shared decision-making in haemato-oncology: A systematic review

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Several interventions known as decision aids (DA) have been developed to support patients with blood cancers to share in the decision-making process during consultations. Whilst Das have been developed robustly, it is unknown whether they would be effective or suitable for different types of consultation.

This review examines the known evidence for efficacy of Das for adult patients with haematological cancers including myeloma, lymphoma and leukaemia and the suitability and potential benefits of using Das in real-world settings (i.e. in-person, video and telephone consultations).

The review was registered on PROSPERO and the design of the review was based upon the PRISMA guidelines. Patient representatives and haematologists provided their opinions on the purpose of the review, the design and the Results.

Medline, Embase, PsychInfo, ERIC and PubMed, were searched to retrieve articles. Additional papers were identified by screening reference lists and hand searches. Eligibility assessment was performed in a standardised manner to capture details about the study, the samples and 'quality indicators' as defined by the STROBE checklist.

104 titles were identified by database searches. A further 9 titles were identified through a PubMed search. Fifty-nine titles were fully screened with 9 titles meeting the selection criteria. Hand searches identified a further 19 titles. After assessing for eligibility, 7 titles were included in the review.

Das evaluated in the real-world setting were typically designed for end-of-life decisions, with few targeting critical treatment decisions. Few studies used validated questionnaires to assess the decision-making of patients. Whilst Das could be easily adapted to different types of consultation (i.e. face-to-face to telemedicine), further research is needed to compare their efficacy in each setting. Taken together, more real-world studies using validated questionnaires that measure decision-making are needed to establish the efficacy of decision aids during treatment decision-making in haemato-oncology consultations.

BSH23-PO55 | Thrombopoietin receptor agonists in older patients with primary immune thrombocytopenia. From UK adult ITP registry

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Background: Adult primary immune thrombocytopenia (ITP) is a rare acquired immune-mediated disorder. It may have a more aggressive course in older patients with the potential for more complex management due to comorbidities and possible drug interactions. Thrombopoietin receptor agonists (TPO-Ras) have reported overall response rates 50%–90% in ITP. Data from this abstract is being presented on behalf of all contributors to the UK ITP registry.

Aims: To assess outcomes after TPO-RA in patients >60 years.

Methods: Data analysed from the UK adult primary ITP registry (https://www.qmul.ac.uk/itpregistry/) for response and events of special interest.

Results: There were 4810 participants as of 9 December 2022. Of these 569 patients (296M; 273F, median age 73 years (IQR 67.4–78.3) ≥ 60 years) had received TPO-RA therapy (307 eltrombopag and 262 romiplostim) (Table 1). The mean time from diagnosis to first TPO-RA treatment was 1.74 years, with median 3 lines of treatment prior to TPO-RA. The median platelet count up to 2 weeks prior to starting TPO-RA was 25×10^9/L; from 2–4 weeks after TPO-RA was 93.67×10^9/L and at 3–4 months was 97×10^9/L. The complete response (CR) rates at 3–4 months were 48.34%; at 6–7 months were 49.79% and at 1–2 years were 60.97%.

Bleeding events were reduced among all groups following TPO-RA treatment. Prior to starting TPO-RA there were 557 mucocutaneous bleed events and 151 organ bleeds, after there were 352 and 86, respectively. Cardiovascular (CV) risk factors before ITP diagnoses are shown in Table 2. Over 50% had at least one risk factor for CV disease. Before/after TPO-RA there were 66 and 18 arterial thrombotic events (ATE); 24 and 19 venous thrombotic events (VTE), respectively.

This data confirms efficacy of TPO-RA in this group with comparable response rates than in the literature based on platelet count response. There does not appear to be an increase in reported thrombotic events after therapy in this group, despite a large number of comorbidities.

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BSH23-PO56 | What does existing data tell us about care experiences of minority ethnic blood cancer patients?

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Background: There is limited evidence on the care experiences of blood cancer patients from minority ethnic backgrounds. This acts as a barrier to planning developments in services and patient care that equally benefit all.
Method: In this research, we analysed data from the National Cancer Patient Experience Survey (NCPES) 2019, looking at results for blood cancer patients and by ethnicity. We complimented this with qualitative research in the form of 1-1 interviews with healthcare professionals involved with blood cancer care. This was to glean their experiences of meeting the needs of multiethnic patient communities. Interviews were transcribed verbatim and analysed using reflexive thematic analysis.
Results: Data from the NCPES were analysed by 5 ethnic groups: White, Mixed, Asian, Black or other ethnicity. The survey overall received a 61% response rate, of which 88.3% of respondents were of white heritage, with the main minority ethnic respondents identifying as Asian (2.7%) and Black (1.7%). Minority ethnic patients are, therefore, underrepresented in the NCPES overall as compared to cancer prevalence data. Analysis of individual items for blood cancer patients highlighted a trend towards minoritized groups reporting poorer experience of blood cancer care overall, waiting times, understanding of their health condition, involvement in health decision-making, and aspects of personalised care. There were equally several shared areas of positive experiences across ethnic groups such as trust in medical doctors and knowing who to contact with concerns post hospital discharge.
Qualitative interviews with seven professionals identified two themes: language and communication, and clustering of disadvantage. The first theme captured explicit language barriers and how this affects care, as well as the language around blood cancer being complex and interfering with health involvement. The second theme considered the interaction of ethnicity, cultural factors, or education; all of which could help explain some of the care disparities noted in the NCPES data analysed.
Conclusion: Findings suggest that developments in blood cancer care will benefit from robust input from patients from distinct minority ethnic communities. First, to better understand the disparities that emerge in responses to the NCPES, and second, to support co-production of actions aimed at narrowing the care experience gap.

BSH23-PO57 | Outcomes with bosutinib vs. imatinib by Charlson Comorbidity Index in patients with chronic myeloid leukaemia

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Introduction: Impact of comorbidities on efficacy and safety of bosutinib (BOS) vs. imatinib (IMA) in patients with newly diagnosed chronic-phase chronic myeloid leukaemia (CML) was analysed.
Methods: In the BFORE trial (NCT02130557), patients were randomised to 400 mg QD BOS or IMA. Outcomes were assessed after 5 years of follow-up (database lock: 12 June 2020) according to the Charlson Comorbidity Index score without the age component (mCCI). Patients were grouped by mCCI scores of 2 (BOS, $n = 195$; IMA, $n = 196$ [2 untreated]) and >2 (BOS, $n = 73$; IMA, $n = 72$ [1 untreated]).
Results: In patients with mCCI 2 and >2, median treatment duration was 55.1 and 53.4 months with BOS vs. 55.0 and 55.1 months with IMA. Permanent treatment discontinuations in patients with mCCI 2 and >2, respectively, were reported in 36.4% and 50.7% of patients with BOS vs. 45.4% and 32.4% with IMA. Most common primary reasons for permanent discontinuation in patients with mCCI 2 and >2 were adverse events (Aes) with BOS (BOS: 21.0% and 35.6%; IMA: 13.9% and 8.5%) and lack of efficacy (suboptimal response, treatment failure, or disease progression) with IMA (BOS: 6.7% and 2.7%; IMA: 20.6% and 9.9%). In both treatment arms, most discontinuations due to Aes occurred in year 1. Among patients with mCCI 2 and >2, respectively, the proportion who achieved an early molecular response (BCR::ABL1 ≤ 10% at 3 months) was 79.4% and 83.8% with BOS vs. 56.5% and 71.0% with IMA. Major molecular response rates by 60 months were 76.4% and 67.1% with BOS vs. 61.7% and 72.2% with IMA. Corresponding MR4 rates were 61.0% and 50.7% vs. 44.9% and 56.9%. Treatment-emergent Aes (TEAEs) occurred in ≥98.5% of patients in each treatment arm and across mCCI subgroups. Among patients with mCCI 2 and >2, respectively, grade 3/4 TEAEs were observed in 69.2% and 84.9% with BOS vs. 55.2% and 62.0% with IMA. Most common (≥30% in either subgroup) any grade TEAEs with BOS were diarrhoea (79.5% and 63.0%), thrombocytopenia (37.4% and 31.5%), nausea (35.9% and 41.1%), increased ALT (34.4% and 31.5%), and fatigue (17.9% and 30.1%). Most common with IMA were diarrhoea

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(37.1% and 49.3%), nausea (41.2% and 45.1%), muscle spasms (33.0% and 23.9%), and anaemia (17.5% and 36.6%).

Conclusions: Results support the use of BOS in patients with newly diagnosed CML irrespective of baseline comorbidities. mCCI stratification may identify patients who are at higher risk of developing TEAEs and require careful monitoring.

BSH23-PO58 | Clinicopathological features and treatment responses in elderly onset CML in a single tertiary-level hospital in Malaysia

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The clinicopathological features and treatment responses of elderly onset chronic myeloid leukaemia (ECML) patients in resource-limited countries, is not well-described. The aim of this study was to determine the clinicopathological parameters at diagnosis and responses to imatinib in ECML patients (aged ≥65 years old).

A retrospective analysis of consecutive adult patients diagnosed with CML who received imatinib from March 2001 till March 2020 was performed.

Out of 70 consecutive patients with newly diagnosed CML at our institution, there were 10 ECML patients. The median time from diagnosis to last follow-up or death was 71.3 months (range, 20.7–175.2). Six patients were male, and all except one had at least one comorbidity. Six patients had high Sokal score. The median haemoglobin level, white cell counts and platelet counts at presentation were 9.2 g/dL (range, 6.9–14.5), 146×10^9/L (range, 12–477), and 533×10^9/L (range, 405–1297), respectively. Median duration from time of diagnosis to imatinib initiation was 2.2 months (range, 0.3–57.4). All patients received oral hydroxyurea for cytoreduction prior to confirmation of diagnosis and imatinib initiation. Five patients started imatinib at the standard daily dose of 400 mg whilst the other five at the reduced daily dose of 100 mg ($n = 2$), 200 mg ($n = 1$) and 300 mg ($n = 2$), as they were deemed frail. Within the first year of treatment, three of the five patients at 400 mg had dose reduction due to gastrointestinal disturbance and Grade 2 haematologic toxicity. Seven patients achieved complete cytogenetic response (CcyR) with median time to achieve CcyR was 12.2 months (range, 2.9–61.1). Five patients achieved major molecular response (MMR) with median time to achieve MMR was 28.5 months (range, 5.3–128.7). One patient who achieved CcyR had a cytogenetic relapse 12 months later, and subsequently developed blastic phase and died. Two patients died of unrelated causes.

In conclusion, ECML patients in our study had the tendency to take a longer time to achieve CcyR and MMR. This was possibly contributed by the lower starting doses, and subsequent dose reductions in patients who were given initial standard starting dose. Future trials on the most appropriate imatinib dose for this cohort of patients may be beneficial in our local setting to overcome the challenge in managing ECML.

BSH23-PO59 | Rates of hospital admission and mortality in patients with MGUS and COVID-19 infection

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It was established early in the COVID-19 (SARS-CoV-2) pandemic that patients with cancer, in particular haematological malignancy, have worse outcomes than non-cancer patients. Monoclonal gammopathy of undetermined significance (MGUS) is a premalignant condition, which may be linked with immune dysregulation, raising the possibility that patients with MGUS could also have worse outcomes from COVID-19 infection.

Our aim was to identify all local patients with MGUS who had contracted COVID-19 since the start of the pandemic. 120 cases of COVID-19 infection in 114 MGUS patients were identified. Pillar 1 (patients with clinical needs and healthcare workers, $n = 35$) and pillar 2 (wider population, $n = 85$) test results were compared to our patient database to identify cases. Patients were subdivided based on the time period when they acquired COVID-19: wave 1 (February 2020–June 2020, $n = 2$), wave 2 (July 2020–May 2021, $n = 31$) and wave 3 (June 2021–current [data to end April 2022], $n = 87$). No subgroup analysis was carried out for the wave 1 data due to small numbers. Our outcomes were hospital admission and 4-week mortality. We compared our patient outcomes to regional, age-matched, population data for the same time periods.

For rates of hospital admission, we looked at patients aged 65+ years. When looking at all MGUS patients from waves 2 and 3, the rate of hospital admission was not significantly different to the regional population (30.1% vs. 24.1%, $p = 0.11$). In wave 3, the hospital admission rate fell in both groups but was significantly higher in the MGUS patients compared to the regional population (26.9% vs. 16.4%, $p = 0.02$).

For patient mortality, we looked at patients aged 50+ years. The overall 4 -week mortality rate for MGUS patients was higher than the age-matched regional population (5.4% vs. 2.4%, $p = 0.02$). A higher mortality rate was seen in waves 2 and 3 of the pandemic, with wave 3 also showing a statistically significant difference (3.7% vs. 1.2%, $p = 0.02$).

Comparing MGUS patient outcomes from wave 2 to wave 3, there was a modest reduction in hospital admission over this time period (38.1% to 26.9%), with a more notable reduction in COVID-related mortality (9.7% to 3.7%).

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In conclusion, patients with MGUS and COVID-19 seem to have higher rates of mortality, and possibly also hospital admission, compared to the age-matched regional population.

BSH23-PO60 | Dapsone monotherapy is efficacious in paediatric chronic ITP: Experience from LMIC

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Background: Immune thrombocytopenia (ITP) in children is self-limiting with 80%–90% recovering completely within 6 months. Chronic ITP (thrombocytopenia >12 months from diagnosis) affects 10%–20% of patients, some of whom require treatment. Several therapeutic agents are available with varying response rates. Long-term outcome for children with chronic ITP is more favourable when compared with adults.[1]
Objectives: To study the efficacy of Dapsone in the treatment of Paediatric Chronic ITP.
Methodology: Retrospective cross-sectional study was conducted at our centre with a review of case-records of children with Chronic ITP managed at our centre, between 1 January 2012 to 31 December 2017.
Inclusion criteria:
All children aged <18 years with chronic ITP (2019 Updated international consensus report), treated with Dapsone and follow-up data available for >12 months.[2]
Exclusion Criteria:
Children with inadequate follow-up and treatment with other therapeutic agents were excluded.
Results: Of 103 children diagnosed with Chronic ITP, data was not available for 2, 45 had <12 month follow-up, 11 did not receive Dapsone and were excluded from the analysis. Of the remaining 45 who received Dapsone, 34 (75.5%) had an initial response with Platelet increment of $>50\times10^9$/L, whilst 11 (24.5%) did not respond and received other modalities of treatment.
Of 34 children who fulfilled the inclusion criteria, 12 (35%) were boys and 22 (65%) were girls. Median follow-up was 49.6 months (IQR: 31.3, 64.7). Overall 16 children (47.1%) had sustained remission of Platelet count $>50\times10^9$/L for 6 or more weeks without additional rescue therapy (PETIT 2 trial).[3] Whilst the remaining 18 (52.9%) lost response to Dapsone during long-term follow-up, of whom 2 succumbed to intracranial haemorrhage.
Conclusion: Dapsone offers good initial response rates and sustained remission in Paediatric Chronic ITP.
Limitations of the study:

- Retrospective data was analysed and data was not available to assess the mean time to response with Dapsone.
- Cost benefit and Quality of life analysis could not be performed due to a lack of data.

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BSH23-PO61 | Safe and efficacious use of low dose rituximab in postpubertal paediatric patients with immune thrombocytopenia

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Background: Rituximab, a monoclonal antibody targeted against CD20 on B cells, is a potential treatment for patients with persistent or chronic Idiopathic Thrombocytopenia (ITP) who require a rapid response without the need for repeated hospital attendances when used in conjunction with oral dexamethasone. Previous studies in adults have compared the use of rituximab at 375 mg/m^2 weekly for 4 weeks (previous standard dose) against a lower dose of 100 mg weekly for 4 weeks. (Dong et al 2021; Zaja et al 2010). Other studies (Bussel et al 2017) have also used higher doses of dexamethasone at 28 mg/m^2. Few studies have looked at the paediatric population.
Methods: Since 2010, patients at the Royal Manchester Children's Hospital who were postpubertal and were treated for ITP with rituximab have received a lower dose of rituximab 100 mg weekly for 4 weeks. One week adjunct courses of dexamethasone 10 mg/m^2 at week one and week three were administered alongside the rituximab. Data was audited from patient records including platelet counts at baseline, 4 weeks, 8 weeks and 12 weeks, duration of response and any side effects from the rituximab. Partial response was defined as a platelet count $>30\times10^9$/L at 3 months; a complete response was defined at platelet count $>100\times10^9$/L at 3 months.
Results: Eight patients (aged 10–17 years; Female:Male ratio—5:3) were identified who met the criteria. Time since diagnosis ranged from 0.5 to 4.5 years. Six patients had a complete response; two had no response. Of the six responsive patients, three remain in remission >2 years after. The other three patients had response durations of 7 months, 14 months and 1 year 10 months. There were no reported side effects from the rituximab.
Conclusion: Low dose rituximab of 100 mg weekly for 4 weeks combined with adjunct low dose dexamethasone in

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post pubertal patients with ITP showed a response rate of 75% (6/8). This was comparable to efficacy in the adult population, which showed non-inferiority between the two dosing regimens. Advantages of low dose rituximab with low dose dexamethasone include fewer side effects and greater cost effectiveness than with high dose regimens whilst still maintaining equivalent efficacy. This is a small study—ideally the findings would be confirmed with data from a larger cohort. However, it appears likely that low dose rituximab combined with low dose dexamethasone may be considered as a safe and efficacious treatment option for post pubertal paediatric patients with ITP.

BSH23-PO62 | Paediatric ITP at a district hospital: Review of local practice and tertiary paediatric haematology involvement

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Background: Primary immune thrombocytopenia (ITP) is an acquired immune-mediated disorder, which affects 4 in 100 000 children per year in the UK. It is characterised by an isolated platelet count and easy bruising or mucosal bleeding, in the absence of obvious underlying pathology. In most cases, there is spontaneous recovery within 6–8 weeks of diagnosis.
Aims: To audit the clinical presentation, acute management, tertiary referral trends and follow-up arrangements of children with immune thrombocytopenia (ITP) in our District General Hospital (DGH).
Methods: Retrospective observational study of all ITP cases diagnosed between December 2017–2022, identified through our paediatric haematology database.
Results: In the 5-year period, 29 patients (18 male) were diagnosed with ITP (incidence: 6/100 000 children per year). Mean age at diagnosis was 6.7 ± 4.4 years. The commonest initial symptom was easy bruising (90%), followed by petechial rashes (48%) and epistaxis (14%). Safeguarding pathways due to bruising were initiated in 1 case (3.4%). At first presentation, 34% were admitted and 28% had specific treatment. A blood film was always performed, but was reviewed by a haematologist in only 31% of cases. Although all patients were given verbal advice prior to discharge, 45% were not given information leaflets. After the initial presentation, patients had a mean 1.76 attendances in our paediatric assessment unit. In a total of 16 episodes of specific treatment administration within the study period (13 IVIG, 3 Steroids), appropriate indication was documented in 80%, and satisfactory response in 69%.
Follow-up for the majority (72%) of patients was by paediatric consultants with haematology interest, whereas for 28% was in a general paediatric clinic. Paediatric Haematology Clinical Nurse Specialist involvement was higher in the former group (100% vs. 50% $p = 0.02$). Our tertiary paediatric haematology teams were involved in 62% of cases; 55% were reviewed at a tertiary outpatient clinic (first review at a mean 5.1 months), and in 45% tertiary involvement was limited to verbal advice.
Conclusion: We note significant variability in several aspects of ITP management, including blood film reviews, written information provision, specific treatments administration and outpatient follow-up arrangements. To address this, we propose a standardised clinical care pathway (including a predischarge checklist) for paediatric ITP. When possible at a DGH setting, ITP management should be overseen by paediatricians with haematology interest, with clear communication links and referral pathways to the regional paediatric haematology services.

BSH23-PO63 | Predicting relapsed/refractory disease in Thai childhood haemophagocytic lymphohistiocytosis: 13-years single institute retrospective study

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Background: Haemophagocytic lymphohistiocytosis (HLH) is one of the life-threatening diseases in paediatric patients. In Thailand, recent multicentre study revealed that 20.5% of cases died within 1 month after diagnosis. Although there were several proposed prognostic factors of early mortality in paediatric HLH, risk factors of relapsed or refractory disease in children at the time of diagnosis were scarce. This study aimed to determine the prognostic indicators for relapsed or refractory disease in paediatric HLH (R/R HLH).
Methods: Infants and children under 18 years of age who were diagnosed with HLH according to HLH-2004 criteria, MAS-HLH criteria for rheumatologic diseases, or H-score undergoing treatment in Chiang Mai University hospital between 2010 and 2022 were included. Demographic data, clinical characteristics, and laboratory parameters were retrospectively reviewed.
Results: Relapsed or refractory HLH were found in 25 patients (39.0%) of total 64 childhood HLH. All patients with primary HLH developed R/R HLH. The most common secondary HLH in this study was infection-associated hemophagocytic syndrome (IAHS), which comprised of 35 cases. 31.4% of patients with IAHS had relapsed or refractory disease. Although Epstein Barr Virus was the dominant pathogen in IAHS, R/R HLH occurred in only 1 out of 8 patients. R/R HLH was found in 60% and 31.3% of patients with malignancy-associated hemophagocytic syndrome (MAHS) and macrophage activation syndrome (MAS-HLH), respectively. 2 patients with primary HLH and 1 patient with MAHS had a molecular diagnosis of RAB27A, LYST, and HAVCR2

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mutation, which solely had R/R HLH. In terms of prognostic indicators, the univariable analysis showed that younger age, overt DIC, higher paediatric Sequential Organ Failure Assessment score (pSOFA) score, hyponatremia, elevated AST, total and direct bilirubin were possible risk factors of R/R HLH. In multivariable logistic regression analysis, there were no significant predicting factors demonstrated in this cohort. In contrary, subgroup analysis of childhood IAHS revealed that pSOFA score of ≥10 was an independent risk factor for R/R IAHS with odds ratio of 44.06 (95% confident interval 1.43–1354.28; $p = 0.03$).

Conclusions: Children with IAHS who initially had pSOFA score of ≥10 were risk of relapsed or refractory disease. This finding was consistent with recently published consensus-based guidelines from The Histiocyte Society in terms of severity grading, which using pSOFA score. Further management should be evaluated in this setting.

BSH23-PO64 | Children immune thrombocytopenia: Patient information quality improvement project

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Introduction: Immune Thrombocytopenia (ITP) is characterised by a transient or persistent reduction in platelet count, which increases the risk of bleeding. It affects 4/100 000 children annually with 25% of that developing chronic ITP. Whilst symptoms monitoring and trauma prevention are crucial steps in improving the prognosis of ITP patients, there are currently insufficient resources catered for children and teenagers. This potentially limits young patients' understanding of ITP and subsequently hinders them from making appropriate health decisions when not under supervision. This quality improvement project aimed to create patient information leaflets specifically designed for paediatric patients. This in hope can provide accurate whilst easily comprehensible information about ITP to improve young patients' understanding of the disease, treatment, and red flags to be aware of.

Methods: Questionnaires were sent to 48 ITP patients aged 8–17. Patients were given a preintervention survey to evaluate their baseline knowledge regarding the disease. After reading the information leaflets, the patients completed a postintervention survey to evaluate the effectiveness of leaflets in providing new information.

Results & Discussion: Among the 8 respondents, more than one-third stated that there is an insufficient amount of ITP information catered for them. In addition, one respondent (aged 12) did not know the differences between acute, persistent and chronic ITP. This shows the need for additional patient resources among the young age group. Regarding the leaflets, 75% of respondents felt that they have gained at least a satisfactory amount of additional knowledge from them. Moreover, all respondents felt that the content has high comprehensibility. This shows that the leaflets are providing a good amount of information effectively. Although the majority of respondents find them attractive, a few older respondents (aged 11–17) felt that they are not motivated to take the leaflets with the cover design. A possible reason might be that teenagers do not find the cartoon-style layout attractive.

Conclusion: This cycle of quality improvement project identified the lack of ITP patient resources for children and teenagers. It has also shown that the customised leaflets are useful measures to provide information for the targeted patient group. Lastly, it is worthwhile to consider adopting a different version of leaflets for teenagers specifically.

BSH23-PO65 | Daratumumab in refractory immune thrombotic thrombocytopenia purpura: The UK experience

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Immune-mediated thrombotic thrombocytopenia purpura (TTP) is a life-threatening thrombotic microangiopathy, defined by severe deficiency of ADAMTS13 activity due to ADAMTS13 antibodies. Some patients are refractory to standard treatment with plasma exchange, corticosteroids, rituximab and caplacizumab, with persistent inhibitors and no improvement in ADAMTS13 activity. This represents a challenging clinical situation, where the optimal treatment is not clear.

In 2022, a Swiss case series was published of two patients with relapsed/refractory immune TTP who were treated with daratumumab (an anti-CD38 antibody). In both cases, eradication of the patients' ADAMTS13 inhibitors and normalisation of their ADAMTS13 activity was reported within 2 weeks of starting treatment with daratumumab. We report the outcomes for the first two patients treated with daratumumab for refractory TTP in the UK.

Both patients had previously been treated with plasma exchange, corticosteroids, rituximab, caplacizumab and mycophenolate. Patient one had also received bortezomib, and patient two ciclosporin. In both cases the patients still had persistent high titre inhibitor and undetectable ADAMTS13 activity more than 3 months after reaching clinical remission from TTP.

Patient one was given two doses of weekly subcutaneous daratumumab 1800 mg (fixed dose). Patient two was given four doses of weekly intravenous daratumumab 16 mg/m^2.

Patient one reached a partial ADAMTS13 remission 32 days after daratumumab. However, the patient's ADAMTS13 inhibitor was falling prior to starting daratumumab and we

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consider it likely that the improvement in ADAMTS13 activity was a delayed response to previous immunosuppressant therapy rather than a response to daratumumab. Patient two had no improvement in ADAMTS13 activity following daratumumab.
These results stand in contrast to those previously published, where both patients had a rapid improvement in ADAMTS13 activity following daratumumab administration.
Further data will be required to understand the role of this approach in the treatment of relapsed/refractory immune-mediated TTP.

BSH23-PO66 | Oral and intravenous iron replacement therapy on maternal ferritin and hepcidin levels

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Anaemia is seen in 30% of pregnant women. Almost 1/3rd fail to respond to oral iron due to: gastrointestinal side effects, compliance, dose inadequacies or lack of response. Lack of response could be improved with a better understanding of maternal iron metabolism including the role of hepcidin, which is the central iron storage regulator. This could enable novel personalised treatment pathways.
Objectives: (1) Assess maternal response to oral versus intravenous (IV) iron therapy by testing a range of haematological/iron measures; (2) Evaluate the predictive effect of biomarkers of iron (hepcidin) for outcomes.
Pregnant women (n = 61) with iron deficiency anaemia (IDA) were prescribed iron based on British Society of Haematology 2012 guidelines. Response was determined by normalisation of gestation adjusted haemoglobin (HB) and/or a 10 g/L increment in HB at a follow-up visit. Serum from each visit was analysed for ferritin and hepcidin by quantitative-sandwich-enzyme-linked-immunosorbent assay. Paired t-test was used and $p < 0.05$ was considered statistically significant. Linearity was determined by Pearson's correlation coefficient.
Results were divided into two groups:

1. R1—responder to oral iron treatment. This group was then subdivided into R1X for participants who attended all four visits (n = 15) these women failed to respond at the first follow-up visit and required advice on administration or dose adjustment; and R1Y for participants who attended three visits (V1, V2 and puerperium) (n = 35) these women responded at the first follow-up visit.
2. R2—those who did not respond to oral treatment and received IV iron at V3 (n = 11).

There was a significant change in R1Y hepcidin and ferritin levels between all visits ($p < 0.005$), associated with a positive correlation (r = 0.99, $p \leq 0.01$). In R1X, only hepcidin levels had significant increase at puerperium compared to V1 and V2 ($p < 0.005$). Both analytes had a strong positive correlation (r = 0.99, $p \leq 0.01$). For R2 group, there was a statistically significant increase in hepcidin and ferritin levels at puerperium ($p < 0.05$) and ($p < 0.001$), respectively, compared to V1–V3 for hepcidin and all visits for ferritin. Interestingly, there was a significant ($p < 0.05$) decrease in hepcidin levels at V3 versus V2 ($p < 0.05$) with no correlation between these (r = 0.756, p = 0.455).
Our work provides insight into the problems of poor responses to oral iron. We identified that with a decrease in hepcidin associated with an increase or constant levels of ferritin may indicate non-responsiveness, and suggest the need for modification to the treatment pathway. Further studies are necessary to distinguish changes between trimesters.

BSH23-PO70 | Sero-prevalence of transfusion-transmittable infections among blood donors in Sudan: A national study

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Blood transfusion is an important procedure that saves the lives of millions of people. However, this procedure can lead to the transmission of infectious diseases such as hepatitis B virus (HBV), hepatitis C virus (HCV), human immunodeficiency virus (HIV), malaria, syphilis, and others. The aim of this study was to identify the Sero-prevalence of transfusion-transmittable infections among blood donors in Sudan.
This is a cross-sectional study, it was conducted in different areas in Sudan (Khartoum state, Al Gezira state, Kassala state, Kordofan state, and White Nile state) during the period from April 2021 to September 2022. Nine blood banks (including National Central Laboratory) have participated in this study and 2981 blood donors were included. A structured questionnaire was used to collect the data along with blood screening. The Statistical Package for Social Sciences (SPSS) version 26 was used to analyse the data and bivariate and multivariate logistic regression models were used. A p-value <0.05 was considered statistically significant. Ethical approval was obtained from the Sudanese Federal Ministry of Health.

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This study found that the total sero-prevalence of transfusion-transmittable infections was 10.2%. The most common transfusion-transmittable disease was HBV (5.8%) followed by syphilis (3.7%), malaria (2.1%), HCV (0.9%), and HIV (0.8%). The following factors were found to be significantly associated with transfusion-transmittable infections: giving care to a patient with HIV or hepatitis (adjusted odd ratio (AOR) = 3.11; 95% confidence interval (CI): 1.52, 7.3), unsafe sex (AOR = 2.81; 95% CI: 1.47, 4.22), dental procedure (AOR = 2.64; 95% CI: 1.21, 4.54), and illiteracy (AOR = 2.31; 95% CI: 1.27, 4.23).

This is the first national study conducted in Sudan regarding the Sero-prevalence of transfusion-transmittable infections among blood donors. This study confirmed that HBV is the most common transfusion-transmittable infection among blood donors in Sudan. In addition, the following factors were found to be associated with transfusion-transmittable infections: giving care to a patient with HIV or hepatitis, unsafe sex, dental procedures, and illiteracy. The awareness of the population should be raised regarding the control and prevention of sexually transmitted diseases.

Keywords: Sero-prevalence, transfusion-transmittable infections, blood donors.

BSH23-PO71 | Serum extracellular vesicle miRNA profiling to determine extracorporeal photopheresis response in graft-versus-host disease

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Graft-versus-host disease (GvHD) is a life-threatening complication of allogeneic haematopoietic stem cell transplantation (HSCT), where donor material recognises host tissue as foreign, resulting in host tissue destruction. Of those receiving HSCT 40%–70% develop GvHD, leading to post-treatment mortality in 40%–60% of patients.

With steroids being the first-line treatment, it comes with significant adverse effects as well as the potential for steroid refractoriness. Extracorporeal photopheresis (ECP) therapy is a safe alternative, though the precise mechanism behind it is unknown, and there are no biomarkers to determine response. Our aim was to investigate the role of serum extracellular vesicles (Evs) in GvHD and ECP therapy.

Serum samples were acquired from n = 12 paediatric patients undergoing ECP therapy at the Great North Children's Hospital before every weekly cycle of treatment. Evs were isolated from 4 sequential time points (TP) by precipitation (Total Exosome Isolation Reagent) for 6 complete (CR) and 6 partial/non-responders (PR). Particle size and concentration were measured using Nanoparticle Tracking Analysis (NanoSight). Total RNA was extracted, concentrated, quantified, and microRNA expression profiling was performed using NanoString technology (n = 800 microRNA).

A significant decrease in EV modal size was observed for PR from TP1 to TP2. Furthermore, preliminary analysis using DESeq2 showed that in a PCA plot three groups formed clusters; PR and two separate clusters of mostly CR with some PR. In this analysis it could also be observed that treatment cycle was not a driving factor as there was no treatment cycle-based clustering. After pulling out the miRNAs, which caused the difference between these groups there were 8 that had a p-adjusted value <0.05 namely, I-miR-23a-3Ihsa-miR-144-3p, I-miR-548aIIhsa-miI78e, hsaIR-21-5p, I-let-7i-5p, hsIiR-579-3pInd hsa-miR-1283. A mix of predicted and experimentally validated target databases will be used to investigate the downstream effect of the up- or downregulation of these miRNAs. Preliminary analysis shows the potential for miRNA biomarkers as an indicator of ECP response as well as shedding light on the mechanisms behind response.

BSH23-PO72 | Efficacy of subcutaneous epcoritamab versus chemoimmunotherapy and axi-cel in R/R LBCL: An indirect comparison

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Introduction: Epcoritamab is a novel, subcutaneous CD3×CD20 bispecific antibody showing potent single-agent efficacy in patients with relapsed/refractory (R/R) large B-cell lymphoma (LBCL) and a manageable safety profile. The phase 1/2 EPCORE NHL-1 trial (NCT03625037) evaluated epcoritamab in R/R LBCL patients who received ≥2 prior lines of systemic therapy. No head-to-head data have compared epcoritamab with chemoimmunotherapy (CIT) or chimeric antigen receptor T-cell (CAR-T) therapy. We conducted an unanchored, matching-adjusted, indirect comparison of the efficacy of epcoritamab with that of CIT and axicabtagene ciloleucel (axi-cel).

Methods: Published data on overall response rate (ORR), complete response rate (CRR), and overall survival (OS) for CIT (SCHOLAR-1 trial) and axi-cel (ZUMA-1 trial) were used to conduct the comparisons versus individual patient-level data (IPD) for epcoritamab (EPCORE NHL-1 trial). Patients with prior CAR-T therapy enrolled in EPCORE NHL-1 were excluded because SCHOLAR-1 and ZUMA-1 included only patients without prior CAR-T therapy. Analyses adjusted for imbalances in baseline characteristics between IPD from EPCORE NHL-1 and aggregate data from SCHOLAR-1 and ZUMA-1. Propensity score weights resulting from the adjustment were applied to estimate risk differences for ORR

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and CRR, and weighted Cox proportional-hazards models were used to estimate OS hazard ratio (HR).

Results: Patients without prior CAR-T therapy enrolled in the EPCORE NHL-1 trial (N = 96) were mostly men ≥65 years who had stage III or IV disease, primary refractory disease, and were refractory to ≥2 consecutive lines of therapy. After adjustments, patients receiving epcoritamab had significantly higher response rates than CIT (ORR: 74.5% vs. 34.1%; difference in ORR, 40.4%; $p < 0.001$; CRR: 52.4% vs. 12.1%; difference in CRR, 40.4%; $p < 0.001$). No statistically significant differences in response rates were observed between patients treated with epcoritamab and axi-cel. Patients treated with epcoritamab also demonstrated statistically significant improvements in OS versus CIT (HR: 0.283; 95% CI: 0.161, 0.497; $p < 0.001$). Median OS for CIT was 5.4 months, whereas median OS for epcoritamab had not been reached (median follow-up: 10.7 months). There was no statistically significant difference in OS between epcoritamab and axi-cel.

Conclusions: Findings from this cross-study comparison suggest that epcoritamab has significantly higher response rates, including CRR, and OS versus CIT and no statistically significant difference in efficacy versus axi-cel. As an unmet need exists for additional therapies in R/R LBCL, these findings are encouraging and underscore the potential of epcoritamab as a core therapy for these patients.

BSH23-PO73 | Re-staging bone marrow biopsy is unnecessary in the management of follicular lymphoma

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Aims: Bone marrow trephine biopsy (BMA/T) is a standard part of restaging for follicular lymphoma patients with marrow involvement at baseline. In this study, we aimed to assess whether this procedure was being performed appropriately, and whether it prompted changes in treatment.

Methods: 230 patients were diagnosed with follicular lymphoma between 2010 and 2019 at our institution. 118 were excluded since, 31% had no restaging marrow, 29% had not received chemotherapy and 28.5% had an alternative diagnosis on bone marrow histology with 16.9% showing diffuse large B-cell lymphoma. Moreover, 9% lacked the necessary imaging and 2.5% passed away mid-treatment. For the remaining 112 patients we assessed:

- Findings on pretreatment BMA/T
- Presence of signs suggestive of marrow involvement on initial imaging
- Whether or not a repeat BMA/T was performed after 4 cycles, and the findings
- Whether or not the repeat BMA/T resulted in a change in treatment
- End of treatment imaging result
- Any follow-up imaging.

Results: When looking at patients with marrow involvement who had both a pretreatment PET/CT and BMA/T we found that 51.3% were detected by BMA/T but not PET/CT, 12.8% were detected by PET/CT but not BMA/T, and 35.9% were detected by both.

42.9% of included patients underwent a repeat BMA/T after 4 cycles. Of these, 83% had involvement at baseline, with 12.8% of these showing persistent involvement. 17% of patients undergoing the procedure did not have initial involvement. Conversely 30% of patients who showed initial involvement did not have a repeat BMA/T. In no case did the findings on repeat BMA/T result in a change in management. On assessment of relapse/progression of disease we found that 80.8% of cases occurred in patients with baseline marrow involvement, of which 61.9% showed no involvement on repeat, 9.5% showed persistent involvement, and 28.6% did not undergo repeat marrow. Only 68.8% of patients had a follow-up PET/CT scan. The remaining 31.2% were not scanned due to there being no evidence of relapse on follow-up outpatient visits.

Conclusion: We found that repeat BMA/T did not influence treatment choices. Whilst infiltration at baseline was seen to be more prevalent in patients who subsequently relapsed, there was no clear evidence that persistence of marrow involvement on interim BMA/T made this more likely. Lastly, in follicular lymphoma BMA/T is more sensitive than PET/CT for detecting marrow involvement.

BSH23-PO74 | Association of metabolic tumour volume and clinical outcomes in second line relapsed/refractory large B-cell lymphoma (ZUMA-7)

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Background: In ZUMA-7, axicabtagene ciloleucel (axi-cel) was superior to standard of care (SOC) across common prognostic subgroups, including tumour burden (TB, calculated

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by the sum of product diameters [SPD]) and lactate dehydrogenase (LDH; Locke et al. ASCO 2022. Abstract 7565). TB can also be measured by metabolic tumour volume (MTV). Here, we present clinical outcomes in ZUMA-7 by MTV.

Methods: MTV was obtained from the attenuation-corrected whole-body FDG PET scans at screening. Whole tumour volumes of interest were placed on tumours using a predefined, semiautomated approach, with subsequent radiologist-defined adjustments. MTV was calculated as the number of volumetric pixels with standardised uptake value (SUV) measurements between 41% and 100% of tumour SUVmax and reported as MTVtotal (mL) per patient. Associations were assessed using descriptive statistics ($p < 0.05$ was considered significant).

Results: MTV was evaluated in 341 patients (axi-cel: 176 and SOC: 165). Median MTV was 228.1 mL in axi-cel patients and 231.9 mL in SOC patients. Median MTV was higher in patients with elevated (n = 185; >median) versus normal (n = 156; ≤median) LDH (371.2 versus 123.9 mL). MTV was positively associated with SPD (n = 308; Spearman correlation, 0.523) and LDH (n = 341; Spearman correlation, 0.452). Axi-cel event-free survival (EFS) was superior to SOC for low (≤median) and high (>median) MTV (HR, 0.423 and HR, 0.417, respectively). Axi-cel EFS trended shorter in patients with high MTV (HR, 1.441) and EFS was shorter in SOC patients with high MTV (HR, 1.486). Median MTV was higher for axi-cel patients who experienced Grade ≥3 neurologic events (NEs; n = 36) versus patients who experienced Grade 1–2 or no NEs (n = 134; 320.9 versus 193.9 mL). Median MTV was higher for axi-cel patients who experienced Grade ≥3 cytokine release syndrome (CRS; n = 11) compared with patients who experienced Grade 1–2 or no CRS (n = 159; 582.9 versus 203.3 mL).

Conclusions/Summary: Similar to associations of efficacy with SPD and LDH in second line LBCL (Locke et al. ASCO 2022. Abstract 7565), high MTV was associated with superior outcomes in patients treated with axi-cel versus SOC. Whilst tumour burden per SPD did not seem to impact axi-cel outcomes in ZUMA-7 (Locke et al. ASCO 2022. Abstract 7565), high MTV was associated with poorer outcomes with axi-cel versus low MTV, and rates of Grade ≥3 NEs and CRS were associated with higher MTV, suggesting that MTV is a more accurate and sensitive measure of TB. Nevertheless, axi-cel was superior to SOC irrespective of MTV subgroup, including among patients with high MTV.

BSH23-PO75 | Immune response In Waldenström macroglobulinaemia patients after BTK inhibition

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The aim of this study, funded by an Early-Stage Research Grant from the British Society of Haematology, was to evaluate the immune microenvironmental changes in the bone marrow, following BTK-inhibitor (BTKi) treatment in Waldenström Macroglobulinaemia (WM) and correlate with molecular profiling. The bone marrow tumour microenvironment (TME) in WM has been shown to have a role in WM disease progression and therapy resistance; however, the effect of BTKi on the TME has been under explored.

The targeted gene expression Oncomine™ Immune Response Research assay, was performed using an Ion S5™ sequencing system (Thermo Fisher Scientific) to interrogate 395 genes involved in the tumour microenvironment. RNA was extracted from FFPE bone marrow trephine biopsies for 31 samples from 15 patients plus 1 control. Each patient had a sample prior to BTKi treatment and 12 months after starting BTKi treatment. Correlation with existing MYD88 mutation status and differential analysis between the two time points was performed using the Affymetrix™ Transcriptome Analysis Console software.

Of the 14 samples that passed the quality control threshold for this analysis, there were 9 MYD88L265P positive samples and 5 MYD88L265P negative samples. Comparative analysis of MYD88L265P positive samples before and after BTKi treatment, showed upregulation of genes involved in the lymphocyte infiltrate pathway, checkpoint pathway and NK cell markers. There was downregulation of genes associated with type 2 interferon signalling and TCR co-expression. Further analysis of all MYD88L265P positive pre and post BTKi samples showed the most significant upregulation in NK cell marker, NCR1 and downregulation of CXCL13 and CD8B post BTKi treatment (p = 0.03). Differential expression of NCR1, which encodes for a cytotoxicity-activating receptor on NK cells, has not previously been associated with Waldenström macroglobulinaemia but is consistently upregulated post BTKi in our cohort. Downregulation of CXCL13, which acts as a B lymphocyte chemoattractant, has previously been shown in both WM and chronic lymphocytic leukaemia after Ibrutinib treatment, minimising the migration of tumour cells and disrupting the TME.

This pilot study has shown regions of the WM immune microenvironment with significant changes in gene expression after BTKi therapy, namely in pathways involving NK cells and type 2 interferon signalling. A larger cohort with paired samples pre and post BTKi treatment is under investigation.

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BSH23-PO76 | Improved overall survival in stage III/IV classical Hodgkin lymphoma: An updated analysis of ECHELON-1

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Background: Trials involving various first-line treatment strategies for classical Hodgkin Lymphoma (cHL) have not shown an overall survival (OS) advantage compared with doxorubicin, bleomycin, vinblastine and dacarbazi'e (ABVD). A '5-years' follow-up of ECHELON-1 (NCT01712490), first-line brentuximab vedotin, doxorubicin, vinblastine, and dacarbazine (A+AVD), demonstrated long-term progression-free survival (PFS) benefits versus ABVD with a manageable long-term safety profile, in patients with previously untreated stage III/IV cHL. We report an OS analysis from ECHELON-1 after a median of 6 years of follow-up (data cut-off 1 June 2021).

Materials and Methods: A total of 1334 patients were randomly assigned to receive up to 6 cycles of A+AVD (n = 664) or ABVD (n = 670) on days 1 and 15, every 28 days. The key secondary endpoint was OS (prespecified, event-driven, alpha-controlled analysis) in the intent-to-treat population. PFS per investigator was assessed at 6-years' follow-up. Deaths during follow-up, including reported investigator-assessed causes of death, were summarised.

Results: At a median follow-up of 73 months, 39 and 64 deaths had occurred in the A+AVD and ABVD arms, respectively, (hazard ratio [HR] 0.59; 95% confidence interval [CI] 0.40–0.88; p = 0.009); 6-year OS estimates were 93.9% (95% CI 91.6–95.5) and 89.4% (95% CI 86.6–91.7), respectively. In a multivariable analysis adjusting for prespecified baseline demographic and disease factors, OS benefit with A+AVD versus ABVD was preserved (HR 0.53; 95% CI 0.34–0.83). PFS was longer with A+AVD versus ABVD (HR 0.68; 95% CI 0.53–0.86). Subsequent therapy use was less frequent with A+AVD versus ABVD (135 [20%] versus 157 [24%]), including fewer autologous (44 [7%] versus 59 [9%]) and allogeneic stem cell transplants (4 [<1%] versus 12 [2%]), but use of radiation was similar (55 [8%] versus 58 [9%]). There were fewer second malignancies (23 versus 32) and fewer deaths related to cHL or treatment complications (32 versus 45) or second malignancies (1 versus 11) reported with A+AVD than ABVD; treatment-related deaths were comparable (8 versus 7). Fertility was not formally assessed; however, a total of 195 pregnancies were reported among patients and their partners (A+AVD 114; ABVD 81). Peripheral neuropathy improved or resolved in most patients at last follow-up; incidence was higher with A+AVD (incidence: A+AVD 443 [67%], ABVD 286 [43%]; improvement/resolution: A+AVD 379 [86%], ABVD 249 [87%]).

Conclusions: Patients with stage III/IV cHL who received A+AVD had a survival benefit over those who received A+AVD with a manageable long-term safety profile, consistent with prior reports.

BSH23-PO77 | Outcomes of full-body computed tomography (CT) In patients referred with sweats

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Referrals to clinical haematology are frequently made for patients with a history of night or drenching sweats. We assessed the outcomes of full-body computed tomography (CT) requests where patients' referral histories included 'sweats'.

Methods: We retrospectively reviewed CT reports where 'sweats' were mentioned in the clinical request to rule out malignancy in 2016 and 2019. The following parameters were recorded: sweats in isolation, sweats with B symptoms,

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sweats with lymphadenopathy or splenomegaly, sweats with 'other' symptoms, diagnosis, age and sex. B symptoms were defined as fever above 38°C and weight loss of >10% of body mass in the past 6 months.

Results: A total of 140 patients were identified, 70 from 2016 and 70 from 2019. 71 were male and 69 were female with a mean age of 56 years. 42 cases (30%) were 2 week wait referrals. Malignancy was detected in 17 patients (12%); 11 (8%) were haematological (Lymphoma (n = 8), myeloid neoplasms (n = 2), acute leukaemia (n = 1)). The 8 lymphoma diagnoses included 3 diffuse large B-cell lymphomas (DLBCL), 2 classical Hodgkin lymphoma (cHL), 2 hairy cell leukaemias (HCL) and 1 low-grade mature B-cell lymphoma. Referral information for the 8 lymphomas included palpable lymphadenopathy or spleen (n = 5, 1 DLBCL, 1 cHL, 1 low grade, 2 HCL), weight loss, fevers and cytopenias (2 DLBCL and1 cHL). Non-haematological malignancies (lung, pancreas, liver, bladder and melanoma) were identified in 6 patients (4%). Lymphadenopathy was palpable in half: supraclavicular lymphadenopathy in 2 small cell lung cancers and an inguinal node in a patient with bladder cancer. Significant weight loss was present in the other 3 cases.

99 CTs (71%) were performed in patients with sweats plus other symptoms. The most common accompanying symptoms were fatigue/tiredness, itching, rash, cough, breathlessness and abdominal pain. Isolated sweats accounted for 16 patients (11%). No malignancies were identified at the time or have subsequently been diagnosed in these patients.

Conclusion: Haematological malignancies, principally lymphoma, were identified in 8% of patients referred through for cross-sectional imaging with sweats in conjunction with lymphadenopathy/splenomegaly and/or cytopenias. Non-haematological cancers were less commonly identified but still only found with clinical signs or B symptoms. Clinical history, examination and appropriate blood tests may adequately screen patients with a history of sweats preventing unnecessary radiological investigations. This supports NICE guidance, which stipulates that unexplained lymphadenopathy or splenomegaly should form the cornerstone for suspecting haematological malignancy, which is then supported by constitutional symptoms.

BSH23-PO78 | Subcutaneous epcoritamab versus polatuzumab-based regimens in R/R LBCL: An indirect comparison

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Introduction: Epcoritamab is a novel, subcutaneous CD3×CD20 bispecific antibody showing single-agent efficacy in patients with relapsed/refractory (R/R) large B-cell lymphoma (LBCL), and a manageable safety profile. The single-arm phase 1/2 EPCORE NHL-1 trial (NCT03625037) evaluated epcoritamab in patients with R/R LBCL who received ≥2 prior lines of systemic therapy. No head-to-head data are available comparing the efficacy of epcoritamab with polatuzumab vedotin combined with bendamustine and/or rituximab (pola+B/R). We conducted an unanchored, matching-adjusted indirect comparison (MAIC) of the efficacy of epcoritamab reported in the EPCORE NHL-1 trial with that of pola+B/R reported in Liebers et al, 2021, which is a real-world observational study in the 3 L setting.

Methods: Published data on overall response rate (ORR), complete response rate (CRR), and overall survival (OS) for pola+B/R were used from the salvage treatment cohort (Liebers et al) to conduct the MAIC versus individual patient-level data (IPD) for epcoritamab (EPCORE NHL-1 trial). Most patients in Liebers et al were treated with pola-based therapy (pola+BR, 59.3%; pola+R without bendamustine, 37%; other pola-based regimens, 3.7%) in 26 German centres. For epcoritamab, response was based on Lugano criteria and assessed by an independent review committee. For pola+B/R, response was investigator-assessed using CT/clinical judgement, with CT scans available in 75.9% of patients. Analysis was conducted by adjusting for imbalances in baseline characteristics between IPD from EPCORE NHL-1 and aggregate data from Liebers et al. Propensity score weights resulting from the adjustment were applied to estimate differences in risk for ORR/CRR, and weighted Cox proportional-hazards models were used to estimate the OS hazard ratio (HR).

Results: LBCL patients in the EPCORE NHL-1 trial (N = 157) were younger and a greater proportion had prior chimeric antigen receptor T-cell therapy compared with the Liebers cohort (N = 54); other baseline characteristics were comparable between the two groups. After adjustment, compared with pola+B/R, patients treated with epcoritamab (n = 96) had higher response rates than pola-based regimens (ORR: 67.6% vs. 48.1%; difference, 19.5%; p = 0.0191; CRR: 44.2% vs. 14.8%; difference, 29.4%; p < 0.001). Patients treated with epcoritamab had improved OS versus pola+B/R (HR: 0.379; 95% CI: 0.239, 0.600; p < 0.001).

Conclusions: These findings show that epcoritamab provides statistically significantly higher response rates and OS compared with pola-based regimens as used in the real world, and epcoritamab has the potential to be a core LBCL therapy.

Funding and Sponsorship: Epcoritamab is jointly developed by Genmab A/S and AbbVie Inc.; AbbVie and Genmab are sponsoring this study.

BSH23-PO79 | Intensive chemotherapy and autologous stem cell consolidation for Bing-Neel syndrome—The UCLH experience

Dr Jahanzaib Khwaja[1], Nicole Japzon, Dr Ali Rismani, Dr Aisling Carr, Professor Michael Lunn, Dr Charalampia

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Background: Bing-Neel syndrome (BNS) is a complication of lymphoplasmacytic lymphoma (LPL) comprising LPL infiltration in the central nervous system (CNS). There is no standard of care; both chemoimmunotherapy (CIT) and BTK inhibitors are effective. The use of intensive CNS-penetrating CIT is well-described in aggressive lymphomas, but there are limited data in BNS and the role of BTKi is uncertain, particularly in the setting of progressive neurological compromise.
Aims: To review the outcomes in our BNS cohort who have received intensive CIT and ASCT consolidation.
Methods: Data from BNS patients referred between 2013 and 2021 to our academic Neurohaematology centre were retrospectively reviewed.
Results: Of 43 symptomatic patients, median age at BNS diagnosis was 66 years (range 43–85). All had leptomeningeal disease. Seven patients with progressive clinical features had ASCT consolidation (median age 59 years [range 43–68]): three patients had additional parenchymal intracranial disease on MRI and two patients also had peripheral nerve involvement, including one patient with suspicion of high-grade transformation (HGT) on CSF. Most patients (4/7) were treatment-naïve and three patients had prior therapy. All patients had 2–4 cycles of intensive CIT (six: rituximab/methotrexate/cytarabine ± thiotepa; one: IDARAM/ESHAP) due to clinical severity.
All patients showed clinical and imaging response. In six patients, consolidation ASCT was delivered in first response at a median of 11 months (range 4–22) from BNS diagnosis. The other patient achieved complete response after intensive CIT so no further treatment was planned; however, this patient relapsed after 2 years so commenced Ibrutinib with CNS progression and, therefore, proceeded with ifosfamide-based salvage (rituximab/ifosfamide/etoposide) and underwent ASCT thereafter.
In the ASCT group, 6/7 had BCNU thiotepa and one had cyclophosphamide-TBI conditioning. Engraftment occurred at a median of 12 days (range 9–13) for neutrophils and 11 days (range 10–14), for platelets. Duration of hospitalisation was 21 days (range 19–41), critical care admission and 100-day non-relapse mortality was 0%. At a median follow-up was 11 months (1–96) from ASCT, 2/7 died with relapsed lymphoma >18 months from ASCT (1/7 HGT), which was refractory to salvage chemotherapy. 5/7 patients were alive: 1/7 experienced systemic relapse (post cyclophosphamide-TBI-ASCT) and commenced ibrutinib; 1/7 experienced CNS progression and commenced zanubrutinib.
Summary: In our selective cohort of relatively young BNS patients with adverse clinical features (extensive CNS burden; high-grade suspicion; BNS post-BTKi/CIT failure) who proceeded with intensive chemotherapy and ASCT consolidation, treatment-related morbidity and non-relapse mortality were low and clinical response was notable.

BSH23-PO80 | Pirtobrutinib, a highly selective, non-covalent BTK inhibitor in Richter transformation: Results from the BRUIN study

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Background: Richter transformation (RT) is an aggressive diffuse large B-cell lymphoma that can occur in 10% of patients with chronic lymphocytic leukaemia (CLL). RT has no approved therapies, poor prognosis, and an estimated median overall survival (OS) of 3–11 months. Pirtobrutinib is a well-tolerated highly selective, non-covalent (reversible) Bruton tyrosine kinase inhibitor (BTKi) with favourable oral pharmacology and promising efficacy in patients with poor prognosis B-cell malignancies following prior covalent BTKi therapy. We now report on RT patients from BRUIN.
Methods: Patients previously treated for B-cell malignancies were eligible for pirtobrutinib monotherapy in dose escalation/expansion within the phase 1/2 BRUIN study. Key endpoints: investigator-assessed overall response rate (ORR) and duration of response (DoR) per Lugano 2014 criteria,

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and safety. The response-evaluable cohort consisted of patients with RT who completed either first response assessment or discontinued therapy. The safety cohort consisted of patients who received ≥1 dose of pirtobrutinib monotherapy (n = 725). Data cut was 31 January 2022.

Results: Of 57 RT patients (median age: 67 years), 91% (n = 52) received at least 1 prior RT-directed therapy. Median lines of prior RT therapy were 2: (anti-CD20 antibody [86%, n = 49], chemotherapy [79%, n = 45], BCL 2 inhibitor [35%, n = 20], BTKi [28%, n = 16], PI3K inhibitor [12%, n = 7], CAR-T [9%, n = 5], immunomodulator [5%, n = 3], stem cell transplant [4%, n = 2], other systemic therapy [33%, n = 19]) and median lines of prior CLL were 2: (BTKi [60%, n = 34], anti-CD20 antibody [60%, n = 34], chemotherapy [47%, n = 27], BCL 2 inhibitor [42%, n = 24], immunomodulator [9%, n = 5], PI3K inhibitor [7%, n = 4], stem cell transplant [7%, n = 4], CAR-T [2%, n = 1], other systemic therapy [11%, n = 6]). ORR for patients within phase 2 (n = 56, 200 mg once/day starting dose) was 54% (95% CI, 39–68): 5 complete responses, 22 partial responses. Median DoR was 8.6 months (95% CI, 1.9-NE, 63% censored) and median OS was 13.1 months (95% CI, 7.1-NE) at a median follow-up time of 5.5 months and 9.7 months, respectively. Six patients electively discontinued pirtobrutinib to pursue curative-intent therapy (allogeneic transplant, n = 6, 3.8 months median time on therapy). The most frequent TEAEs were fatigue (26%, n = 191), diarrhoea (22%, n = 160), and contusion (19%, n = 138). Of Grade ≥3 TEAEs, neutropenia (20%, n = 143) was the most frequent and hypertension (3%, n = 20), haemorrhage (2%, n = 16), and atrial fibrillation/flutter (1%, n = 7) were of low grade. Fifteen (2%) patients discontinued due to a treatment-related AE.

Conclusions: Pirtobrutinib demonstrated promising preliminary efficacy and was well-tolerated for patients who received prior RT-directed chemoimmunotherapy and covalent BTKi.

Presented: ASH2022

BSH23-PO81 | Pirtobrutinib efficacy in covalent BTK-inhibitor pre-treated relapsed/refractory mantle cell lymphoma: Extended follow-up from BRUIN

Dr David John Lewis[1], Dr. Michael L. Wang[2], Dr. Nirav N. Shah[3], Dr. Wojciech Jurczak[4], Dr. Pier Luigi Zinzani[5], Dr. Toby A. Eyre[6], Dr. Chan Y. Cheah[7], Dr. Chaitra S. Ujjani[8], Dr. Youngil Koh[9], Dr. Koji Izutzu[10], Dr. James N. Gerson[11], Dr. Ian Flinn[12], Dr. Benoit Tessoulin[13], Dr. Alvaro J. Alencar[14], Dr. Shuo Ma[15], Dr. Ewa Lech-Maranda[16], Dr. Joanna Rhodes[17], Dr. Krish Patel[18], Dr. Jennifer Woyach[19], Dr. Nicole Lamanna[20], Dr. Yucai Wang[21], Dr. Constantine Tam[22], Dr. John F. Seymour[22], Dr. Talha Munir[23], Dr. Hirokazu Nagai[24], Dr. Francisco Hernandez-Ilizaliturri[25], Dr. Anita Kumar[26], Dr. Andrew Zelenetz[26], Dr. Preteesh Jain[27], Dr. Binoj Nair[28], Dr. Donald E. Tsai[28], Dr. Minna Balbas[28], Dr. Richard A. Walgren[28], Dr. Paolo Abada[28], Dr. Chunxiao Wang[29], Dr. Junjie Zhao[28], Dr. Anthony R. Mato[26]

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Background: Covalent Bruton tyrosine kinase inhibitors (BTKi) have transformed the treatment of mantle cell lymphoma (MCL), though treatment failure occurs through development of resistance/intolerance. Pirtobrutinib is a highly selective, non-covalent (reversible) BTKi. We report updated mantle cell lymphoma (MCL) results from BRUIN.

Methods: Patients with previously treated B-cell malignancies, including MCL, were eligible for pirtobrutinib monotherapy in dose-escalation/expansion portions of BRUIN. Key endpoints: ORR per Lugano 2014 criteria, DOR, and safety. The reported cohort was the first 90 MCL patients enrolled to phase 1/2 who had measurable disease, had

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received a prior covalent BTKi-containing regimen, and had no known central nervous system involvement. Data cut was 31 January 2022. The safety cohort was patients with B-cell malignancies who received at least one dose of pirtobrutinib monotherapy (n = 725).

Results: Of 90 patients with MCL who had received a prior BTKi (median age 70 [46–87] years), the median number of prior lines of therapy was 3 (1–8). Additional prior therapies included anti-CD20 antibody (96%), chemotherapy (88%), immunomodulator (21%), stem cell transplant (21%), BCL-2 inhibitor (16%), CAR-T cell therapy (4%), and PI3K inhibitor (3%). Most patients (82%, n = 74) discontinued prior BTKi therapy due to disease progression. 86% (n = 77) received the recommended phase 2 dose (200 mg once daily) as starting dose. The ORR by IRC was 58% (95% CI, 47–68): 18 complete responses (20%), 34 partial responses (38%). Responses were observed in 6/8 pts (75%) with blastoid MCL and in 6/12 pts (50%) with pleomorphic MCL. At a median follow-up time of 12 months, the median DOR among the 52 responding patients was 22 months (95% CI, 7.5-not estimable [NE]). 12-month and 18-month estimated DOR rates were 57% (95% CI, 39–72) and 52% (95% CI, 34–68), respectively. Patients who discontinued prior BTKi due to disease progression (n = 74) had an ORR of 50% (95% CI, 38–62) and a median DOR of 14.8 months (95% CI, 5.6-NE). In the safety cohort the most common TEAEs were fatigue (26%), diarrhoea (22%), and contusion (19%). The most frequent Grade ≥3 TEAE was neutropenia (20%). Low rates of Grade ≥3 TEAEs of hypertension (3%), haemorrhage (2%), and atrial fibrillation/flutter (1%) were observed. 15 (2%) patients discontinued due to a treatment-related AE.

Conclusion: Pirtobrutinib continues to demonstrate promising, durable efficacy in heavily pretreated R/R MCL patients who have been treated with a prior covalent BTKi. Pirtobrutinib was well-tolerated, with low rates of discontinuation due to drug-related toxicity.

Presented:ASH2022.

BSH23-PO82 | Efficacy of pirtobrutinib, a highly selective, non-covalent (reversible) BTK inhibitor in relapsed/refractory Waldenström macroglobulinemia

Dr David John Lewis[1], Dr. M Lia Palomba[2], Dr. Manish R. Patel[3], Dr. Toby A. Eyre[4], Dr. Wojciech Jurczak[5], Dr. Thomas Gastinne[6], Dr. Shuo Ma[7], Dr. Jonathon B. Cohen[8], Dr. Krish Patel[9], Dr. Jennifer R. Brown[10], Dr. Lydia Scarfò[11], Dr. Talha Munir[12], Dr. Ewa Lech-Maranda[13], Dr. Marc S. Hoffmann[14], Dr. Chaitra S. Ujjani[15], Dr. Bita Fakhri[16], Dr. Michael Wang[17], Koji Izutsu[18], Hirokazu Nagai[19], Constantine S. Tam[20], John F. Seymour[20], Joanna M. Rhodes[21], Julie Vose[22], Matthew McKinney[23], James N. Gerson[24], Minal A. Barve[25], Bryone Kuss[26], Youngil Koh[27], Wei Gao[28], Amy S. Ruppert[28], Richard A. Walgren[29], Donald E. Tsai[29], Binoj Nair[29], Katherine Bao[29], Anthony R. Mato[2], Chan Y. Cheah[30]

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Background: Covalent Bruton tyrosine kinase inhibitors (BTKi) have been an important advancement of Waldenström macroglobulinemia (WM) treatment, but they are non-curative, and treatment effectiveness can be limited by intolerance and resistance. Pirtobrutinib is a highly selective, non-covalent (reversible) BTKi. We report the first report on Waldenström macroglobulinemia (WM) patients from BRUIN.

Methods: Patients with previously treated B-cell malignancies, including WM, were eligible for pirtobrutinib monotherapy in dose-escalation/expansion portions of BRUIN. Key endpoints: investigator-assessed ORR and DoR, per modified IWWM6 criteria, and safety. Major response was CR, very good partial response (VGPR)/PR. The response-evaluable cohort was relapsed/refractory (R/R) WM patients in phase 1/2 who had undergone first response assessment/discontinued therapy. The safety cohort was patients with B-cell malignancies receiving at least one pirtobrutinib monotherapy dose (n = 725). Data cut: 31 January 2022.

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Results: Of 78 WM patients (median age: 68 [42–84] years), the median number of prior therapies was 3 (1–11). 66 (85%) patients had received chemotherapy+anti-CD20 antibody (CIT), 61 (78%) ≥1 prior BTKi (≥2: 13/61, 21%), and 50 (64%) CIT + BTKi. Of 61 patients who received ≥1 prior BTKi, 40 (66%) discontinued BTKi due to disease progression. 91% (71/78) received the recommended phase 2 dose (200 mg once daily) as starting dose. The major response rate for 72 response-evaluable patients was 68% (95% CI, 56–79); 17 VGPRs (24%), 32 PRs (44%). At a median response follow-up time of 7.7 months, median DoR among 49 responding patients was not reached (95% CI, 10-NE); 6-month estimated DoR rate: 86% (95% CI, 69–94). Of 55 response-evaluable patients who had received ≥1 prior BTKi, the major response rate was 64% (95% CI, 50–76); 13 VGPRs (24%), 22 PRs (40%) (median DoR also not reached [95% CI, 8-NE], 6-month estimated DoR rate: 83% [95% CI, 60–93]). Of 60 response-evaluable patients who had received CIT, the major response rate was 68% (95% CI, 55–80); 13 VGPRs (22%), 28 PRs (47%). Of 44 response-evaluable patients who had received CIT+BTKi, the major response rate was 64% (95% CI, 48–78); 9 VGPRs (21%), 19 PRs (43%). In the safety cohort of all pirtobrutinib-treated patients with B-cell malignancies (n = 725), the most frequent TEAEs were fatigue (26%), diarrhoea (22%), contusion (19%); most frequent Grade ≥3 TEAE: neutropenia (20%). Low rates of Grade ≥3 TEAEs hypertension (3%), haemorrhage (2%), atrial fibrillation/flutter (1%) were observed. Overall, 15 (2%) patients discontinued due to treatment-related AE.

Conclusions: Pirtobrutinib was highly active in WM patients, regardless of prior therapy. Pirtobrutinib was well-tolerated with low discontinuation rates.

Presented: ASH2022

BSH23-PO83 | DNA hybridisation for diagnosis of T and B-cell malignancies

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Background: Current multiplex polymerase chain reaction (PCR)-based clonality assays are useful gold-standard tests for diagnosis of T and B-cell malignancies, but have several limitations including their time-consuming nature and the need for in-depth training in interpretation. We have previously developed a method for assessment of biopsy sample clonality involving sequential targeted DNA capture hybridisations of all T-cell receptor (TCR) and B-cell receptor (BCR) loci, which offers several advantages over PCR clonality assays. However, a high proportion of non-recombined TCR and BCR DNA sequences may limit efficiency and sensitivity.

Objective: We examined the effect of adding a novel subtractive DNA hybridisation step to our hybridisation-based method for clonality assessment. We reasoned that using probes to deplete non-recombined DNA fragments may increase the proportion of recombined sequences for analysis.

Method: Hybridisation capture reactions used biotinylated probes targeting all V and J segments across all seven TCR and BCR loci. Subtraction probes were designed to target sequences immediately downstream of V segments and upstream of J segments, which would have been excised if the adjacent segment had undergone V(D)J recombination. DNA extracted from bone marrow, blood, and lymph node biopsy samples surplus to diagnostic requirements underwent capture hybridisations with and without a following subtractive hybridisation step. Samples derived from 28 patients with T or B-cell malignancies and 24 healthy potential bone marrow donors as controls. Rearranged TCR and BCR sequences were identified from sequencing data using MiXCR v3.0.13.

Results: Subtractive hybridisation did not improve the proportion of recombined TCR and BCR sequences detected (2.04% vs. 2.25%, p = 0.06). Surprisingly, sequence diversity as measured by the number of distinct segment rearrangements increased following subtractive hybridisation (median 623 vs. 522, p = 0.0001). However, we found that our method correctly identified malignant samples as exhibiting significantly higher percentages of clonal sequences than benign controls regardless of whether subtractive hybridisation was performed.

Conclusion: Our results suggest that V/J segment capture with subtractive hybridisation of unrearranged sequences is unlikely to be a viable approach for increasing sequencing efficiency or accuracy in TCR/BCR sequence analysis. Nonetheless, our results indicate that sample clonality can be inferred based on DNA hybridisation-based techniques both with and without the use of subtractive hybridisation. Further work is required to explore the usefulness of hybridisation-based assays for immunoreceptor profiling beyond cancer diagnosis and to optimise the bioinformatic pipeline for analysis of B-cell malignancies where somatic hypermutation has occurred.

BSH23-PO84 | Pirtobrutinib efficacy in covalent BTK-inhibitor pre-treated relapsed/refractory CLL/SLL: Additional patients and extended follow-up from BRUIN

Dr Talha Munir[1], Dr. Anthony R. Mato[2], Dr. Jennifer A. Woyach[3], Dr. Jennifer R. Brown[4], Dr. Paolo Ghia[5], Dr. Krish Patel[6], Dr. Toby A. Eyre[7], Dr. Ewa Lech-Maranda[8], Dr. Nicole Lamanna[9], Dr. Constantine S. Tam[10], Dr. John F. Seymour[10], Dr. Nirav N. Shah[11], Dr. Catherine C. Coombs[12], Dr. Chaitra S. Ujjani[13], Dr. Manish R. Patel[14], Dr. Bita Fakhri[15], Dr. Chan Y. Cheah[16], Dr. Alvaro J. Alencar[17], Dr. Jonathon B. Cohen[18], Dr. James N. Gerson[19], Dr. Ian W.

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Flinn[20], Dr. Shuo Ma[21], Dr. Deepa Jagadeesh[22], Dr. Joanna M. Rhodes[23], Dr. Francisco Hernandez-Ilizaliturri[24], Dr. Pier Luigi Zinzani[25], Dr. Minna Balbas[26], Dr. Binoj Nair[26], Dr. Paolo Abada[26], Dr. Chunxiao Wang[27], Dr. Denise Wang[26], Dr. Donald E. Tsai[26], Dr. William G. Wierda[28], Dr. Wojciech Jurczak[29]

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Background: Covalent Bruton tyrosine kinase inhibitors (BTKi) have transformed the treatment landscape of chronic lymphocytic leukaemia/small lymphocytic lymphoma (CLL/SLL), though failure often occurs through development of resistance or intolerance. Pirtobrutinib is a highly selective, non-covalent (reversible) BTKi. We report updated chronic lymphocytic leukaemia/small lymphocytic lymphoma (CLL/SLL) results from BRUIN.

Methods: Patients with previously treated B-cell malignancies, including CLL/SLL, were eligible for pirtobrutinib monotherapy in dose-escalation/expansion portions of BRUIN. Key endpoints: ORR per 2018 iwCLL response criteria, PFS, and safety. The reported cohort was CLL/SLL patients enrolled to phase 1/2 who had received a prior covalent BTKi-containing regimen and had undergone their first response assessment/discontinued therapy. The safety cohort was patients with B-cell malignancies who received at least one dose of pirtobrutinib monotherapy (n = 725). Data cut was 31 January 2022.

Results: Of 276 patients with CLL/SLL who had received a prior BTKi, (median age: 69 [36–88] years) the median number of prior therapies was 3 (1–11). Additional prior therapies included anti-CD20 antibody (89%), chemotherapy (80%), BCL2 inhibitor (44%), PI3K inhibitor (24%), CAR-T cell therapy (6%), and stem cell transplantation (2%). High-risk features were frequent: del(17p) in 29% (58/197), mutated TP53 in 40% (91/230), unmutated IGHV in 85% (188/220). Most patients (n = 206, 75%) discontinued prior BTKi therapy due to disease progression. Overall, 84% (n = 232) received the recommended phase 2 dose (200 mg once daily) as starting dose. The investigator-assessed ORR was 74% (95% CI, 68–79): 3 complete responses (1%), 174 partial responses (PR; 64%), 23 PRs with lymphocytosis (PR-L; 8%), and 1 nodular PR (<1%). At a median follow-up of 13.9 months, the median PFS was 19.4 months (95% CI, 16.6–22.3). 12-month and 18-month estimated PFS rates were 68% (95% CI, 62–74) and 54% (95% CI, 46–61), respectively. In the safety cohort (n = 725), the most common TEAEs were fatigue (26%, n = 191), diarrhoea (22%, n = 160), and contusion (19%, n = 138). The most frequent Grade ≥3 TEAE was neutropenia (20%, n = 143). Low rates of Grade ≥3 TEAEs of hypertension (3%, n = 20), haemorrhage (2%, n = 16), and atrial fibrillation/flutter (1%, n = 7) were observed. Overall, 15 (2%) patients discontinued due to a treatment-related AE.

Conclusion: Pirtobrutinib continues to demonstrate promising, durable efficacy in heavily pretreated R/R CLL/SLL patients who have been treated with a prior covalent BTKi, regardless of prior therapy, reason for prior BTKi discontinuation, age, high-risk TP53 mutations, C481 mutational status, and/or del(17p). Pirtobrutinib was well-tolerated with low rates of discontinuation due to drug-related toxicity.

Presented: ASH2022

BSH23-PO85 | Hodgkin lymphoma: The role of EBV plasma viral load testing in an HIV endemic setting

Assoc. Prof. Jessica Opie[1,2], Associate Professor Estelle Verburgh[2,3], Dr Dharshnee Chetty[1,2], Dr Zainab Mohamed[2,3], Ms Jenna Oosthuizen[2,3], Associate Professor Diana Hardie[1,2]

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Background: In South Africa (SA) more than 8 million people are living with HIV in 2022. The incidence of classical Hodgkin

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lymphoma (cHL) in HIV is markedly increased, including for those receiving antiretroviral therapy (ART). HIV positive (+) cHL patients are known to have EBV + tumour cells, compared to about 40% of HIV negative (–) cHL patients. Apoptotic EBV+ tumour cells shed cell-free (cf) EBV DNA into the plasma, and plasma EBV DNA levels have shown concordance with EBER-ISH tumour status. Thus, plasma EBV-DNA has great potential utility as a biomarker in a resource restricted HIV endemic context where access to PET-CT scanning is limited.

Aim: In newly diagnosed cHL HIV+ and HIV– patients:

- To measure whole blood (WB) and plasma EBV DNA using quantitative PCR and correlate these levels with EBV tumour status.
- To establish the sensitivity and specificity of WB and plasma EBV DNA levels in EBV tumour + cHL.

Methods: Prospective study. The EBV viral load was measured (Abbott Real Time) in WB and plasma on EDTA samples from newly diagnosed cHL patients prior to chemotherapy.

Results: 54 patients were studied, with a median age of 35 (IQR 26–51). 31 (57%) were female and 17 (31%) HIV+ve; 65% of these were virally suppressed on ART. 23/54 (43%) of all patients had EBV+ cHL based on diagnostic tissue staining; 17/17 (100%) of HIV+ and 4/38 (10.5%) of HIV–.

An EBV viral load of >50 IU/mL in WB showed a sensitivity of 100% (95% CI 85.7–100) and specificity of 58.06% (95% CI 40.77–73.58) for predicting EBV+ tumour status, whilst a plasma viral load of >50 IU/mL had a sensitivity of 95.65% (CI 79.01–99.78) and specificity of 80.65% (CI 63.72–90.81). Receiver operative curve (ROC) analysis showed that cut offs of: >952 IU/mL in WB, or >762 IU/mL in plasma best predicted EBV+ tumour status.

Conclusions: All HIV+ cHL patients were EBV tumour + compared to 10.5% of HIV–. Detectable WB EBV DNA is a sensitive indicator of EBV+ tumour status but lacks specificity. Plasma EBV DNA has greater specificity, but lower sensitivity. In patients with EBV+ cHL, EBV DNA plasma and WB levels were closely matched, reflecting that circulating EBV DNA is tumour derived cfDNA circulating in plasma. EBV DNA plasma testing is, therefore, a potential biomarker of particular significance in HIV endemic settings where virtually all cHL cases are EBV tumour +ve.

BSH23-PO86 | The role of MYC homology domains in the control of in vitro B-cell differentiation

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MYC proto-oncogene dysregulation contributes to the molecular pathogenesis of multiple subsets of B-cell lymphoma (BCL): with clinically relevant implications for disease phenotype and classification. In aggressive B-cell lymphoma, co-deregulation of MYC and B-cell lymphoma 2 (BCL2) by translocation (double hit), or through strong co-expression without translocation (double expressor), can identify partially overlapping high-risk disease.

The protein interaction networks necessary for MYC effector function are heavily influenced by six highly conserved homology boxes (MB domains), of which MB0, MBI and MBII have been previously linked to the development of cancer. Whilst existing studies have begun to define the unique protein interactory networks of MB0-MBII, this research offers a novel approach to the characterisation of MB domain influence in human B-cell differentiation, with particular relevance to BCL.

Anonymous donor memory B-cell populations were cultured within an established in-vitro model of B-cell differentiation and subsequently transduced with viral vectors encoding full-length or MYC homology box deletion variants alongside BCL2 and a selectable marker. Transduced cells successfully overexpressed both full-length MYC, MYC variants and BCL2 proteins: allowing the specific effects of MYC MB domain deletions to be tracked as B cells differentiated. The differentiating B cells were assessed to determine the extent of cluster formation, taken to be indicative of cellular proliferation, the expansion of the successfully transduced subset and cell surface marker expression (CD19, CD20, CD27, CD38, CD138).

Concurrent overexpression of full-length MYC and BCL2 drove successfully transduced population expansion as expected, with analysis of cell surface markers suggesting the maintenance of an undifferentiated state. Cells with MBI deletion MYC variant protein recapitulated this effect with little variation from full-length MYC overexpression results across the tested assays. In contrast, MB0 deletion and, in particular, MBII deletion, did lead to differences in cellular behaviour. Deletion of MB0 was linked to a marked reduction in cluster formation across donor samples. Meanwhile, flow-cytometric analysis of cell surface marker expression interestingly revealed that MBII deletion was associated with clear indications of physiological plasmablast differentiation, akin to the non-transduced populations, rather than the prolonged undifferentiated state of the wildtype MYC overexpressing cells.

These results suggest distinct roles for MB0 and MBII within MYC-driven B-cell transformation, with the overexpression of MB0 and MBII deletion MYC variant protein resulting in attenuation of typical MYC overexpression phenotype.

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BSH23-PO87 | BGB-11417 (Bcl-2 inhibitor) monotherapy or combination with Zanubrutinib in non-Hodgkin lymphoma or Waldenström macroglobulinemia patients

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Background: The combination of Bcl-2 and Bruton tyrosine kinase (BTK) inhibitors is tolerable with synergistic activity in chronic lymphocytic leukaemia, small lymphocytic lymphoma, and mantle cell lymphoma (MCL). BGB-11417 is a novel Bcl-2 inhibitor that is more potent and selective than venetoclax. Zanubrutinib, a next-generation BTK inhibitor, has favourable activity and safety in B-cell malignancies. BGB-11417-101 is an ongoing phase 1/1b dose-escalation/expansion study (NCT04277637). Data from separate cohorts of MCL, Waldenström macroglobulinemia (WM), and combined non-Hodgkin lymphoma (NHL; follicular lymphoma [FL], diffuse large B-cell lymphoma [DLBCL], marginal zone lymphoma [MZL], transformed FL) are presented.

Methods: Monotherapy: BGB-11417 40, 80, 160, 320, or 640 mg QD with a ramp-up to the intended dose. Combination: zanubrutinib (320 mg QD or 160 mg BID) 8-12 weeks before BGB-11417. Dose-limiting toxicity was evaluated during ramp-up through intended dose day 21. Responses were assessed per Lugano criteria. Adverse events (AEs) were reported per Common Terminology Criteria for AEs v5.0.

Results: As of 15 May 2022, 45 patients with NHL, WM, or MCL received BGB-11417 (34 monotherapy; 11 combination). Monotherapy: $n = 28$ (NHL) and $n = 6$ (WM) received BGB-11417 doses ≤640 mg. Combination: $n = 11$ (MCL) received zanubrutinib; 9 (82%) also received BGB-11417 doses ≤160 mg (includes 2 patients in zanubrutinib pretreatment). Escalation to 640 mg was completed for NHL monotherapy; maximum tolerated dose (MTD) was not reached. Dose escalation is ongoing for monotherapy in WM and combination in MCL. Median follow-up: Monotherapy, 6.5 months (range 0.4–25.3); Combination, 4.8 months (range 0.4–8.9). Most common treatment-emergent AEs (TEAEs): Monotherapy, nausea (38%); grade ≥3, neutropenia (12%); Combination, contusion, and neutropenia (27% each); grade ≥3, neutropenia and thrombocytopenia (9% each). Treatment discontinuations: Monotherapy, $n = 22$ disease progression (PD), $n = 1$ AE, $n = 2$ other reasons; Combination, $n = 2$ PD. No TEAEs lead to death; no tumour lysis syndrome was reported. Of 23 patients who reached the first response assessment time point (most below recommended phase 2 dose [RP2D]), 3 responded ($n = 2$ DLBCL, $n = 1$ MZL) including 1 CR (DLBCL). MCL combination cohort, 6/11 (55%) responded. WM monotherapy cohort, 1/4 (25%) exhibited minor response at 80 mg. Haemoglobin count increases >20 g/L were seen in 3/6 treated patients; all remain on treatment.

Conclusion: Initial data show an encouraging safety profile and evidence of efficacy for BGB-11417 in NHL, MCL, and WM cohorts. MTD was not reached at highest dose (640 mg QD). All low-grade TEAEs and grade ≥3 neutropenia were manageable. Longer follow-up of BGB-11417 monotherapy and combination therapy at the RP2D is needed. Monotherapy MCL data are forthcoming.

BSH23-PO88 | Assessment of durable responses after brexucabtagene autoleucel (brexu-cel) in relapsed/refractory mantle cell lymphoma (R/R MCL)

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Introduction: Brexu-cel is an autologous anti-CD19 CAR-T cell therapy approved in the US for adults with R/R MCL and in the EU for adults with R/R MCL after ≥2 prior therapies, including a Bruton tyrosine kinase inhibitor (BTKi). With 35.6-months median follow-up in ZUMA-2, brexu-cel demonstrated an objective response rate of 91% with median duration of response (DOR) of 28.2 months in R/R MCL (Wang et al. J Clin Oncol. 2022).

Methods: Adults with R/R MCL (≤5 prior regimens including a BTKi) underwent leukapheresis and conditioning chemotherapy followed by one infusion of brexu-cel (2×10^6 anti-CD19 CAR-T cells/kg). Post hoc assessments of patient and product characteristics are reported by response status at 24 months postinfusion (ongoing vs. relapsed).

Results: With 35.6-months median follow-up, 59 patients with response were evaluable, 29 (49%) had ongoing responses at 24 months (ongoing responders) and 30 (51%) had relapsed prior to 24 months (relapsed responders). At baseline, median age was 65 years and median number of prior therapies was 3 in both subgroups. Among ongoing vs. relapsed responders, 66% vs. 43% had ibrutinib as last prior therapy and 14% vs. 13% had acalabrutinib as last prior therapy, with median time from last prior therapy of 63 vs. 64.5 months, respectively. A smaller proportion of ongoing vs. relapsed responders received bridging therapy (21% vs. 53%) and prior platinum therapy (10% vs. 40%); whilst similar proportions received prior bendamustine therapy (45% vs. 53%), prior proteosome inhibitor therapy (41% and 37%) and prior autologous stem cell transplant (48% vs. 37%), respectively.

Among ongoing vs. relapsed responders, 79% vs. 57% had baseline ECOG score of 0, 66% vs. 60% had Ki-67 proliferation index scores ≥30%, 45% vs. 37% had elevated lactate

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dehydrogenase levels and median tumour burden (SPD) was 935.1 and 4233.6, respectively.

Median DOR in ongoing vs. relapsed responders with CR (n = 28 vs. n = 15) was not reached and 8.3 months, respectively.

Median peak (102.4 vs. 59.9) and area under the curve (1487 vs. 688.2) CAR-T cell levels were ~2× higher in ongoing vs. relapsed responders and median total number of infused CCR7+ cells (10^6) was 119.8 vs. 89.1, respectively.

Conclusions/Summary: After ~3-years follow-up, brexucel continues to demonstrate durable responses with 49% of patients in ongoing response at 24 months. Ongoing responders had lower ECOG scores and tumour burden, and less frequent bridging therapy use, suggesting potential for greater benefit with brexu-cel if given in earlier course of disease.

BSH23-PO89 | Relationship between skin, blood and lymph node responses in mogamulizumab-treated patients: MAVORIC trial post-hoc analysis

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The Phase 3 MAVORIC trial (NCT01728805) showed superior overall response rate (ORR; 28.0% vs. 4.8%, $p < 0.0001$) for mogamulizumab versus vorinostat in patients with relapsed or refractory mycosis fungoides (MF) or Sézary syndrome (SS).

This post-hoc analysis evaluated characteristics of responders (complete [CR] and partial [PR]) to mogamulizumab focusing on skin, blood and lymph nodes.

In MAVORIC, treatment response was assessed according to international consensus guidelines (Olsen, 2011). In this analysis, best responses by disease compartment and their inter-relationships were considered alongside patient disease subtype (MF/SS).

A compartmental best response was observed in 97/124 (78.2%), 81/186 (43.5%), and 24/136 (17.6%) patients in blood, skin, and lymph nodes, respectively.

Combined blood and skin CRs were observed in seven patients (MF n = 1; SS n = 6). Skin responses were most frequent in patients with a best blood response of CR (79.4%) versus those with a best blood response of PR (36.4%) or stable disease (SD; 18.8%). Skin responses were statistically associated with blood response ($p < 0.001$).

Combined blood and lymph nodes CRs were observed in seven patients (MF $n = 2$; SS $n = 5$). Responses in lymph nodes were more frequent in patients with blood involvement (B1 or B2) at baseline than those without (B0) (21.1% versus 11.1%, respectively) and more frequent in patients with a best blood response of CR (33.3%) compared to those with blood PR (10.4%) or SD (0.0%). Lymph nodes responses were statistically associated with blood response ($p = 0.011$).

Even in early stages, primary cutaneous T-cell lymphoma may involve blood, lymph nodes, and viscera. Recommended skin-directed therapies do not address other disease compartments. Complete responses in MF/SS in any compartment are rare but often found in the Blood of patients treated with Mogamulizumab. Here we report CRs in skin and LN of Mogamulizumab-treated patients and show that those with blood responses were more likely to have significant skin or lymph nodes responses. Consistent with global ORR and skin responses, mogamulizumab-treated patients with blood involvement were more likely to have a response in lymph nodes.

BSH23-PO90 | Redirecting NK cells to the lymph nodes to augment their lymphoma-targeting capacity

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Natural Killer (NK) cells are cytotoxic lymphocytes, which play an important role in the immune defence against viral infections and cancer. It has previously been shown that adoptive transfer of haploidentical NK cells can induce remission or prevent relapse in some patients with haematological malignancies. However, the benefit of NK cell therapy may be limited by suboptimal in vivo persistence of the cells and poor homing to tumour-bearing tissues. In this work, we have used mRNA electroporation to generate NK cells overexpressing the lymph node homing molecules CCR7, CXCR5 and CD62L, aiming to enhance lymph node infiltration and, therefore, lymphoma targeting.

Electroporation of ex vivo-expanded human NK cells with CCR7, CXCR5 and/or CD62L mRNA resulted in high levels of surface expression of these molecules detectable at 6 h after electroporation. CCR7 and CXCR5 mRNA-electroporated cells displayed up to 20-fold increase in vitro migration capacity towards the corresponding chemokine ligands CCL19, CCL21 or CXCL13. mRNA electroporation did not have any negative effects on the cell viability or on

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NK cell degranulation and cytotoxicity against different types of target cells, when triggered by natural cytotoxicity pathways or via antibody-dependent cellular cytotoxicity. Adoptive infusion of CCR7, CXCR5 and CD62L mRNA-electroporated NK cells into SCID/Beige mice resulted in a significant increase in the number of NK cells in the lymph nodes of these mice, compared to control mice who received sham-electroporated cells. Importantly, NK cells derived from lymphoma patients were expanded and modified equally as efficiently as healthy donor-derived NK cells, highlighting the translational potential of this approach.
In conclusion, our data establish proof of concept that adoptively infused NK cells can be redirected to the lymph nodes by overexpressing key lymph node homing molecules via mRNA electroporation. This approach could potentially be used in synergy with chimeric antigen receptors (CAR), bi- or tri-specific killer engagers (BiKEs or TriKEs), or monoclonal tumour-targeting antibodies to increase the efficacy of NK cell immunotherapy for patients with lymphoma and lymph node metastases.

BSH23-PO92 | Correlation of MYD88 and TNFAIP3 mutations with clinico-haematological profile in diffuse large B-cell lymphoma

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Activated B-cell–like (ABC) subtype of diffuse large B-cell lymphoma (DLBCL) is characterised by chronic active B-cell receptor signalling and a constitutive activation of the nuclear factor kappa B (NF-KB) pathway. As a driving force of NF-KB overactivity, myeloid differentiation primary response gene 88 (MYD88) L265P mutation occurs in ABC-DLBCL. However, the MYD88 L265P mutation has not yet been studied with tumour necrosis factor alpha induced protein3 (TNFAIP3) mutation in DLBCL cases.
Aim: to determine the prevalence of MYD88 and TNFAIP3 mutations and their relationship with the clinical and haematological parameters in DLBCL.
Methods: Fresh lymph nodes samples were taken from 100 DLBCL patients for detection of MYD88 L265P and TNFAIP3 mutation by using real-time polymerase chain reaction.
Results: MYD88 L265P CT genotype was found in 20% of cases. CT mutant genotype was significantly frequent in ABC type, stage IV, higher IPI groups, extranodal infiltration and BM infiltration. It was also associated with shorter overall survival (OS).
TNFAIP3 mutation GA genotype was found in 18% of the patients where 77.8% of them were ABC-DLBCL subtype. GA mutant genotype was frequently associated with stage IV, and extranodal infiltration and shorter OS.
COX Regression analysis conducted for prediction of shorter disease-free survival DFS and OS using age, gender, type, staging, ECOG, IPI, extra nodal, B symptoms, MYD88 L265P and TNFAIP3 mutation as covariates. In multivariate analysis Only the CT was a significant risk factor for shorter DFS and OS whilst GA genotype was a significant risk factor for shorter DFS only.
Conclusion: Our study suggests a role of MYD88 mutation in pathogenesis and prognosis of DLBCL.

BSH23-PO93 | R-CHOP and R-Bendamustine-treated lymphoma patients have an unexpectedly high hospitalisation rate during 2 years post-treatment

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Introduction: Treating lymphoma with either rituximab plus cyclophosphamide, doxorubicin, vincristine and prednisolone (R-CHOP) or bendamustine (R-Bendamustine) commonly causes infections but it remains unclear in real-world practice how common complications requiring hospitalisation are with these regimens. With this study, we aim to quantify this risk.
Methods: This is a single-centre retrospective study of 200 patients (100 standard-dose R-CHOP and 100 R-Bendamustine) treated between 2014 and 2020. Eligible patients all received at least two cycles and admissions were recorded from start of treatment until relapse, death or 2 years post-treatment, whichever occurred first. Lymphocyte count and IgG were recorded pretreatment, at the end of treatment, and 2 years post-treatment. Default primary prophylaxis differed: filgrastim with R-CHOP and aciclovir/co-trimoxazole with R-Bendamustine.
Results: Sex distribution was similar (55% male), but R-Bendamustine patients were significantly older (median age 68.8 (range: 32.6–85.2) vs. 63.1 (range: 18.3–78.2) years, $p < 0.001$) and 25% had received 1+ prior line of chemotherapy (8% for R-CHOP). The R-CHOP patients were predominantly diffuse large B-cell lymphoma (83%) whilst R-Bendamustine patients were 38% follicular lymphoma, 21% CLL, 20% lymphoplasmacytic lymphoma. Significantly more patients in the R-Bendamustine group received rituximab maintenance (44% vs. 20%, $p < 0.001$) and R-Bendamustine patients received significantly more acyclovir/co-trimoxazole prophylaxis (100% vs. 25%, $p < 0.001$).

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The R-CHOP group had an increased number of 'all-cause' hospital admissions over the study period, with 60% of R-CHOP patients having at least 1 admission vs. 46% post R-Bendamustine (p = 0.0652). Infection-related admissions were also more common with R-CHOP (44% vs. 31%, p = 0.0793). Median duration of admission was the same in both groups (7 days 'all-cause'; 6 days 'infection-related'). This is despite the R-CHOP patients having a shorter data collection period as collection stopped before 2 years in 31.7% of R-CHOP and 19.6% of R-Bendamustine patients owing to a shorter R-CHOP 2-year progression-free survival (68.3% vs. 80.4%).

There was no significant difference in IgG levels before, immediately after or 2 years post-treatment between the groups. However, median lymphocyte count was significantly higher pretreatment in the R-Bendamustine group (1.4×10^9/L vs. 1.17×10^9/L, $p < 0.001$) and significantly lower at the end of treatment (0.435×10^9/L vs. 0.65×10^9/L, $p < 0.001$) and 2 years post-treatment (1.05×10^9/L vs. 1.4×10^9/L, p = 0.0113).

Conclusion: Our large real-world dataset revealed an unexpectedly high level of 'all-cause' and 'infection-related' hospitalisation post R-CHOP and R-Bendamustine, although admission rates appeared lower for R-Bendamustine patients, despite these patients being older and more likely to have relapsed disease. The exaggerated lymphopenia seen with R-Bendamustine did not correlate with hospitalisation.

BSH23-PO94 | The B7 family of immune checkpoint proteins in diffuse large B-cell lymphoma (DLBCL)

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Targeting the PD-1/PD-L1 axis has demonstrated unprecedented activity in the treatment of many forms of malignancy but in most subtypes of diffuse large B-cell lymphoma (DLBCL), response rates to immune checkpoint inhibitors are low, suggesting other mechanisms of immune evasion and tumorigenesis (PMID: 27269947, PMID: 30620669). The B7 family of immunomodulatory proteins comprises ten different cell surface proteins that may either stimulate or attenuate immune responses and may regulate anti-tumour immune responses. The expression and possible functional significance of these proteins in DLBCL remain unclear.

We, therefore, sought to determine gene expression levels of B7 family members B7-1 (CD80), B7-2 (CD86), B7-H1 (PD-L1/CD274), B7-DC (CD273/PD-L2), B7-H2 (ICOSLG/CD275), B7-H3 (CD276), B7-H4 B7-H5 (VISTA), B7-H6 (NCR3LG1) and B7-H7 (HHLA2) firstly in a panel of 32 DLBCL cell lines and secondly in publicly available RNA-Seq DLBCL datasets. Cell lines were classified by RNA sequencing. Gene expression analysis data were log2 transformed and analysed using the Mann–Whitney U test. To investigate the impact of B7 family member expression on overall survival (OS), FPKM expression for 200 patient samples with associated clinical data was accessed (GDC Data Portal NCICCR-DLBCL and supplementary data PMID: 2964196).

In the DLBCL cell lines, high level expression of PD-L1/CD274 was seen in two primary mediastinal cell lines (PMBCL) as anticipated; however, expression was also seen in 6 other DLBCL cell lines. In contrast high level expression of ICOSLG/B7-H2/CD275 was seen in 11 DLBCL cell lines; similarly, high level expression of CD80 and CD86 were seen in 44% cell lines. Interestingly high levels of NCR3LG1/B7-H6 were seen in 13, whilst 23 cases expressed low levels of B7-H5/VISTA.

Using publicly available RNA sequencing datasets from patients with DLBCL with matched survival data, B7 gene expression was correlated with overall survival (OS). High level expression of CD276/B7-H3 and ICOSLG were found to be associated with a significant difference in OS at 10 years. In diagnostic samples with high CD276, 10 year OS was 60% versus 40% in those with low (p = 0.047). In tumours with high expression of ICOSLG, 10 year OS was 55% versus 45% in those with low expression (p = 0.006).

These data refer only to RNA and not protein levels. Furthermore, in the patient datasets, expression will have derived at least in part from the tumour microenvironment. However, targeting of B7 proteins may potentially be of relevance for the treatment of DLBCL and further exploration is warranted.

BSH23-PO95 | The relationship between adherence behaviour, adherence barriers, and quality of life in sickle cell disease

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Background: Sickle cell disease (SCD) requires lifelong treatment and management. Whilst hydroxyurea, voxelotor and/or crizanlizumab reduce complications and improves health-related quality of life (HRQOL), adherence remains suboptimal, especially among adolescents and young adults (AYA). Further, patient barriers to adherence are poorly understood.

Objectives: Assess the relationship of patient medication adherence to barriers and HRQOL domain scores and understand factors contributing to variation in medication adherence among AYA with SCD.

Methods: Sixty-one patients (10–25 years old) with SCD (all genotypes), receiving hydroxyurea, voxelotor and/or crizanlizumab participated in this single-institution longitudinal study over 12-months. Participants completed the Adherence to Refills and Medications Scale 7 (ARMS-7),

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Brief Medication Questionnaire (BMQ), and Patient-Reported Outcomes Measurement Information System (PROMIS) HRQOL measures every 3–4 months. Patient characteristics were analysed using descriptive statistics, and Wilcoxon rank-sum test, Kruskal–Wallis test, chi-square and spearman tests to assess the relationships.

Results: Participants (52% Female; 93% Black; median age of 14 years [IQR 12–18]), at enrolment, had a median (IQR) foetal haemoglobin (HbF) of 12.1% (7.9–24.9) and mean corpuscular volume (MCV) of 93.5 fL (80–104.2). Self-report adherence rates (median 12 [IQR 10–14]) were correlated with HbF ($r = -0.44$, $p < 0.001$) and MCV ($r = -0.27$, $p = 0.03$). Participants reported medication recall barriers or forgetfulness (41%), negative beliefs about medication (23%), and difficulty with access to medications, such as paying for refills (32%). Most patients endorsed ≥1 barrier domain ($n = 40$, 66%).

Patients with lower medication adherence were significantly more likely to report forgetfulness ($p \leq 0.01$), access barriers ($p \leq 0.01$) and negative beliefs ($p = 0.04$), compared to patients with higher adherence. Patients with lower adherence reported worse physical stress ($p = 0.01$), psychological stress ($p \leq 0.01$), anxiety ($p \leq 0.01$), depression ($p \leq 0.01$) and self-efficacy managing symptoms ($p = 0.02$). Furthermore, participants with negative beliefs reported significantly worse anxiety ($p = 0.03$) and depression ($p \leq 0.01$) a trend towards significantly more fatigue ($p = 0.06$) and social isolation ($p = 0.08$) than those without negative beliefs. Participants with reported medication access barriers trended towards significantly worse physical stress ($p = 0.06$), anxiety ($p = 0.08$) and pain ($p = 0.08$), compared to those without access barriers.

Conclusion: A significant number of AYA with SCD have at least one medication adherence barrier related to access, forgetfulness, or negative beliefs. Lower medication adherence and negative beliefs were significantly associated with impaired HRQOL. These findings suggest that mitigating barriers to medication access, helping to reduce forgetfulness, and overcoming negative beliefs could contribute to improved HRQOL.

BSH23-PO96 | The impact of illness perception and stigma on patient-reported outcomes in sickle cell disease

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Background: Stigma and illness perceptions of adolescents and young adults (AYA) with sickle cell disease (SCD) are not well-studied. These socio-emotional factors may have implications on patient-reported outcomes, including health-related quality of life (HRQOL) and medication adherence.

Objectives: Describe illness perception and stigma among AYA with SCD and examine the relationship between illness perceptions and stigma of AYA with SCD and HRQOL and medication adherence.

Design/Method: Patients, 10–25 years old, English-speaking, and taking hydroxyurea, voxelotor and/or crizanlizumab for SCD treatment were recruited and enrolled from outpatient haematology clinics at a single academic institution. Participants completed the Brief-Illness Perception Questionnaire (B-IPQ), Measure of Sickle Cell Stigma Questionnaire (MoSCS), Patient-Reported Outcomes Measurement Information System (PROMIS) HRQOL Domains, and Adherence to Refills and Medications Scale 7 (ARMS-7) every 3 months over a 1-year period. Patient characteristics were analysed using descriptive statistics, and Wilcoxon rank-sum, Kruskal–Wallis and spearman's correlations to assess the relationships.

Results: Sixty-one participants completed all assessments (93% Black; 52% Female; median age 14 years [IQR 12–18]). Older participants (18+ years) reported a better understanding of SCD (10 vs. 8, $p = 0.02$) but worse emotional response to SCD (6 vs. 3, $p = 0.03$) compared to younger participants (10–17 years). Patients with higher self-reported medication adherence reported less perceived SCD-related consequences (4 vs. 0, $p = 0.03$) and symptoms (4 vs. 2, $p = 0.01$) and had better treatment control (10 vs. 8, $p = 0.03$). Patients with less perceived personal control of SCD and more emotional response to SCD had worse social isolation ($rs = -0.29$, $p = 0.03$; $rs = 0.33$, $p < 0.01$), mobility ($rs = 0.34$, $p < 0.01$; $rs = -0.26$, $p = 0.04$), anxiety ($rs = -0.26$, $p = 0.04$; $rs = 0.41$, $p < 0.01$), depression ($rs = -0.27$, $p = 0.03$ L; $rs = 0.48$, $p < 0.001$), and fatigue ($rs = -0.39$, $p < 0.01$; $rs = 0.32$, $p = 0.01$), respectively. Further, patients with more perceived personal control had with less physical stress ($rs = -0.50$, $p < 0.001$), pain ($rs = -0.30$, $p = 0.02$), and greater mobility ($rs = 0.34$, $p < 0.01$). Participants with higher negative perceptions of SCD-consequences and concerns reported more physical stress ($rs = 0.30$, $p = 0.02$; $rs = 0.26$, $p = 0.04$), depression ($rs = 0.40$, $p < 0.01$; $rs = 0.52$, $p < 0.001$), and fatigue ($rs = 0.32$, $p = 0.01$; $rs = 0.41$, $p < 0.01$). Most patients (80%) endorsed ≥1 statement related to stigma and those who reported greater stigma had more perceived SCD-related consequences ($p = 0.03$), concerns ($p = 0.02$) and emotional response ($p = 0.04$).

Conclusion: Patients with more perceived stigma, as well as negative perceptions of SCD and related medications reported lower medication adherence and worse HRQOL. These multifaceted and dynamic relationships potentially offer new targets to improve HRQOL and medication adherence in AYA with SCD.

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BSH23-PO97 | Real-World effectiveness of voxelotor for the treatment of sickle cell disease

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Introduction: Voxelotor is a first-in-class sickle haemoglobin polymerisation inhibitor that targets the pathophysiology of sickle cell disease (SCD). Studies have shown that voxelotor increases haemoglobin and reduces markers of haemolysis. Emerging evidence suggests that voxelotor may improve the clinical symptoms of SCD, lower vaso-occlusive crisis (VOC) rates, and reduce transfusion needs.
Objective: To examine the real-world impact of voxelotor on transfusion, VOC, and hospitalisation rates among patients with SCD.
Methods: Medical and pharmacy claims data for patients ≥4 years with SCD who started voxelotor between November 2019 and March 2022 were obtained from the Symphony Health database. Patients with ≥1 year of data before the index date (date of first voxelotor claim) were included. Annualised study outcomes were calculated for patients with ≥1 occurrence of the corresponding event in the 3-month preindex period. Outcomes from a 90-day lookback were reported for the total and paediatric (aged 4 to <18 years) populations.
Results: Of 4023 eligible patients from the Symphony Health database included in the analysis, 596 were <18 years. Compared with the 3-month preindex period, significantly lower annualised rates of transfusions, VOCs, and hospitalizations, and lower annualised mean number of inpatient days, were observed in the total population and paediatric subgroup over the 3-month postindex period. For the total population, the annualised event rates declined by 50.6% for transfusions (n = 248), 23.1% for VOCs (n = 1368), 35.5% for VOC-related hospitalizations (n = 757), and 39.4% for all-cause hospitalizations (n = 928). The annualised mean number of inpatient days declined by 29.6% for VOC-related hospitalizations (n = 757) and by 22.9% for all-cause hospitalizations (n = 928). For paediatric patients, the annualised event rates declined by 79.6% for transfusions (n = 18), 42.4% for VOCs (n = 157), 56.8% for VOC-related hospitalizations (n = 81), and 51.5% for all-cause hospitalizations (n = 106). The annualised mean number of inpatient days declined by 54.1% for VOC-related hospitalizations (n = 81) and by 45.8% for all-cause hospitalizations (n = 106).
Conclusions: Treatment with voxelotor may provide a clinical benefit to patients with SCD by reducing the frequencies of transfusions, VOCs, and hospitalizations and decreasing inpatient days. Greater reductions were observed in the paediatric subgroup, potentially due to the smaller sample size, historically greater treatment compliance in paediatric patients, or younger patients having accumulated fewer SCD-related complications, enabling a greater clinical response. Limitations include the study's non-randomized design, reliance on claims data, and changes in healthcare use during the COVID-19 pandemic confounding the data.

BSH23-PO98 | Improving pain management in sickle cell crisis by developing an educational video

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Background: The APPG report into care of patients with Sickle Cell Disease (November 2021) highlights that emergency care of patients with sickle cell crisis requires improvement. NICE guidelines state that all patients who present with Sickle Cell crisis should be offered analgesia within 30 min of presentation. It is clear that prompt analgesia and assessment improves outcomes.
Rationale for project: Audits locally have identified that our trust has been failing to provide analgesia within 30 min for a significant proportion of patients admitted with sickle cell crisis. Following discussion in our local departmental meeting it was identified that there was a lack of awareness of the NICE guidelines as well as the severity and impact of Sickle Cell pain among medical and nursing staff. This was exacerbated by a lack of consistent medical staff in admissions areas. Another key reason was felt to be the lack of simple and rapidly accessible information. We have, therefore, sought to address this through a short educational video for healthcare professionals.
Methodology: Since 2018 we have audited the pain management of patients presenting with sickle cell crisis against the NICE guideline [CG143]. This includes assessing the proportion of patients being offered analgesia within 30 min. In response, we have created a video involving the Digital Medical Education team, aimed at medical and nursing staff caring for patients with sickle cell crisis. The video provides specific and concise guidance about the management of sickle cell crisis based on the NICE guidelines. It also provides helpful recommendations to facilitate prompt prescribing such as advice on dosing and route. The video contains slides with simple flowcharts as well as film footage of the presenter presenting the slides interspersed throughout to make it more engaging.
Results: We have created a short 5 min video highlighting the key points of the NICE and Trust guidelines aimed at clinical teams working in admission areas. A QR code that links to the video was put up in admission areas so that clinical staff can easily access it in an acute situation. We have received positive feedback from staff and an audit is planned to assess its effectiveness.
Implications: In 2021, our yearly audit showed that only 29% of patients were offered analgesia within 30 min of arrival. We hope that following the introduction of our educational video this proportion will increase significantly.

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